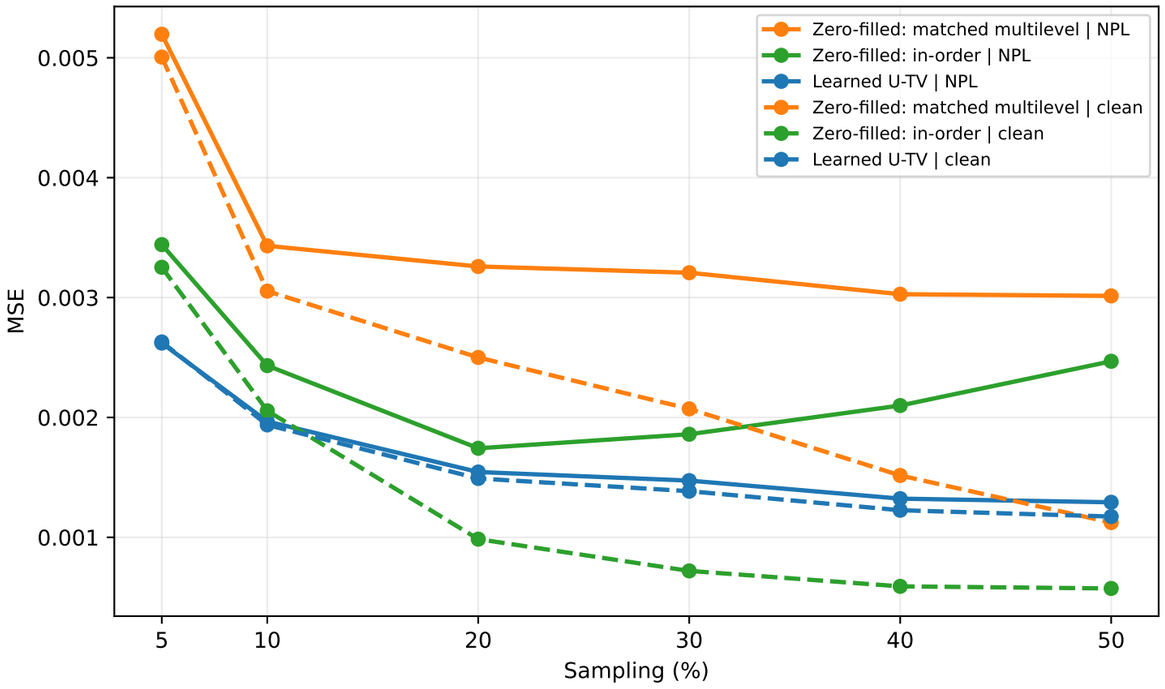
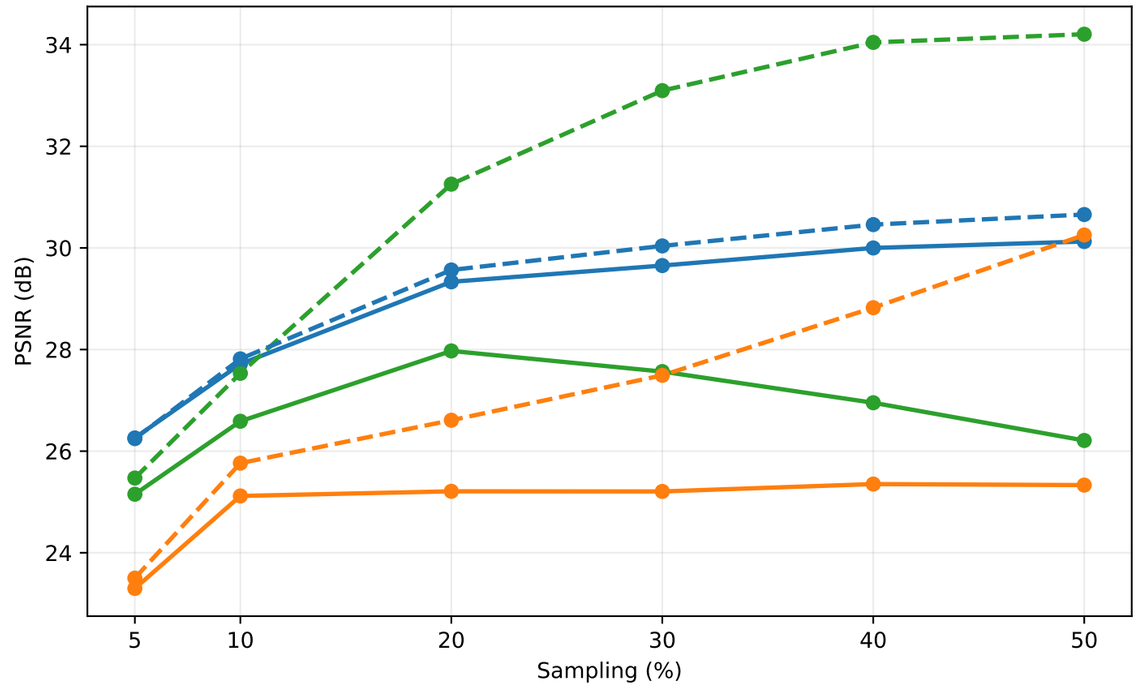
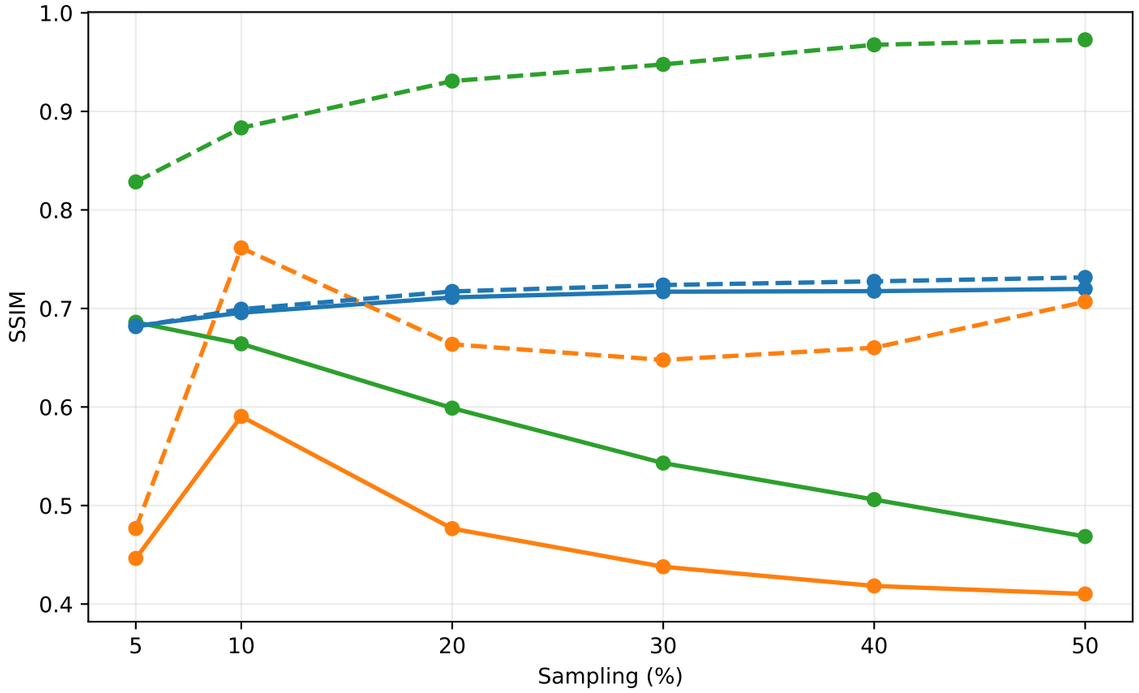
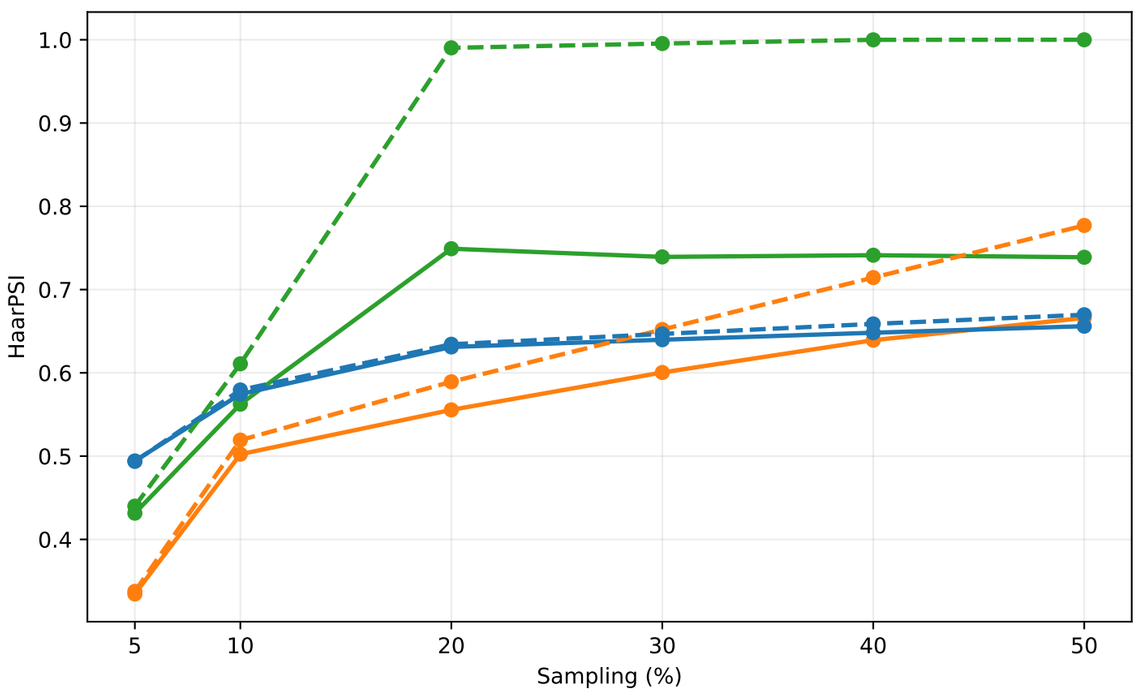
U-TV vs zero-filled baselines | 10 held-out test examples | epoch 5









Methods share original image 256x256; NPL and noiseless masks matched. MSE/PSNR/SSIM/HaarPSI computed on unclipped float images.

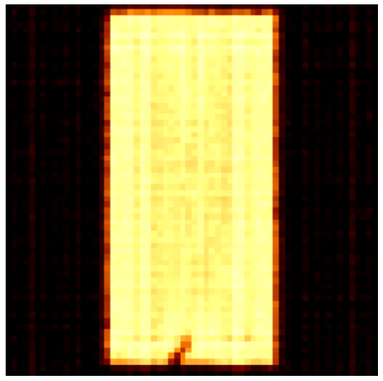
CIGS / image0611 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

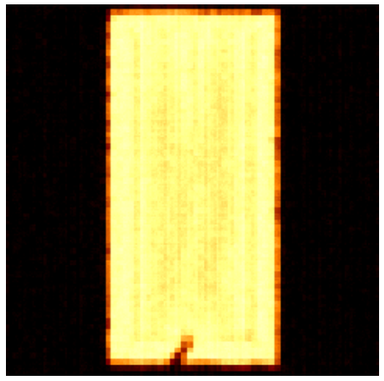


5% sampling



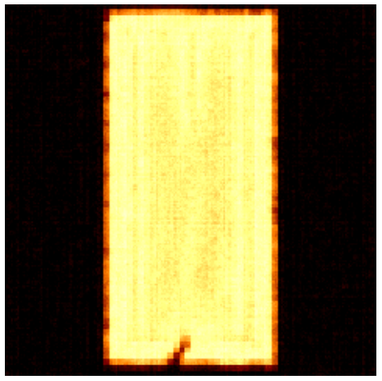
Zero-filled: matched multilevel
MSE 0.00392 | PSNR 24.06 dB
SSIM 0.441 | HaarPSI 0.347

10% sampling



Zero-filled: matched multilevel
MSE 0.00242 | PSNR 26.16 dB
SSIM 0.588 | HaarPSI 0.524

20% sampling



Zero-filled: matched multilevel
MSE 0.00258 | PSNR 25.89 dB
SSIM 0.478 | HaarPSI 0.561

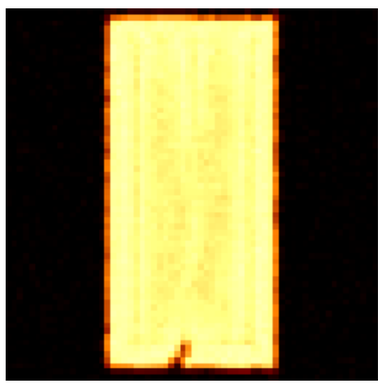
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

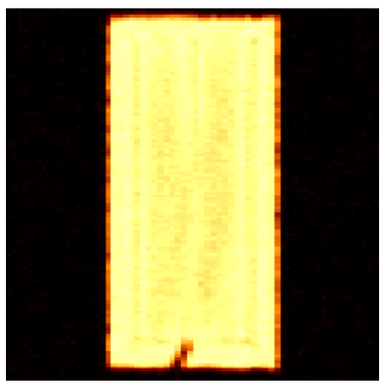
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

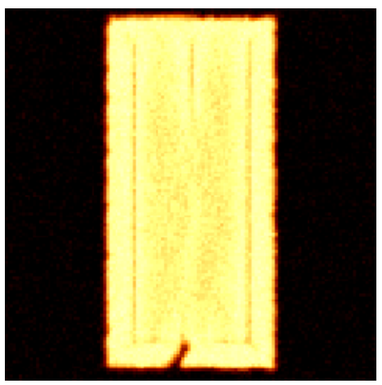
Single learned scale key; coarse J=6; PDHG=512.



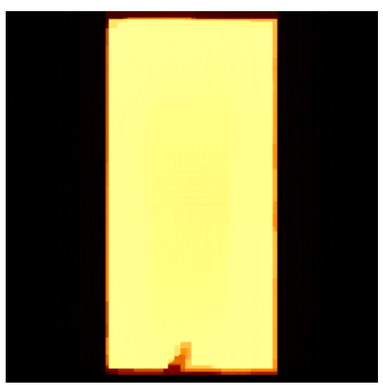
Zero-filled: in-order
MSE 0.00245 | PSNR 26.10 dB
SSIM 0.718 | HaarPSI 0.438



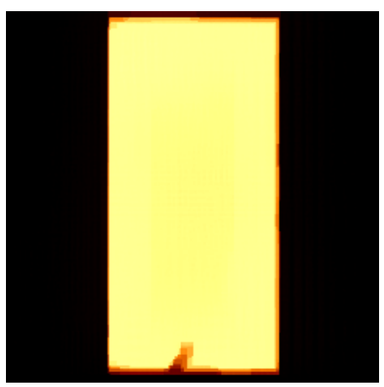
Zero-filled: in-order
MSE 0.00217 | PSNR 26.63 dB
SSIM 0.649 | HaarPSI 0.504



Zero-filled: in-order
MSE 0.00114 | PSNR 29.44 dB
SSIM 0.595 | HaarPSI 0.764



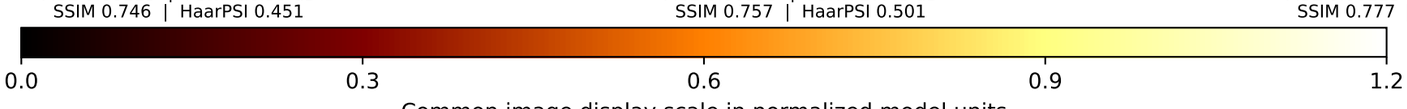
Learned U-TV
MSE 0.00219 | PSNR 26.59 dB
SSIM 0.746 | HaarPSI 0.451



Learned U-TV
MSE 0.00179 | PSNR 27.48 dB
SSIM 0.757 | HaarPSI 0.501



Learned U-TV
MSE 0.00123 | PSNR 29.08 dB
SSIM 0.777 | HaarPSI 0.546



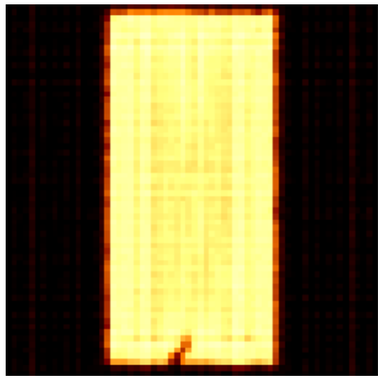
CIGS / image0611 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

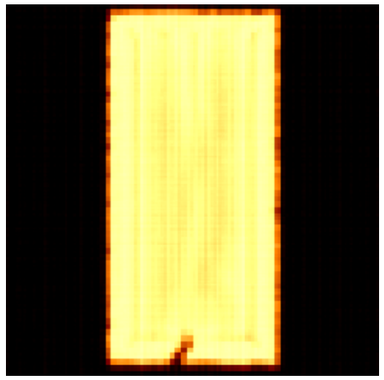


5% sampling



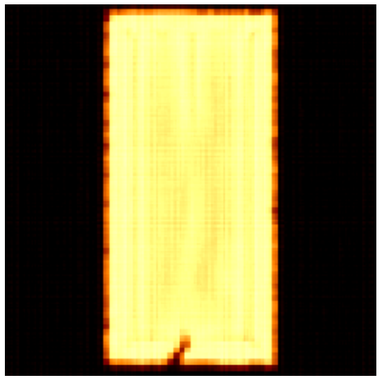
Zero-filled: matched multilevel
MSE 0.00374 | PSNR 24.27 dB
SSIM 0.475 | HaarPSI 0.349

10% sampling



Zero-filled: matched multilevel
MSE 0.00206 | PSNR 26.86 dB
SSIM 0.778 | HaarPSI 0.539

20% sampling



Zero-filled: matched multilevel
MSE 0.00189 | PSNR 27.24 dB
SSIM 0.718 | HaarPSI 0.587

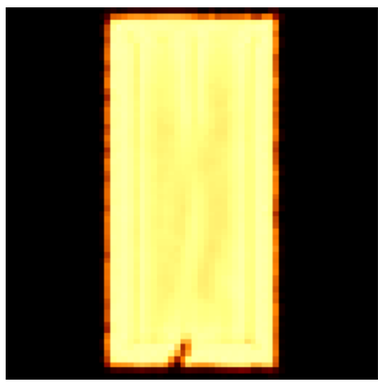
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

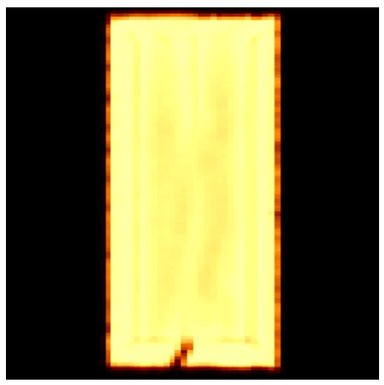
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

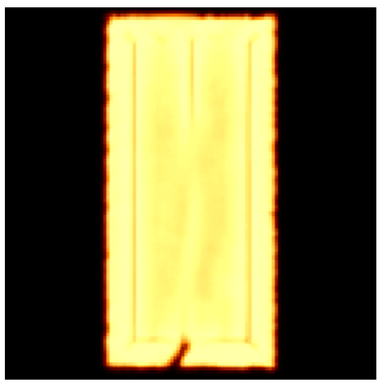
Single learned scale key; coarse J=6; PDHG=512.



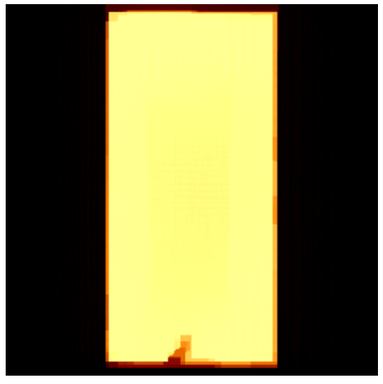
Zero-filled: in-order
MSE 0.00228 | PSNR 26.42 dB
SSIM 0.896 | HaarPSI 0.443



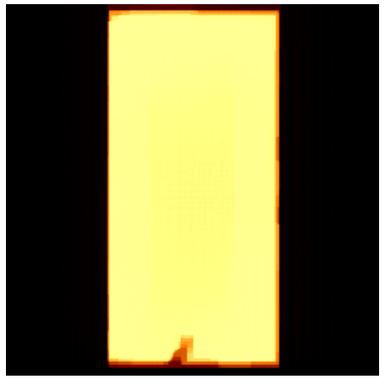
Zero-filled: in-order
MSE 0.00182 | PSNR 27.40 dB
SSIM 0.911 | HaarPSI 0.523



Zero-filled: in-order
MSE 0.00044 | PSNR 33.52 dB
SSIM 0.978 | HaarPSI 0.996



Learned U-TV
MSE 0.00224 | PSNR 26.50 dB
SSIM 0.749 | HaarPSI 0.449



Learned U-TV
MSE 0.00180 | PSNR 27.44 dB
SSIM 0.761 | HaarPSI 0.503



Learned U-TV
MSE 0.00122 | PSNR 29.14 dB
SSIM 0.785 | HaarPSI 0.541

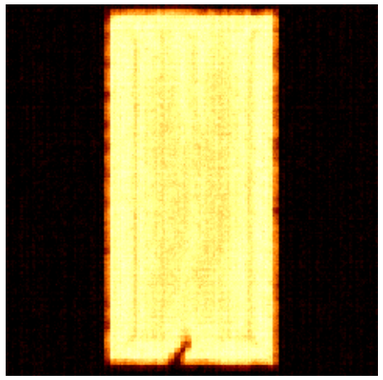
CIGS / image0611 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

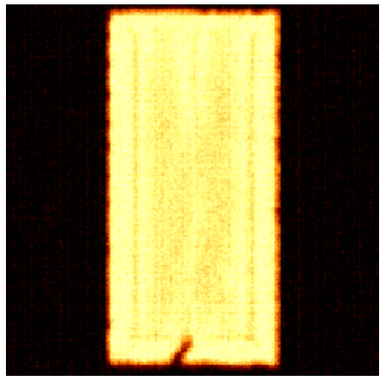


30% sampling



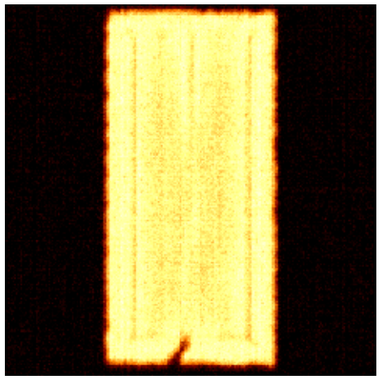
Zero-filled: matched multilevel
MSE 0.00258 | PSNR 25.88 dB
SSIM 0.422 | HaarPSI 0.592

40% sampling



Zero-filled: matched multilevel
MSE 0.00239 | PSNR 26.21 dB
SSIM 0.395 | HaarPSI 0.664

50% sampling



Zero-filled: matched multilevel
MSE 0.00240 | PSNR 26.20 dB
SSIM 0.388 | HaarPSI 0.689

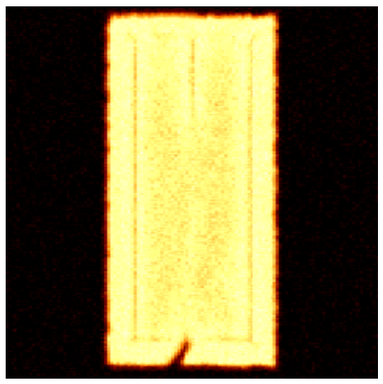
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

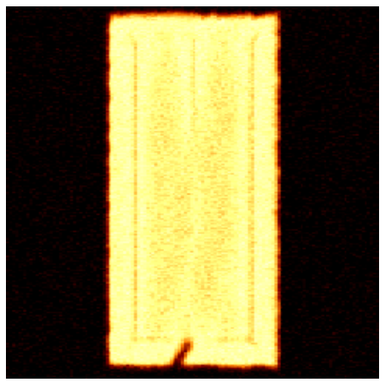
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

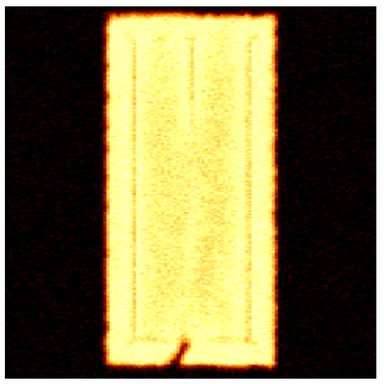
Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00141 | PSNR 28.52 dB
SSIM 0.518 | HaarPSI 0.753



Zero-filled: in-order
MSE 0.00172 | PSNR 27.63 dB
SSIM 0.466 | HaarPSI 0.750



Zero-filled: in-order
MSE 0.00209 | PSNR 26.80 dB
SSIM 0.424 | HaarPSI 0.756



Learned U-TV
MSE 0.00109 | PSNR 29.63 dB
SSIM 0.793 | HaarPSI 0.555



Learned U-TV
MSE 0.00107 | PSNR 29.71 dB
SSIM 0.784 | HaarPSI 0.542



Learned U-TV
MSE 0.00100 | PSNR 30.00 dB
SSIM 0.800 | HaarPSI 0.573

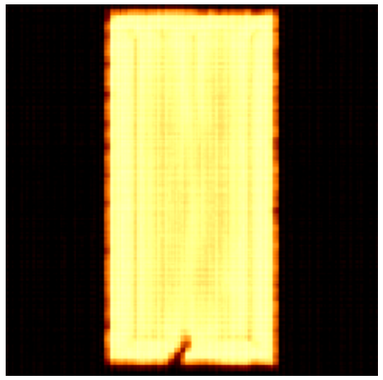
CIGS / image0611 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

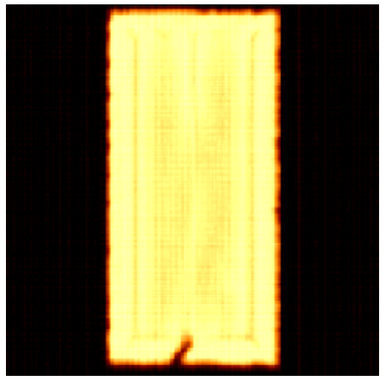


30% sampling



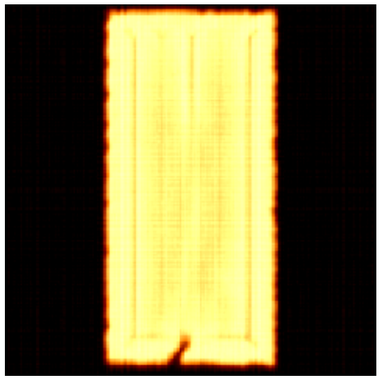
Zero-filled: matched multilevel
MSE 0.00152 | PSNR 28.19 dB
SSIM 0.668 | HaarPSI 0.640

40% sampling



Zero-filled: matched multilevel
MSE 0.00098 | PSNR 30.07 dB
SSIM 0.663 | HaarPSI 0.747

50% sampling



Zero-filled: matched multilevel
MSE 0.00064 | PSNR 31.92 dB
SSIM 0.744 | HaarPSI 0.825

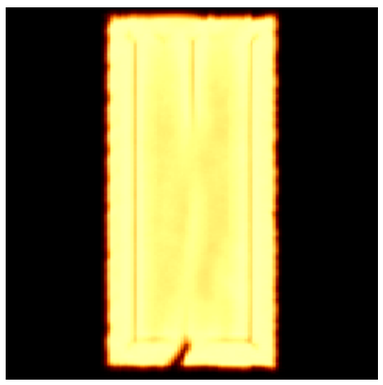
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

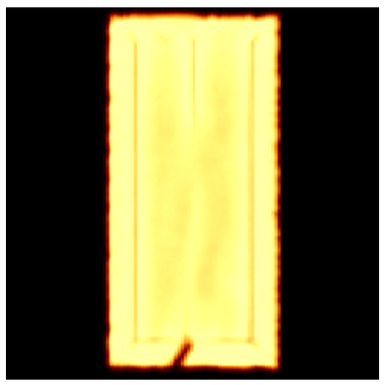
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00034 | PSNR 34.73 dB
SSIM 0.982 | HaarPSI 0.999



Zero-filled: in-order
MSE 0.00032 | PSNR 34.92 dB
SSIM 0.983 | HaarPSI 1.000



Zero-filled: in-order
MSE 0.00032 | PSNR 34.95 dB
SSIM 0.984 | HaarPSI 1.000



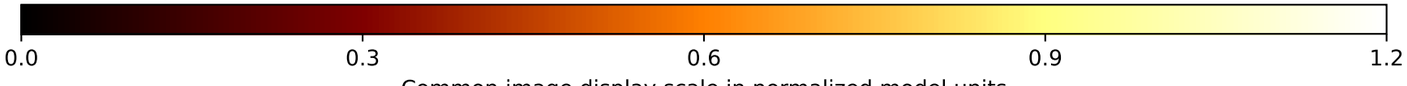
Learned U-TV
MSE 0.00103 | PSNR 29.88 dB
SSIM 0.801 | HaarPSI 0.558



Learned U-TV
MSE 0.00101 | PSNR 29.96 dB
SSIM 0.799 | HaarPSI 0.544



Learned U-TV
MSE 0.00091 | PSNR 30.40 dB
SSIM 0.808 | HaarPSI 0.587



CIGS / image0249 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

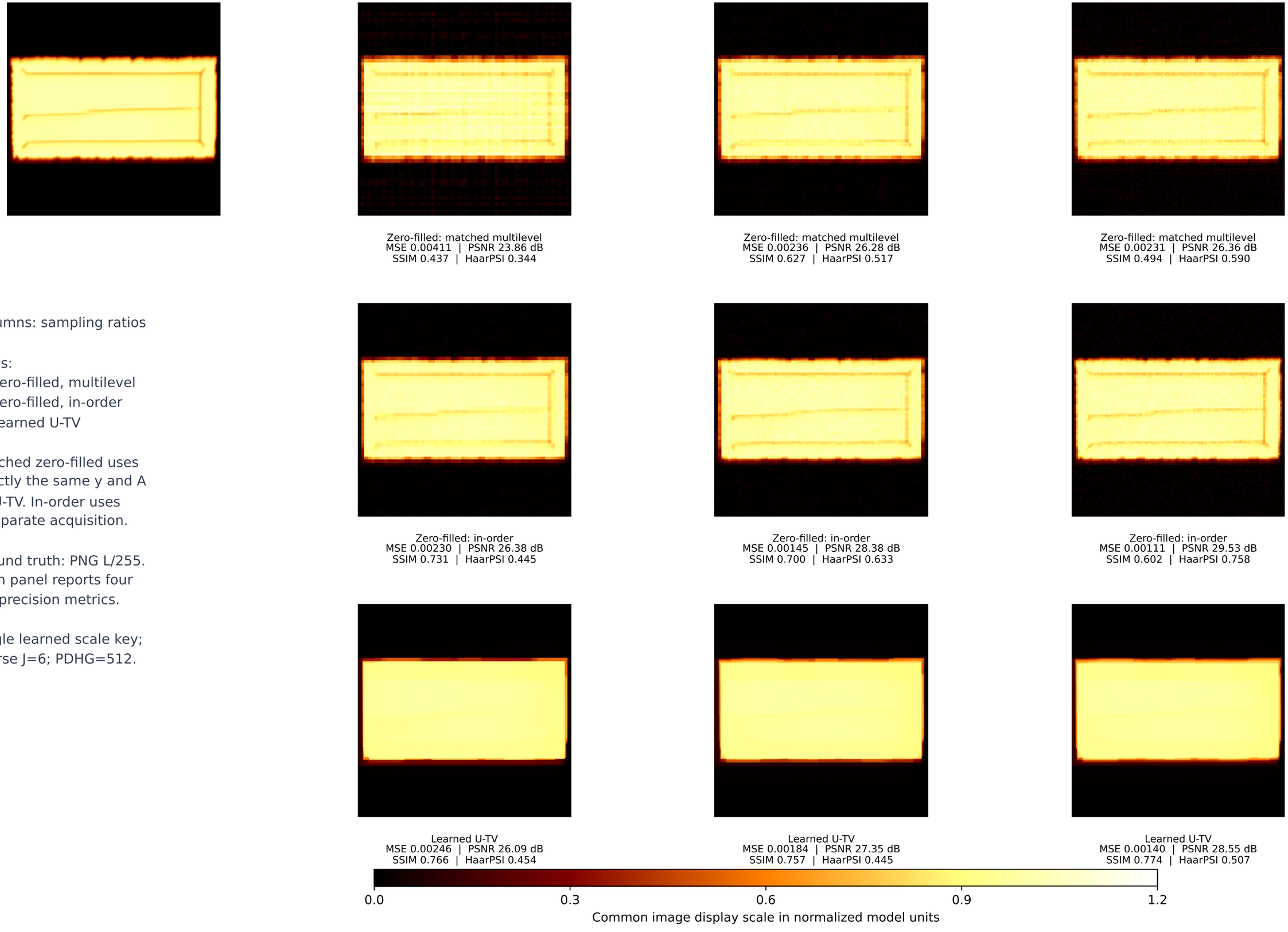
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.



CIGS / image0249 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

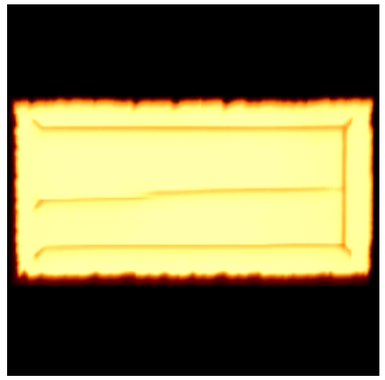
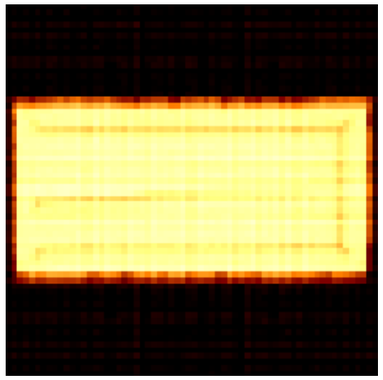
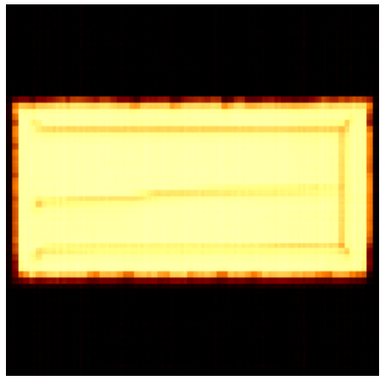
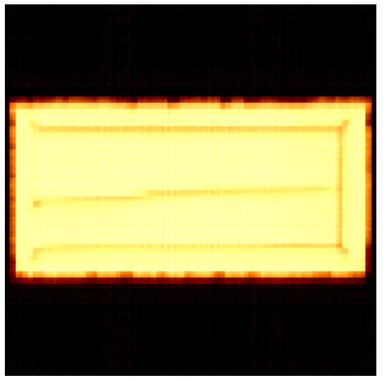
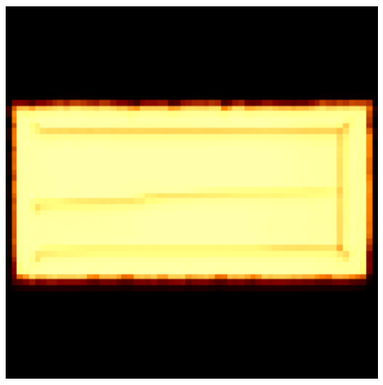
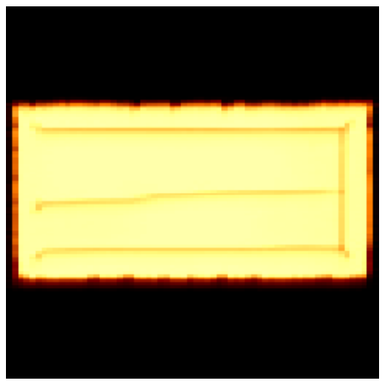
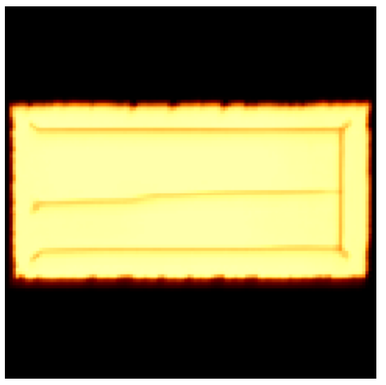
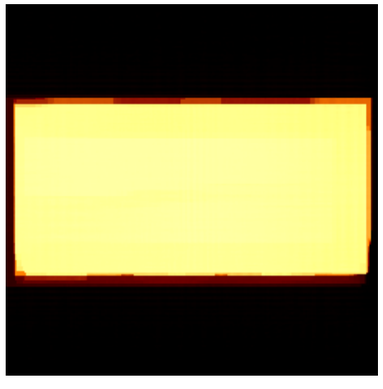
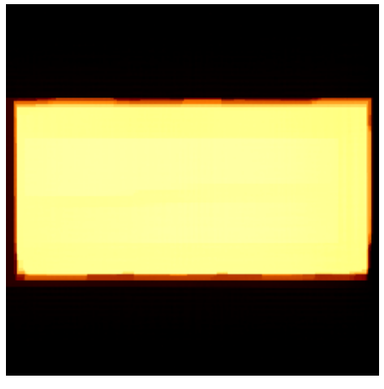
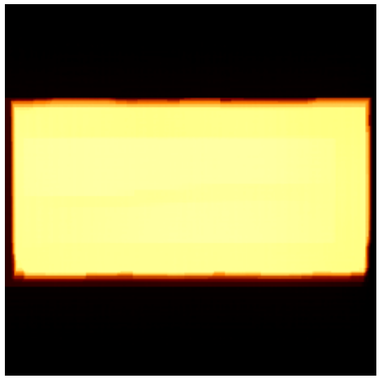
Columns: sampling ratios

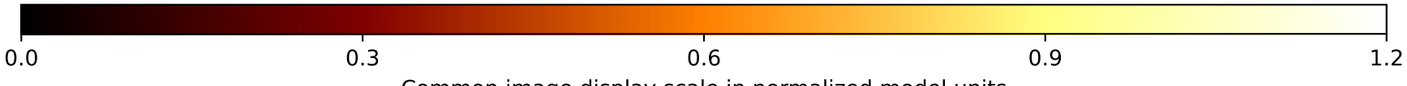
Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

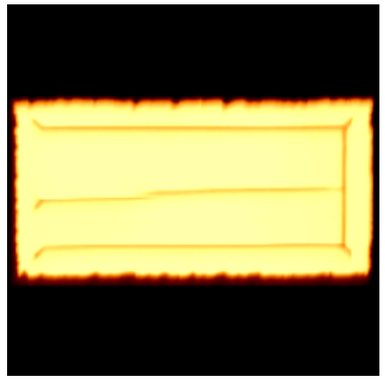
Ground truth	**5% sampling**	**10% sampling**	**20% sampling**
			
	Zero-filled: matched multilevel MSE 0.00393 | PSNR 24.05 dB SSIM 0.461 | HaarPSI 0.350	Zero-filled: matched multilevel MSE 0.00200 | PSNR 26.98 dB SSIM 0.842 | HaarPSI 0.534	Zero-filled: matched multilevel MSE 0.00158 | PSNR 28.00 dB SSIM 0.707 | HaarPSI 0.629
			
	Zero-filled: in-order MSE 0.00212 | PSNR 26.73 dB SSIM 0.913 | HaarPSI 0.451	Zero-filled: in-order MSE 0.00110 | PSNR 29.59 dB SSIM 0.964 | HaarPSI 0.706	Zero-filled: in-order MSE 0.00039 | PSNR 34.05 dB SSIM 0.982 | HaarPSI 0.997
			
	Learned U-TV MSE 0.00244 | PSNR 26.12 dB SSIM 0.763 | HaarPSI 0.453	Learned U-TV MSE 0.00184 | PSNR 27.36 dB SSIM 0.762 | HaarPSI 0.444	Learned U-TV MSE 0.00136 | PSNR 28.66 dB SSIM 0.786 | HaarPSI 0.507



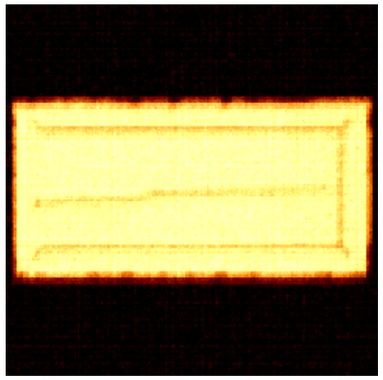
CIGS / image0249 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

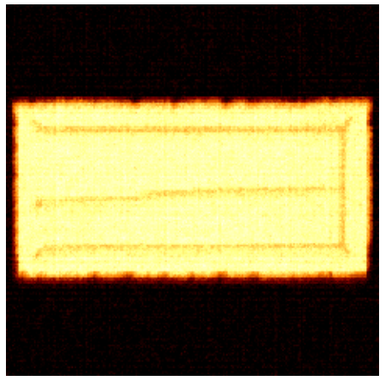


30% sampling



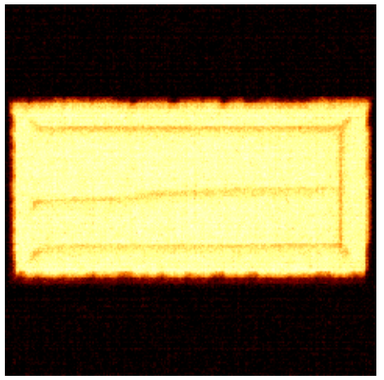
Zero-filled: matched multilevel
MSE 0.00232 | PSNR 26.35 dB
SSIM 0.451 | HaarPSI 0.639

40% sampling



Zero-filled: matched multilevel
MSE 0.00232 | PSNR 26.34 dB
SSIM 0.424 | HaarPSI 0.671

50% sampling



Zero-filled: matched multilevel
MSE 0.00243 | PSNR 26.15 dB
SSIM 0.401 | HaarPSI 0.688

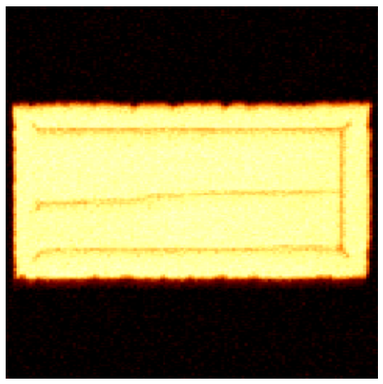
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

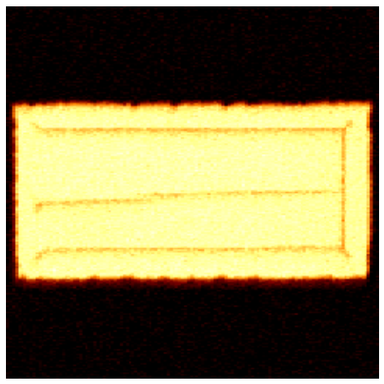
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

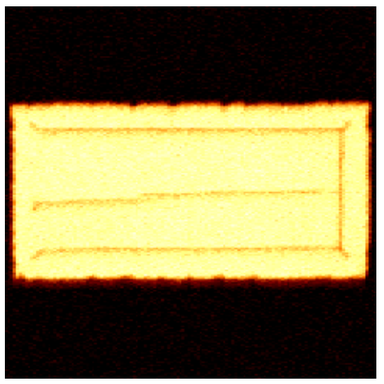
Single learned scale key; coarse J=6; PDHG=512.



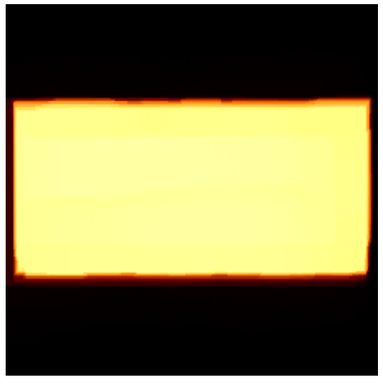
Zero-filled: in-order
MSE 0.00125 | PSNR 29.04 dB
SSIM 0.539 | HaarPSI 0.746



Zero-filled: in-order
MSE 0.00158 | PSNR 28.01 dB
SSIM 0.487 | HaarPSI 0.754



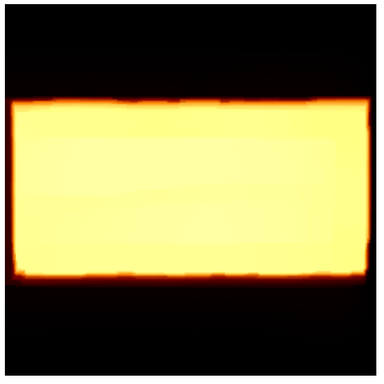
Zero-filled: in-order
MSE 0.00194 | PSNR 27.13 dB
SSIM 0.447 | HaarPSI 0.747



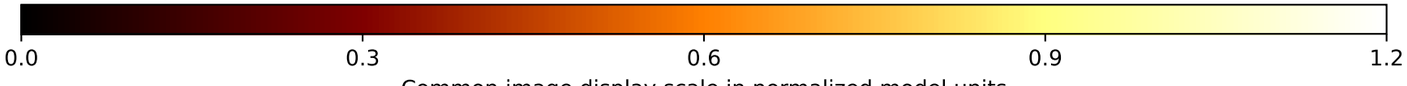
Learned U-TV
MSE 0.00135 | PSNR 28.71 dB
SSIM 0.780 | HaarPSI 0.506



Learned U-TV
MSE 0.00129 | PSNR 28.89 dB
SSIM 0.781 | HaarPSI 0.509



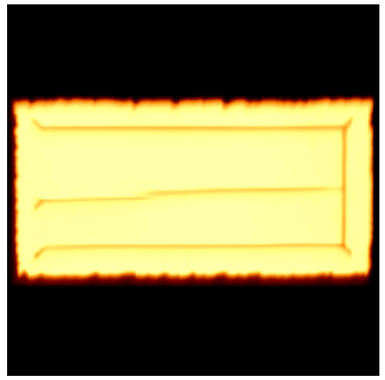
Learned U-TV
MSE 0.00129 | PSNR 28.90 dB
SSIM 0.777 | HaarPSI 0.510



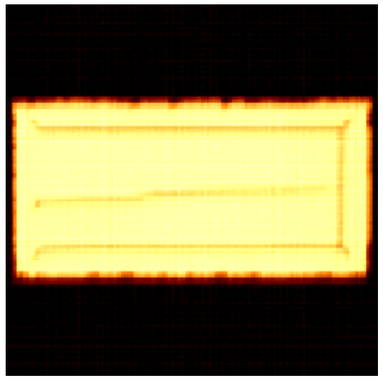
CIGS / image0249 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

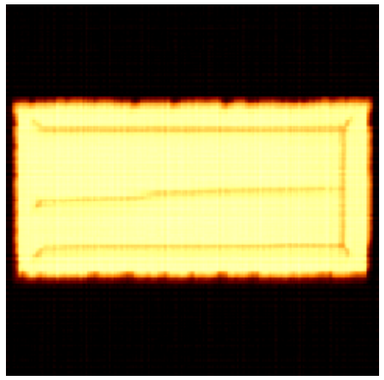


30% sampling



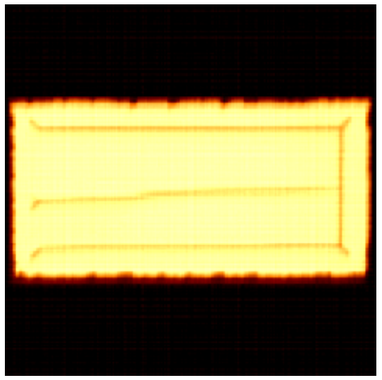
Zero-filled: matched multilevel
MSE 0.00125 | PSNR 29.02 dB
SSIM 0.686 | HaarPSI 0.723

40% sampling



Zero-filled: matched multilevel
MSE 0.00090 | PSNR 30.47 dB
SSIM 0.716 | HaarPSI 0.753

50% sampling



Zero-filled: matched multilevel
MSE 0.00062 | PSNR 32.06 dB
SSIM 0.721 | HaarPSI 0.808

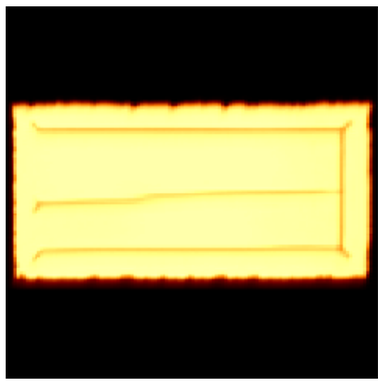
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

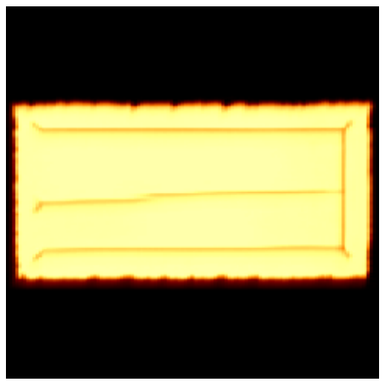
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

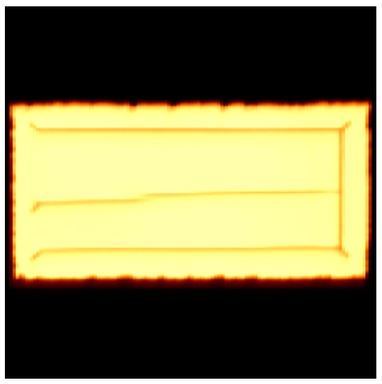
Single learned scale key; coarse J=6; PDHG=512.



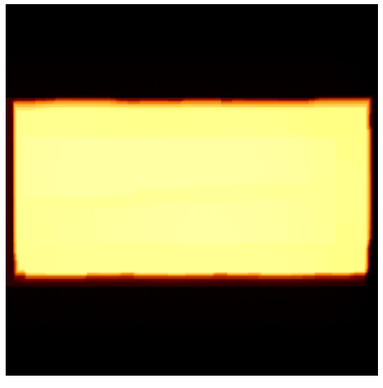
Zero-filled: in-order
MSE 0.00018 | PSNR 37.43 dB
SSIM 0.991 | HaarPSI 1.000



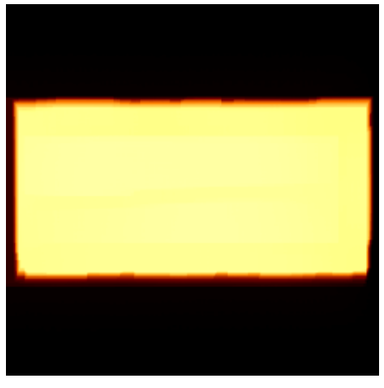
Zero-filled: in-order
MSE 0.00015 | PSNR 38.24 dB
SSIM 0.995 | HaarPSI 1.000



Zero-filled: in-order
MSE 0.00015 | PSNR 38.30 dB
SSIM 0.996 | HaarPSI 1.000



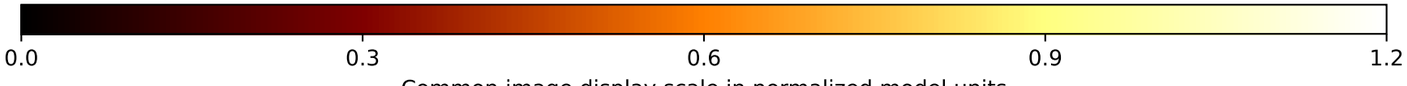
Learned U-TV
MSE 0.00129 | PSNR 28.90 dB
SSIM 0.791 | HaarPSI 0.513



Learned U-TV
MSE 0.00121 | PSNR 29.18 dB
SSIM 0.791 | HaarPSI 0.517



Learned U-TV
MSE 0.00118 | PSNR 29.29 dB
SSIM 0.797 | HaarPSI 0.529



Organic / image0611 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

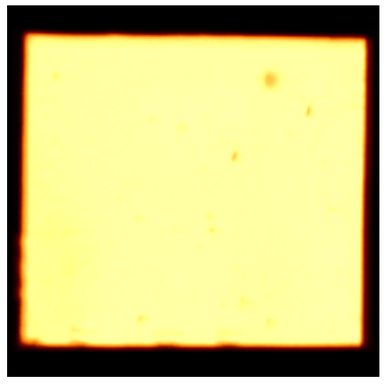
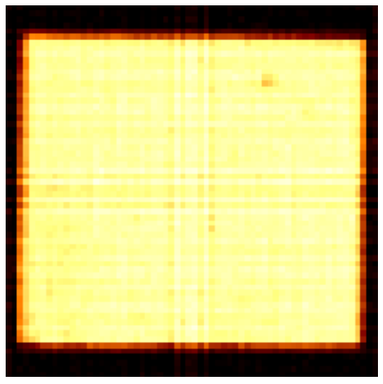
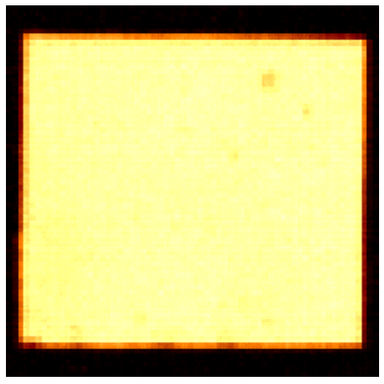
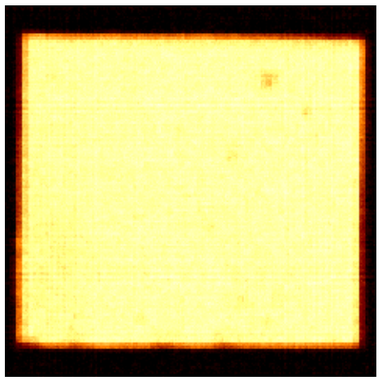
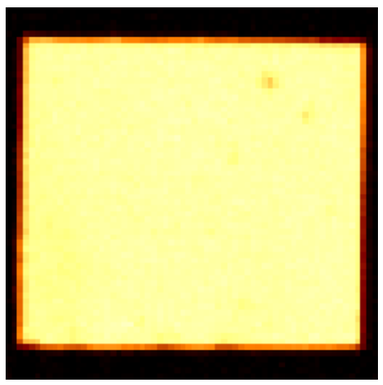
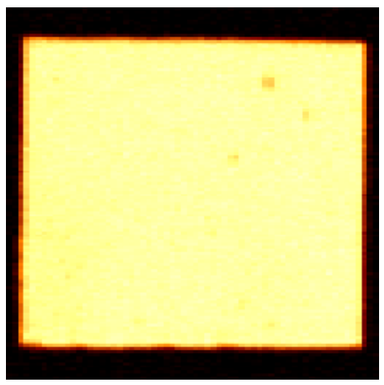
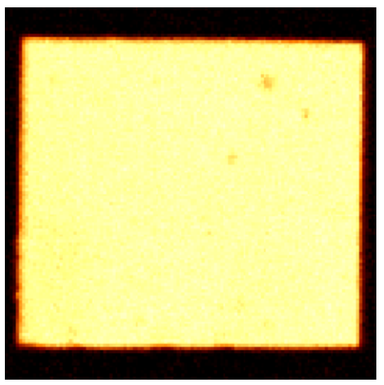
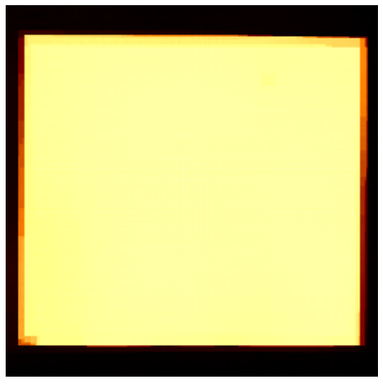
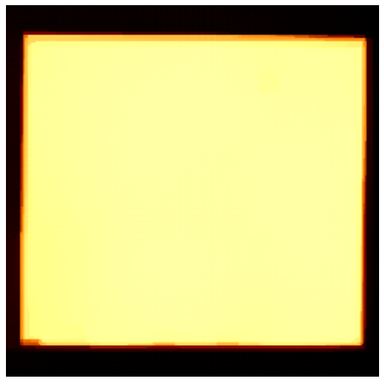
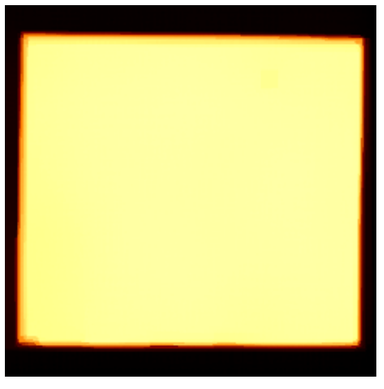
Columns: sampling ratios

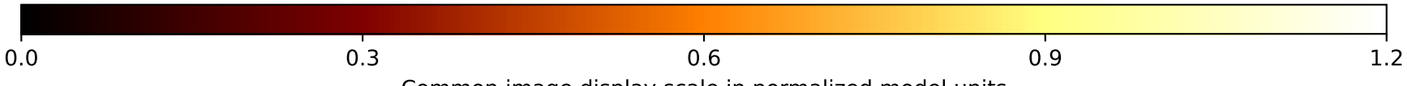
Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

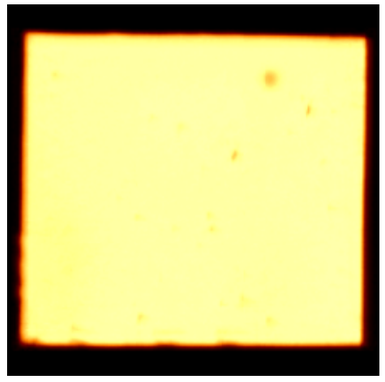
Ground truth	5% sampling	10% sampling	20% sampling
			
	Zero-filled: matched multilevel MSE 0.00516 | PSNR 22.88 dB SSIM 0.485 | HaarPSI 0.311	Zero-filled: matched multilevel MSE 0.00240 | PSNR 26.20 dB SSIM 0.657 | HaarPSI 0.526	Zero-filled: matched multilevel MSE 0.00236 | PSNR 26.28 dB SSIM 0.507 | HaarPSI 0.605
			
	Zero-filled: in-order MSE 0.00234 | PSNR 26.30 dB SSIM 0.793 | HaarPSI 0.444	Zero-filled: in-order MSE 0.00162 | PSNR 27.91 dB SSIM 0.718 | HaarPSI 0.573	Zero-filled: in-order MSE 0.00128 | PSNR 28.91 dB SSIM 0.601 | HaarPSI 0.743
			
	Learned U-TV MSE 0.00253 | PSNR 25.98 dB SSIM 0.826 | HaarPSI 0.483	Learned U-TV MSE 0.00107 | PSNR 29.70 dB SSIM 0.863 | HaarPSI 0.744	Learned U-TV MSE 0.00046 | PSNR 33.42 dB SSIM 0.879 | HaarPSI 0.789



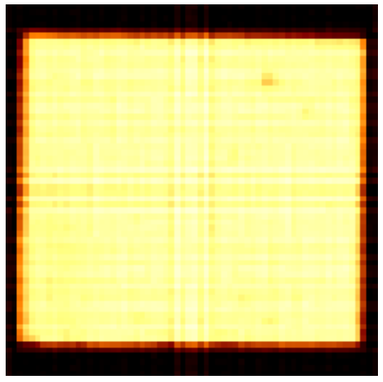
Organic / image0611 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

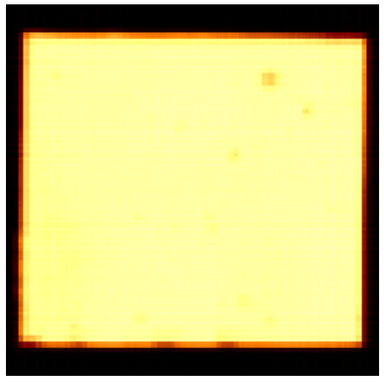


5% sampling



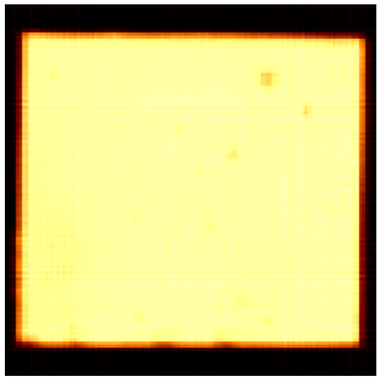
Zero-filled: matched multilevel
MSE 0.00494 | PSNR 23.06 dB
SSIM 0.524 | HaarPSI 0.316

10% sampling



Zero-filled: matched multilevel
MSE 0.00198 | PSNR 27.03 dB
SSIM 0.847 | HaarPSI 0.553

20% sampling



Zero-filled: matched multilevel
MSE 0.00152 | PSNR 28.18 dB
SSIM 0.743 | HaarPSI 0.664

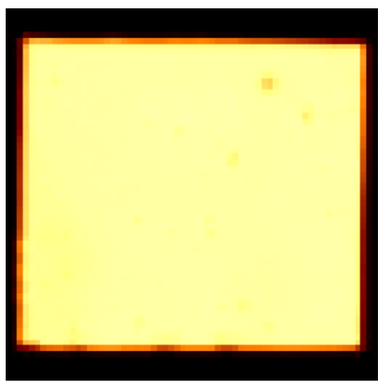
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

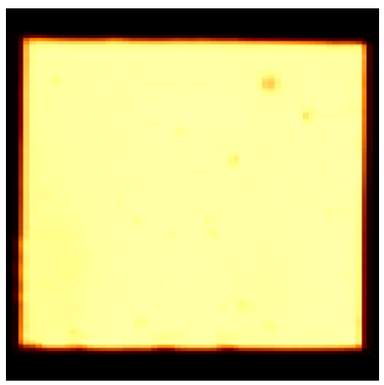
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

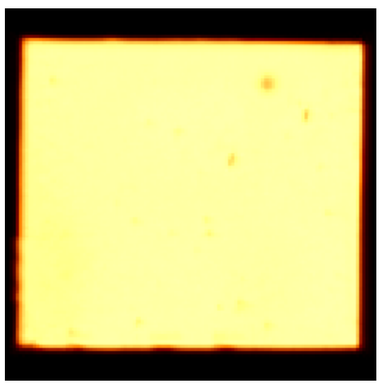
Single learned scale key; coarse J=6; PDHG=512.



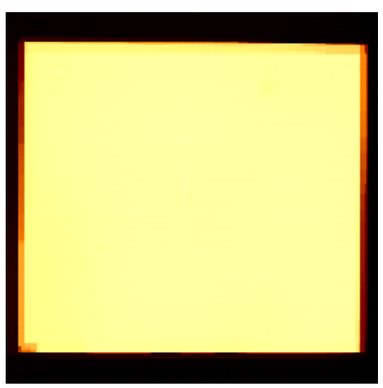
Zero-filled: in-order
MSE 0.00213 | PSNR 26.72 dB
SSIM 0.924 | HaarPSI 0.452



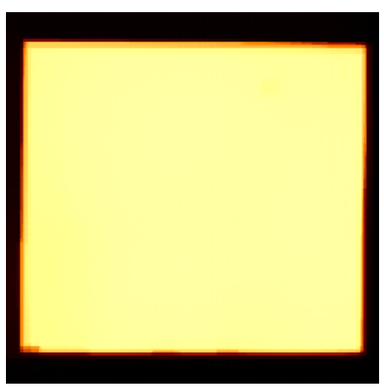
Zero-filled: in-order
MSE 0.00118 | PSNR 29.27 dB
SSIM 0.955 | HaarPSI 0.636



Zero-filled: in-order
MSE 0.00043 | PSNR 33.70 dB
SSIM 0.981 | HaarPSI 0.999



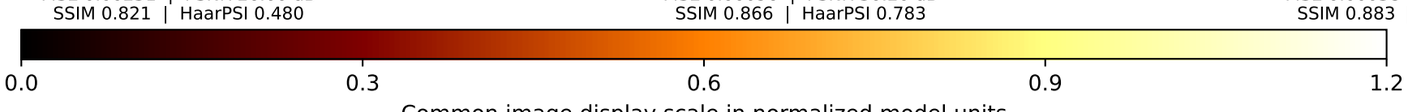
Learned U-TV
MSE 0.00251 | PSNR 26.00 dB
SSIM 0.821 | HaarPSI 0.480



Learned U-TV
MSE 0.00096 | PSNR 30.20 dB
SSIM 0.866 | HaarPSI 0.783



Learned U-TV
MSE 0.00038 | PSNR 34.18 dB
SSIM 0.883 | HaarPSI 0.810



Common image display scale in normalized model units

Organic / image0611 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

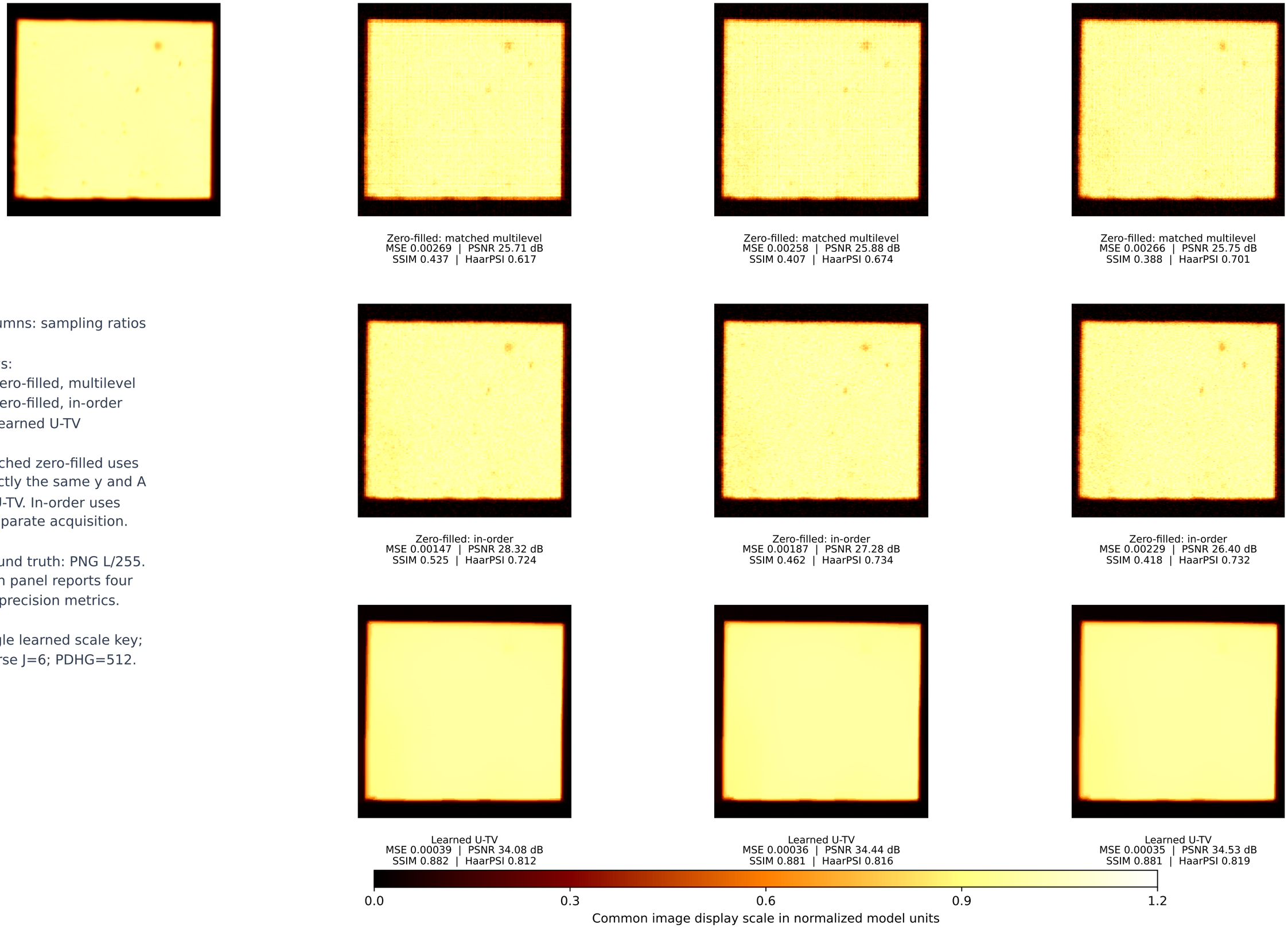
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

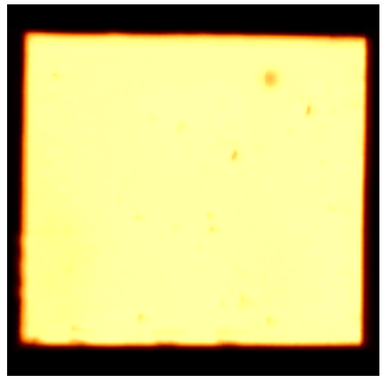
Single learned scale key; coarse J=6; PDHG=512.



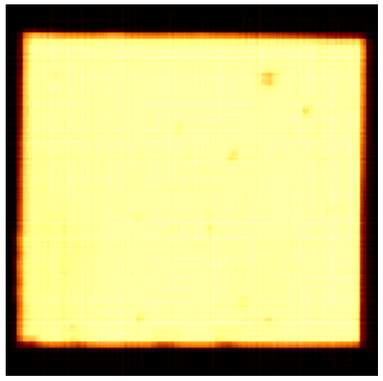
Organic / image0611 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

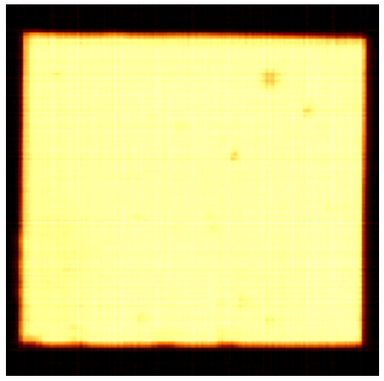


30% sampling



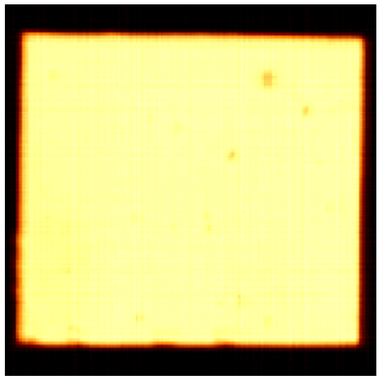
Zero-filled: matched multilevel
MSE 0.00141 | PSNR 28.52 dB
SSIM 0.712 | HaarPSI 0.680

40% sampling



Zero-filled: matched multilevel
MSE 0.00091 | PSNR 30.41 dB
SSIM 0.732 | HaarPSI 0.767

50% sampling



Zero-filled: matched multilevel
MSE 0.00059 | PSNR 32.32 dB
SSIM 0.795 | HaarPSI 0.860

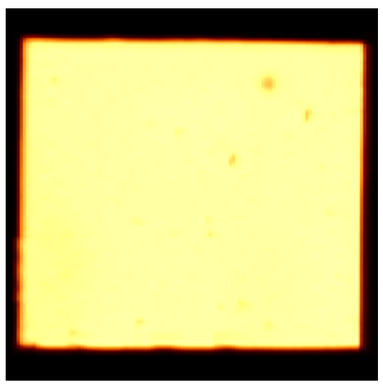
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

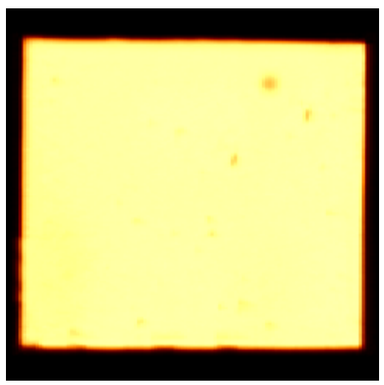
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

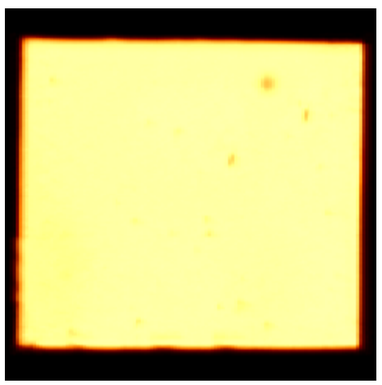
Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00021 | PSNR 36.70 dB
SSIM 0.986 | HaarPSI 0.997



Zero-filled: in-order
MSE 0.00019 | PSNR 37.23 dB
SSIM 0.991 | HaarPSI 1.000



Zero-filled: in-order
MSE 0.00019 | PSNR 37.26 dB
SSIM 0.991 | HaarPSI 1.000



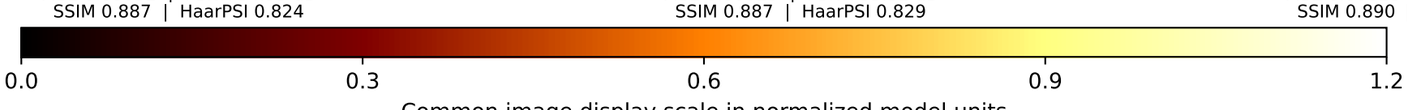
Learned U-TV
MSE 0.00032 | PSNR 34.99 dB
SSIM 0.887 | HaarPSI 0.824



Learned U-TV
MSE 0.00029 | PSNR 35.32 dB
SSIM 0.887 | HaarPSI 0.829



Learned U-TV
MSE 0.00028 | PSNR 35.51 dB
SSIM 0.890 | HaarPSI 0.836



Common image display scale in normalized model units

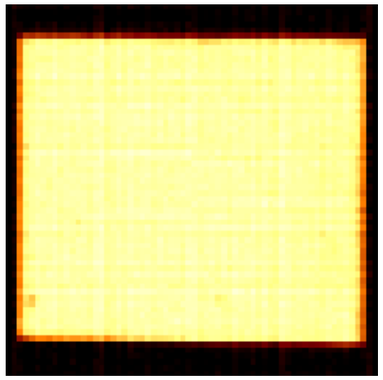
Organic / image0249 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

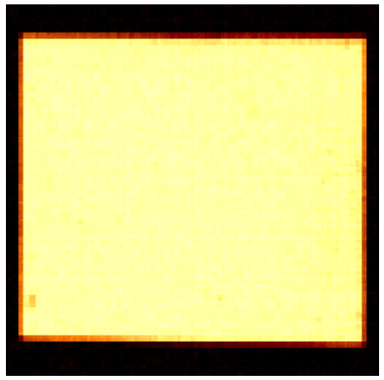


5% sampling



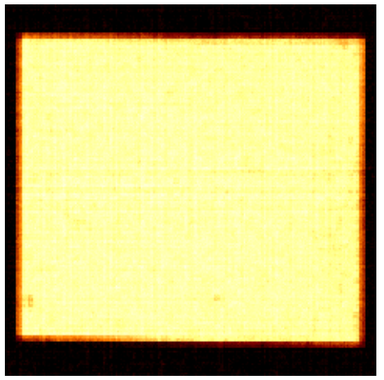
Zero-filled: matched multilevel
MSE 0.00457 | PSNR 23.40 dB
SSIM 0.543 | HaarPSI 0.337

10% sampling



Zero-filled: matched multilevel
MSE 0.00337 | PSNR 24.73 dB
SSIM 0.620 | HaarPSI 0.504

20% sampling



Zero-filled: matched multilevel
MSE 0.00310 | PSNR 25.08 dB
SSIM 0.455 | HaarPSI 0.540

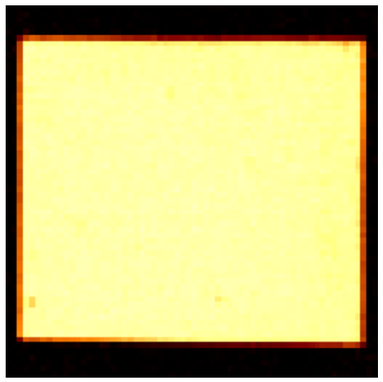
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

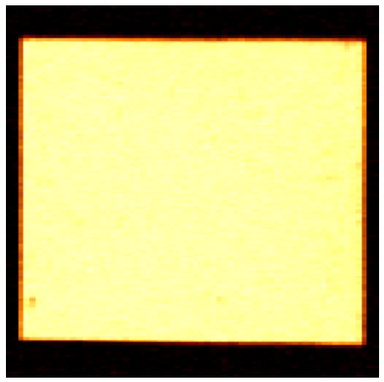
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

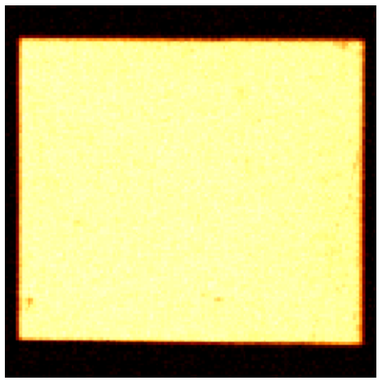
Single learned scale key; coarse J=6; PDHG=512.



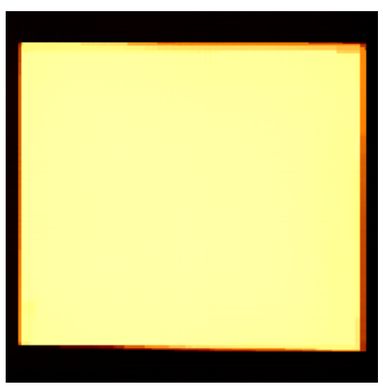
Zero-filled: in-order
MSE 0.00334 | PSNR 24.76 dB
SSIM 0.779 | HaarPSI 0.464



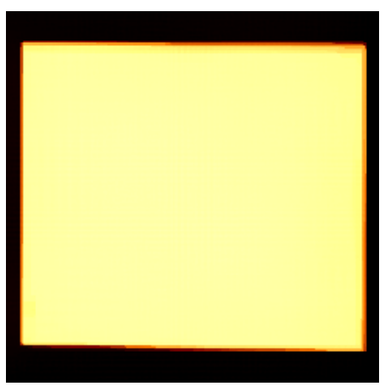
Zero-filled: in-order
MSE 0.00225 | PSNR 26.47 dB
SSIM 0.705 | HaarPSI 0.604



Zero-filled: in-order
MSE 0.00150 | PSNR 28.25 dB
SSIM 0.587 | HaarPSI 0.710



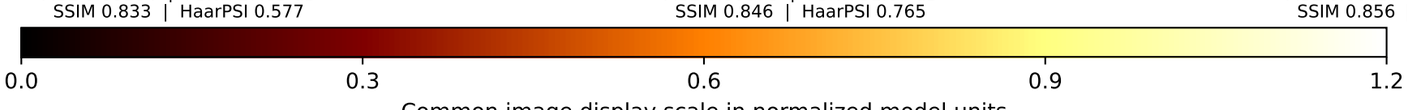
Learned U-TV
MSE 0.00164 | PSNR 27.85 dB
SSIM 0.833 | HaarPSI 0.577



Learned U-TV
MSE 0.00078 | PSNR 31.07 dB
SSIM 0.846 | HaarPSI 0.765



Learned U-TV
MSE 0.00037 | PSNR 34.35 dB
SSIM 0.856 | HaarPSI 0.833



Organic / image0249 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

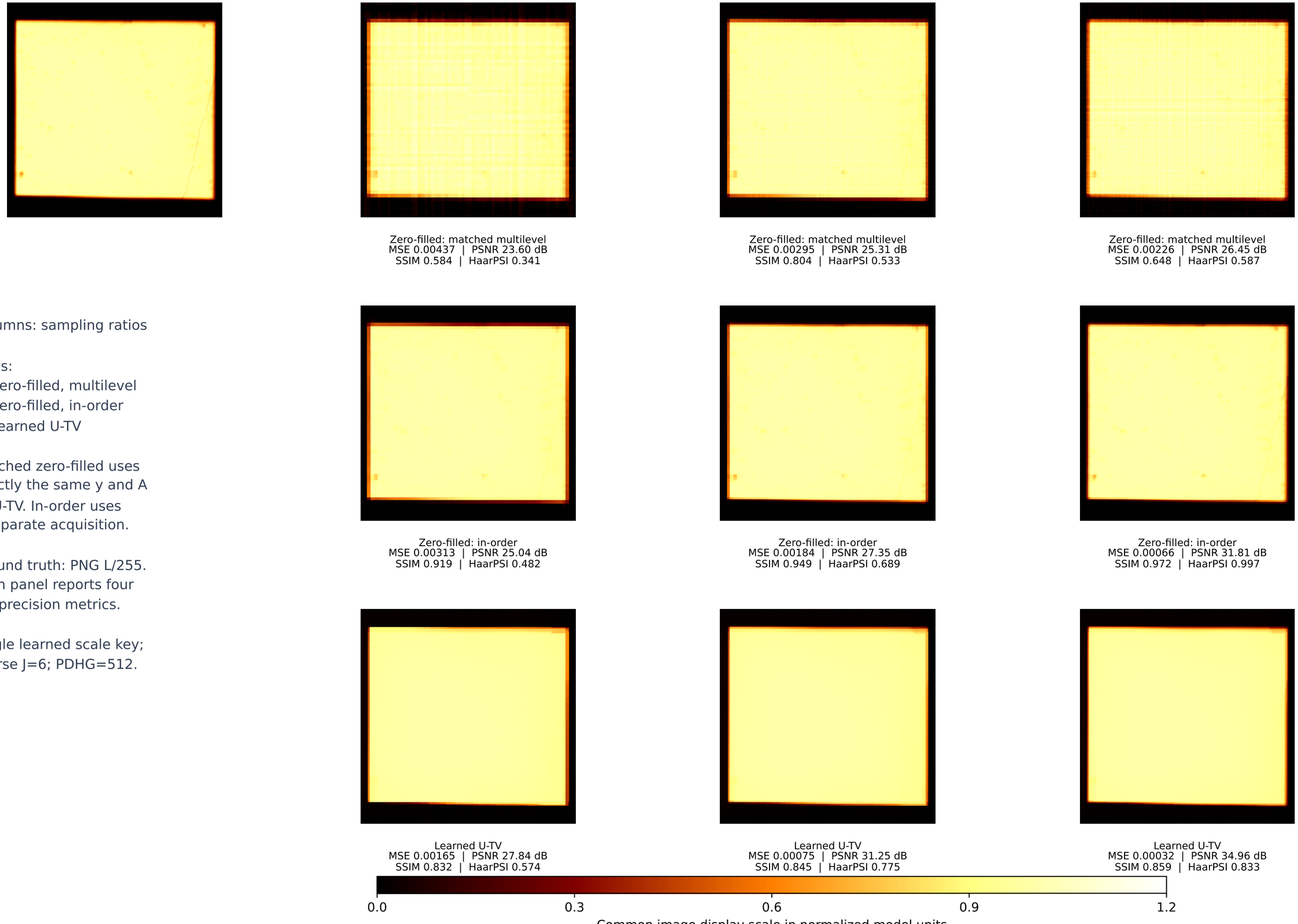
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.



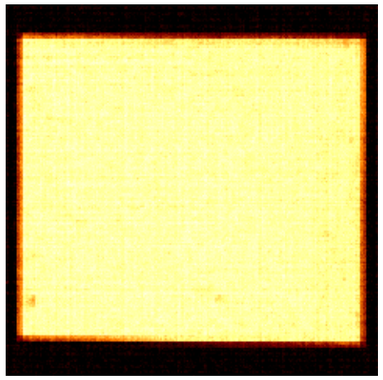
Organic / image0249 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

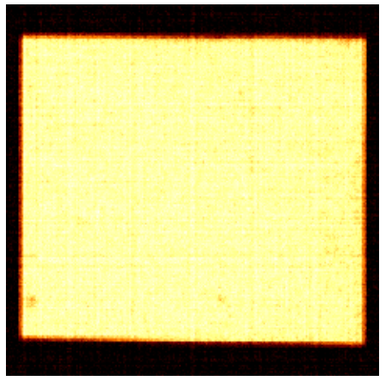


30% sampling



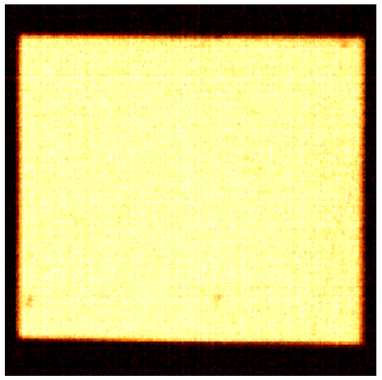
Zero-filled: matched multilevel
MSE 0.00340 | PSNR 24.69 dB
SSIM 0.401 | HaarPSI 0.532

40% sampling



Zero-filled: matched multilevel
MSE 0.00304 | PSNR 25.16 dB
SSIM 0.368 | HaarPSI 0.603

50% sampling



Zero-filled: matched multilevel
MSE 0.00302 | PSNR 25.20 dB
SSIM 0.352 | HaarPSI 0.612

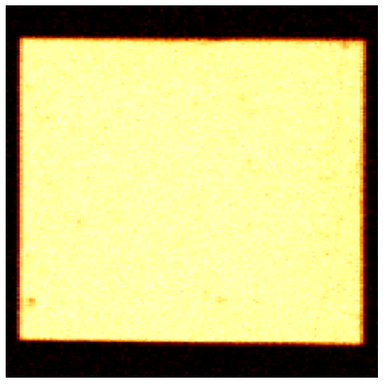
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

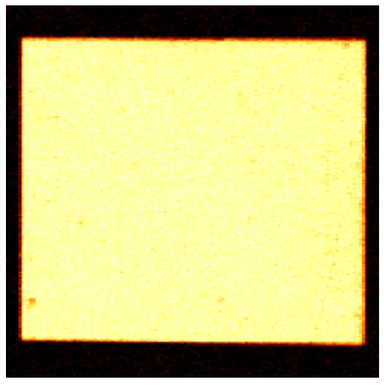
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

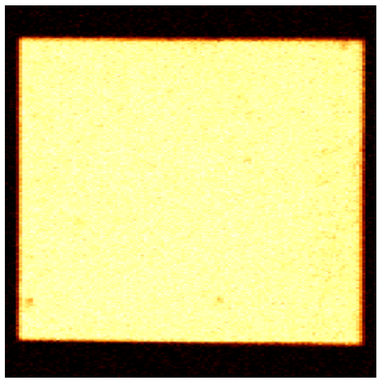
Single learned scale key; coarse J=6; PDHG=512.



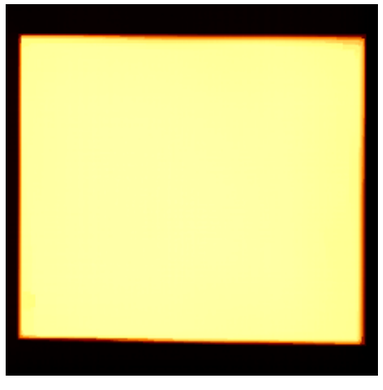
Zero-filled: in-order
MSE 0.00163 | PSNR 27.89 dB
SSIM 0.504 | HaarPSI 0.700



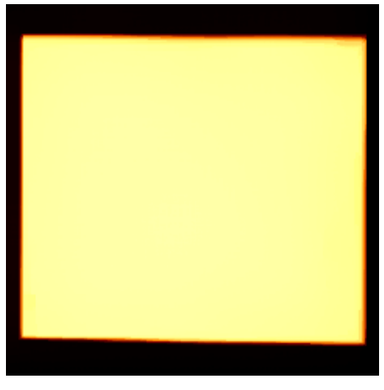
Zero-filled: in-order
MSE 0.00199 | PSNR 27.01 dB
SSIM 0.446 | HaarPSI 0.704



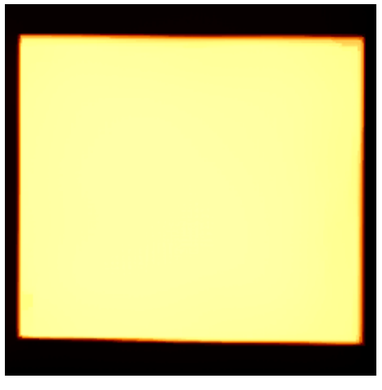
Zero-filled: in-order
MSE 0.00243 | PSNR 26.15 dB
SSIM 0.400 | HaarPSI 0.700



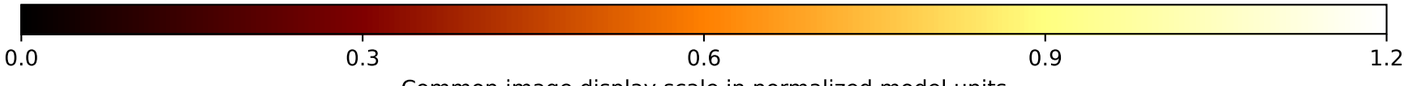
Learned U-TV
MSE 0.00034 | PSNR 34.71 dB
SSIM 0.858 | HaarPSI 0.846



Learned U-TV
MSE 0.00032 | PSNR 35.02 dB
SSIM 0.863 | HaarPSI 0.849



Learned U-TV
MSE 0.00028 | PSNR 35.54 dB
SSIM 0.864 | HaarPSI 0.861



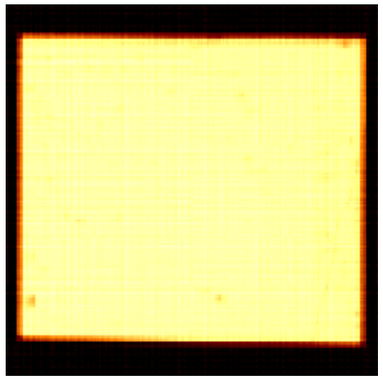
Organic / image0249 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

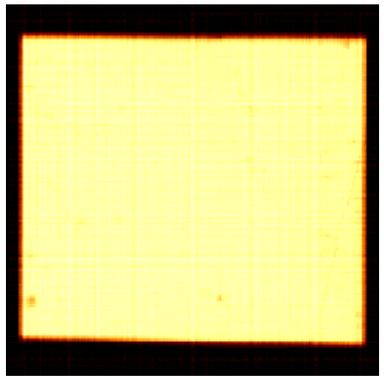


30% sampling



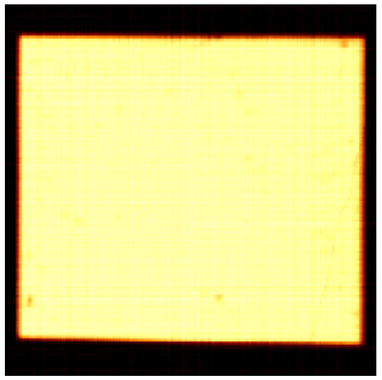
Zero-filled: matched multilevel
MSE 0.00215 | PSNR 26.68 dB
SSIM 0.630 | HaarPSI 0.562

40% sampling



Zero-filled: matched multilevel
MSE 0.00137 | PSNR 28.63 dB
SSIM 0.621 | HaarPSI 0.714

50% sampling



Zero-filled: matched multilevel
MSE 0.00094 | PSNR 30.28 dB
SSIM 0.673 | HaarPSI 0.730

Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

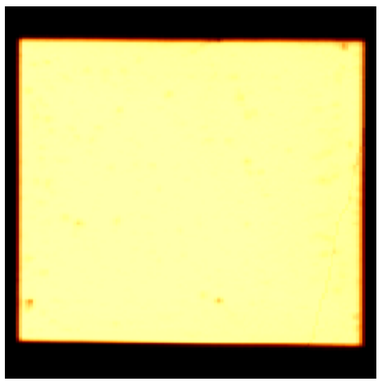
Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00036 | PSNR 34.44 dB
SSIM 0.982 | HaarPSI 0.993



Zero-filled: in-order
MSE 0.00032 | PSNR 34.93 dB
SSIM 0.988 | HaarPSI 1.000



Zero-filled: in-order
MSE 0.00032 | PSNR 34.95 dB
SSIM 0.988 | HaarPSI 1.000



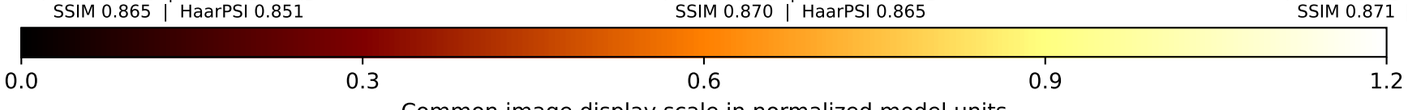
Learned U-TV
MSE 0.00029 | PSNR 35.40 dB
SSIM 0.865 | HaarPSI 0.851



Learned U-TV
MSE 0.00025 | PSNR 36.04 dB
SSIM 0.870 | HaarPSI 0.865



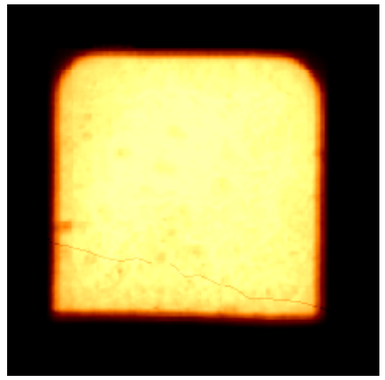
Learned U-TV
MSE 0.00023 | PSNR 36.33 dB
SSIM 0.871 | HaarPSI 0.869



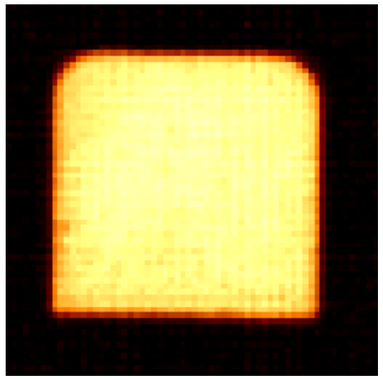
Perovskite / image0611 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

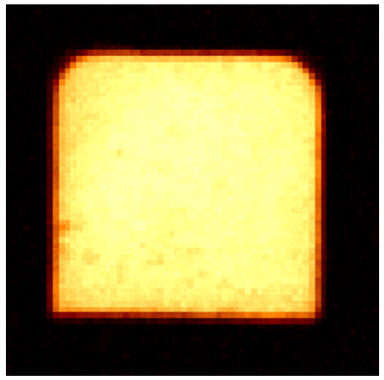


5% sampling



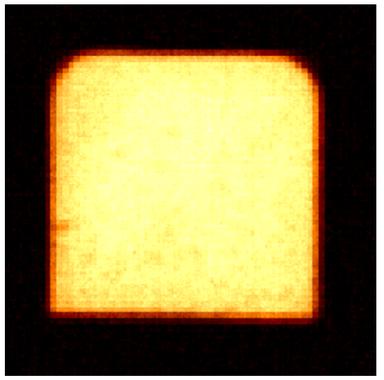
Zero-filled: matched multilevel
MSE 0.00155 | PSNR 28.09 dB
SSIM 0.588 | HaarPSI 0.431

10% sampling



Zero-filled: matched multilevel
MSE 0.00120 | PSNR 29.22 dB
SSIM 0.665 | HaarPSI 0.532

20% sampling



Zero-filled: matched multilevel
MSE 0.00143 | PSNR 28.44 dB
SSIM 0.550 | HaarPSI 0.613

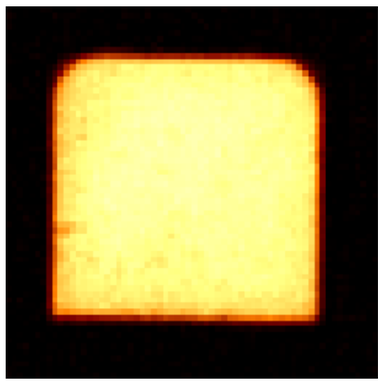
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

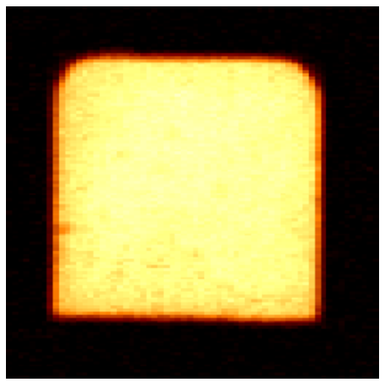
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

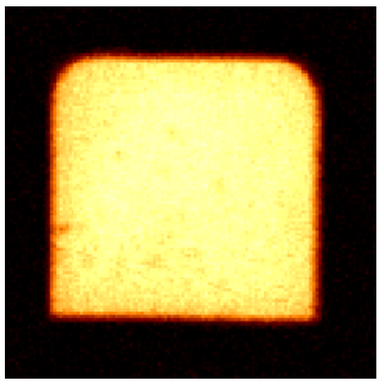
Single learned scale key; coarse J=6; PDHG=512.



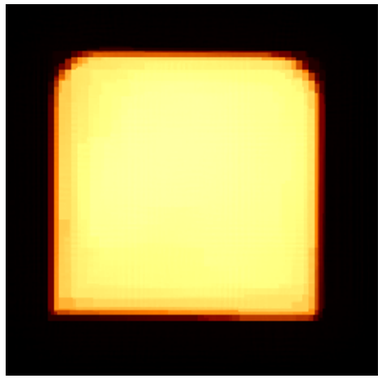
Zero-filled: in-order
MSE 0.00106 | PSNR 29.75 dB
SSIM 0.741 | HaarPSI 0.477



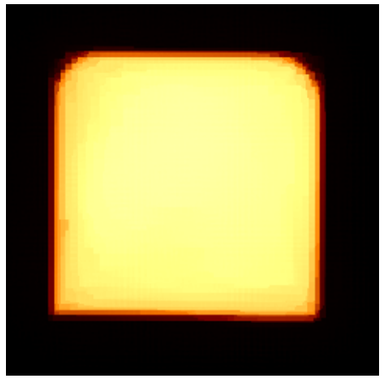
Zero-filled: in-order
MSE 0.00090 | PSNR 30.45 dB
SSIM 0.692 | HaarPSI 0.606



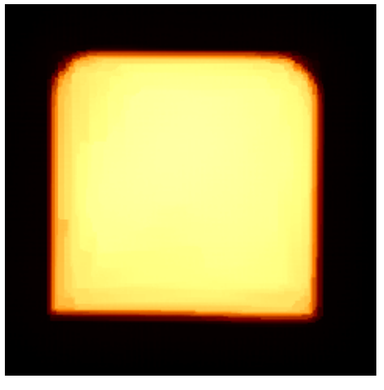
Zero-filled: in-order
MSE 0.00096 | PSNR 30.17 dB
SSIM 0.598 | HaarPSI 0.769



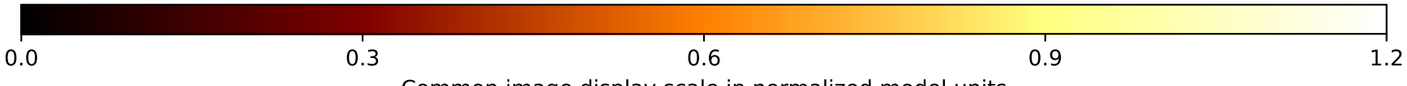
Learned U-TV
MSE 0.00098 | PSNR 30.11 dB
SSIM 0.726 | HaarPSI 0.629



Learned U-TV
MSE 0.00087 | PSNR 30.60 dB
SSIM 0.729 | HaarPSI 0.679



Learned U-TV
MSE 0.00044 | PSNR 33.55 dB
SSIM 0.741 | HaarPSI 0.784



Perovskite / image0611 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

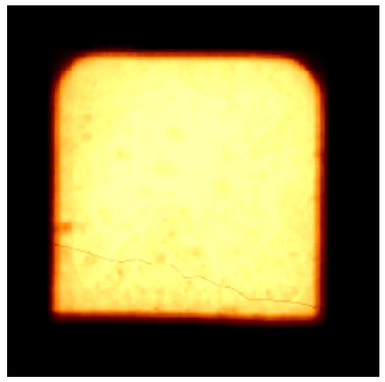
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

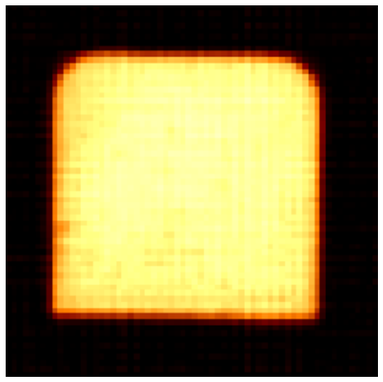
Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.



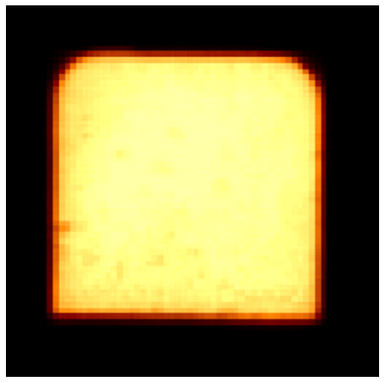
Ground truth

5% sampling



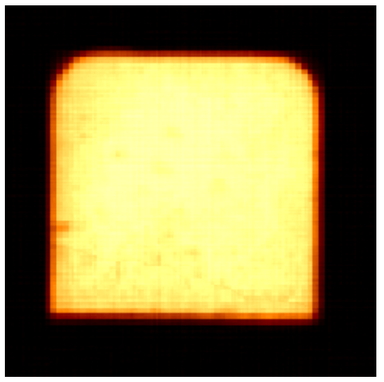
Zero-filled: matched multilevel
MSE 0.00137 | PSNR 28.62 dB
SSIM 0.650 | HaarPSI 0.438

10% sampling

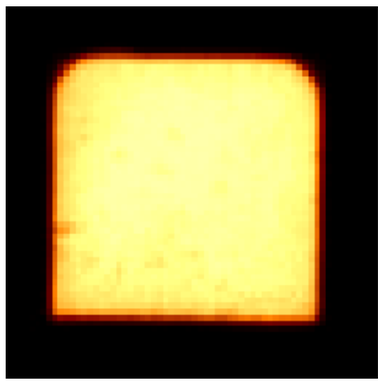


Zero-filled: matched multilevel
MSE 0.00083 | PSNR 30.78 dB
SSIM 0.907 | HaarPSI 0.531

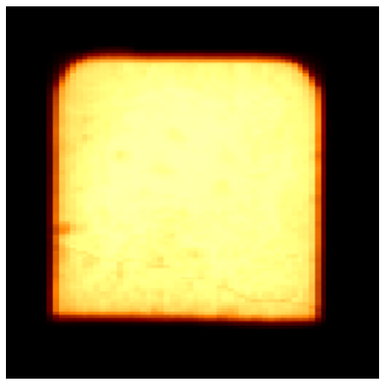
20% sampling



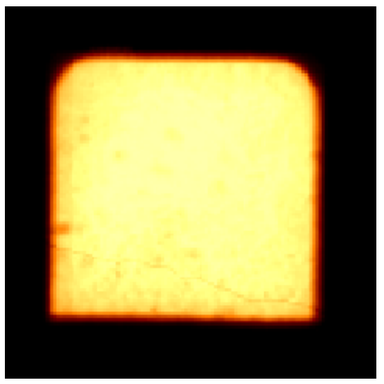
Zero-filled: matched multilevel
MSE 0.00072 | PSNR 31.44 dB
SSIM 0.850 | HaarPSI 0.650



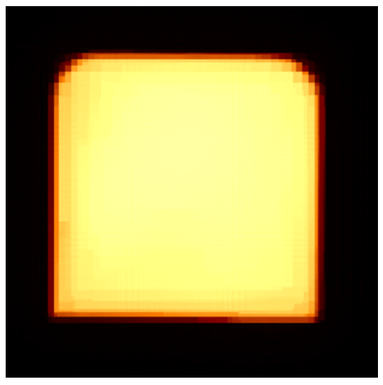
Zero-filled: in-order
MSE 0.00087 | PSNR 30.59 dB
SSIM 0.914 | HaarPSI 0.486



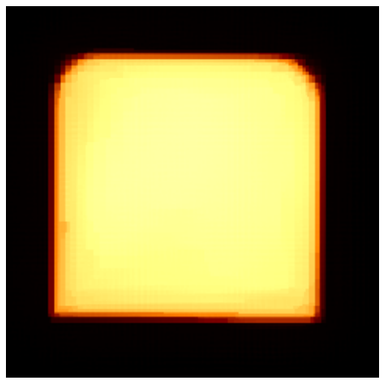
Zero-filled: in-order
MSE 0.00054 | PSNR 32.67 dB
SSIM 0.945 | HaarPSI 0.658



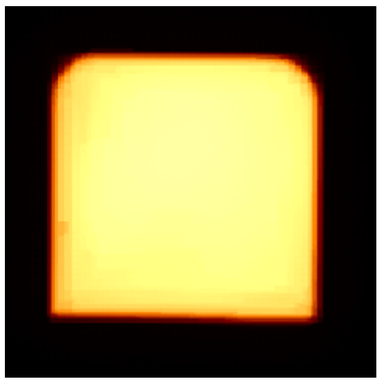
Zero-filled: in-order
MSE 0.00025 | PSNR 36.10 dB
SSIM 0.973 | HaarPSI 0.997



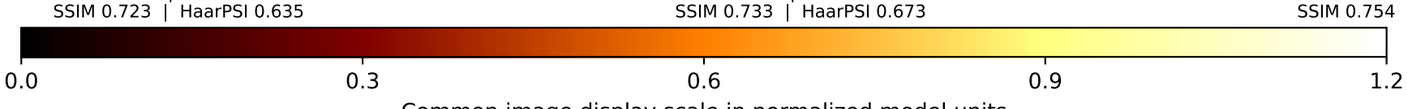
Learned U-TV
MSE 0.00096 | PSNR 30.16 dB
SSIM 0.723 | HaarPSI 0.635



Learned U-TV
MSE 0.00086 | PSNR 30.65 dB
SSIM 0.733 | HaarPSI 0.673



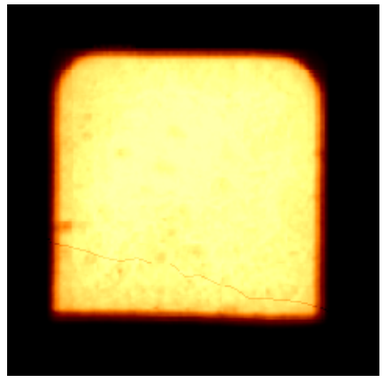
Learned U-TV
MSE 0.00042 | PSNR 33.74 dB
SSIM 0.754 | HaarPSI 0.781



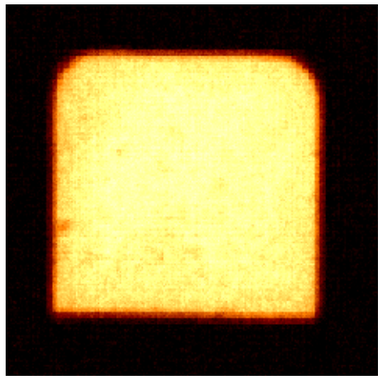
Perovskite / image0611 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

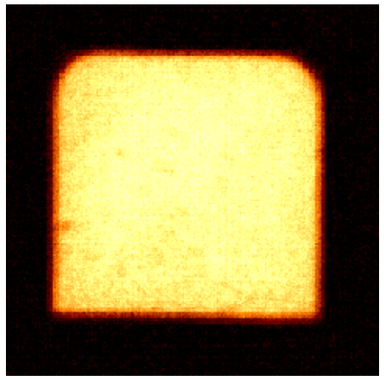


30% sampling



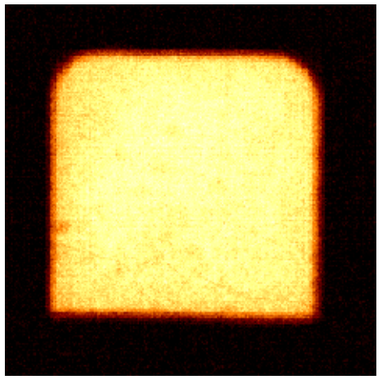
Zero-filled: matched multilevel
MSE 0.00156 | PSNR 28.07 dB
SSIM 0.478 | HaarPSI 0.716

40% sampling



Zero-filled: matched multilevel
MSE 0.00184 | PSNR 27.35 dB
SSIM 0.440 | HaarPSI 0.729

50% sampling



Zero-filled: matched multilevel
MSE 0.00214 | PSNR 26.70 dB
SSIM 0.407 | HaarPSI 0.738

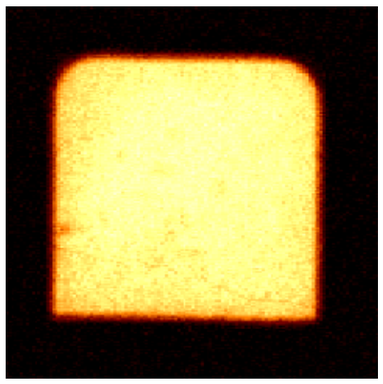
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

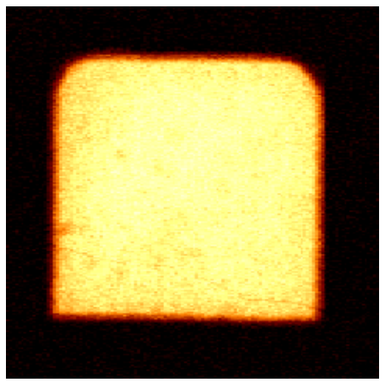
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

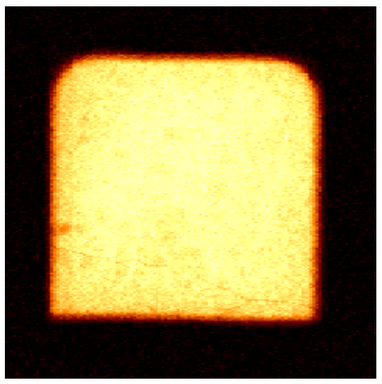
Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00122 | PSNR 29.15 dB
SSIM 0.531 | HaarPSI 0.762



Zero-filled: in-order
MSE 0.00156 | PSNR 28.08 dB
SSIM 0.473 | HaarPSI 0.755



Zero-filled: in-order
MSE 0.00191 | PSNR 27.19 dB
SSIM 0.433 | HaarPSI 0.754



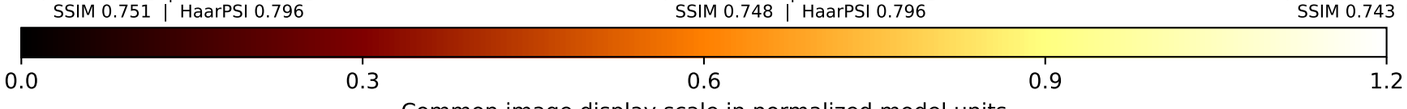
Learned U-TV
MSE 0.00038 | PSNR 34.25 dB
SSIM 0.751 | HaarPSI 0.796



Learned U-TV
MSE 0.00036 | PSNR 34.46 dB
SSIM 0.748 | HaarPSI 0.796



Learned U-TV
MSE 0.00036 | PSNR 34.44 dB
SSIM 0.743 | HaarPSI 0.809



Perovskite / image0611 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

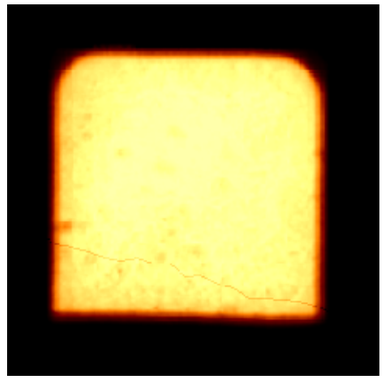
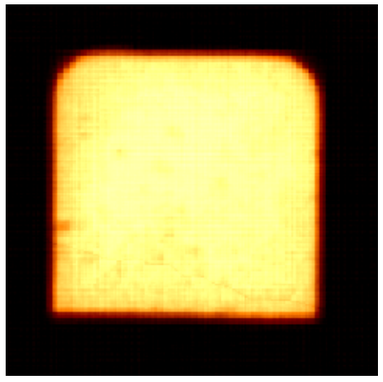
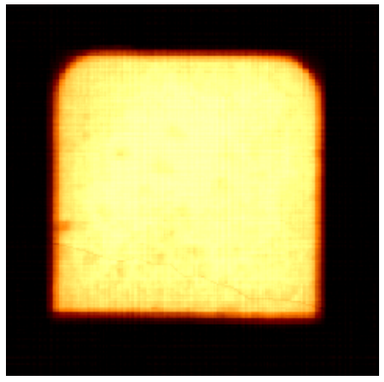
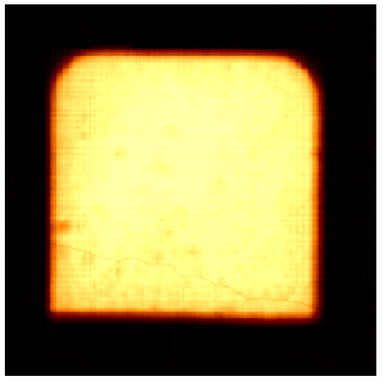
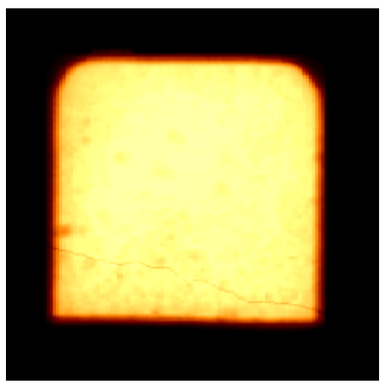
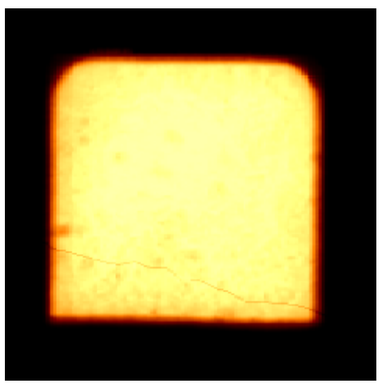
Columns: sampling ratios

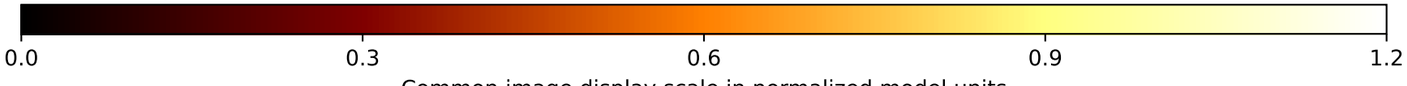
Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

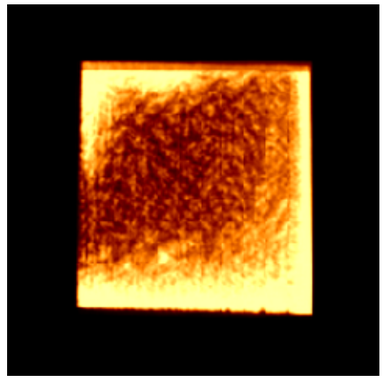
Ground truth	30% sampling	40% sampling	50% sampling
			
	Zero-filled: matched multilevel MSE 0.00049 | PSNR 33.07 dB SSIM 0.820 | HaarPSI 0.820	Zero-filled: matched multilevel MSE 0.00041 | PSNR 33.83 dB SSIM 0.829 | HaarPSI 0.855	Zero-filled: matched multilevel MSE 0.00033 | PSNR 34.84 dB SSIM 0.850 | HaarPSI 0.897
			
	Zero-filled: in-order MSE 0.00015 | PSNR 38.23 dB SSIM 0.980 | HaarPSI 0.996	Zero-filled: in-order MSE 0.00012 | PSNR 39.12 dB SSIM 0.987 | HaarPSI 1.000	Zero-filled: in-order MSE 0.00012 | PSNR 39.36 dB SSIM 0.989 | HaarPSI 1.000
			
	Learned U-TV MSE 0.00032 | PSNR 34.99 dB SSIM 0.761 | HaarPSI 0.811	Learned U-TV MSE 0.00030 | PSNR 35.24 dB SSIM 0.764 | HaarPSI 0.819	Learned U-TV MSE 0.00028 | PSNR 35.49 dB SSIM 0.768 | HaarPSI 0.828



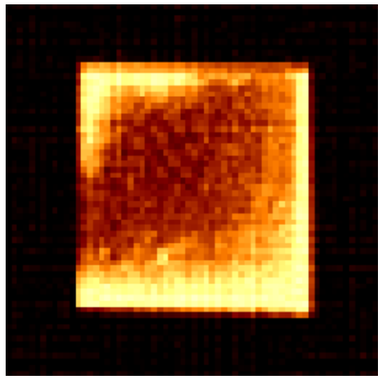
Perovskite / image0249 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

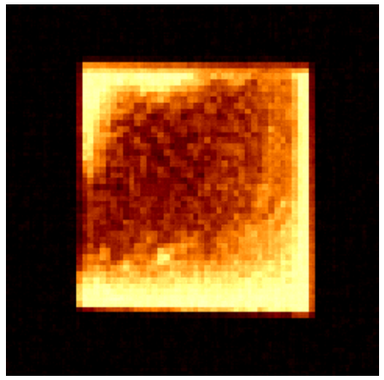


5% sampling



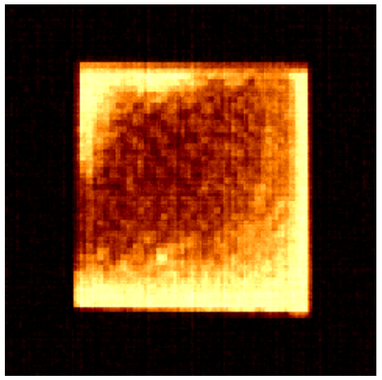
Zero-filled: matched multilevel
MSE 0.00518 | PSNR 22.85 dB
SSIM 0.405 | HaarPSI 0.333

10% sampling



Zero-filled: matched multilevel
MSE 0.00374 | PSNR 24.28 dB
SSIM 0.576 | HaarPSI 0.474

20% sampling



Zero-filled: matched multilevel
MSE 0.00327 | PSNR 24.85 dB
SSIM 0.471 | HaarPSI 0.546

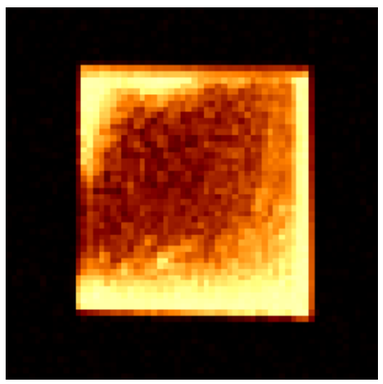
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

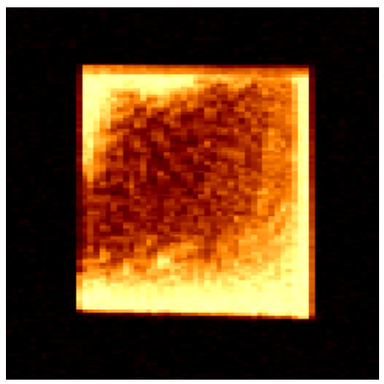
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

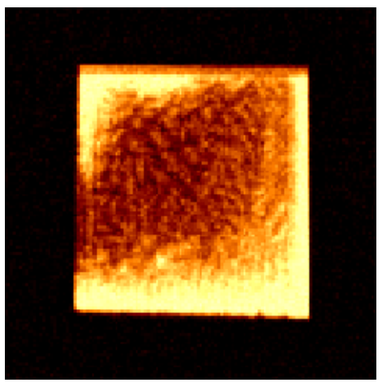
Single learned scale key; coarse J=6; PDHG=512.



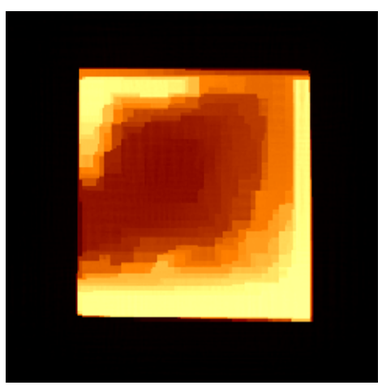
Zero-filled: in-order
MSE 0.00384 | PSNR 24.16 dB
SSIM 0.648 | HaarPSI 0.407



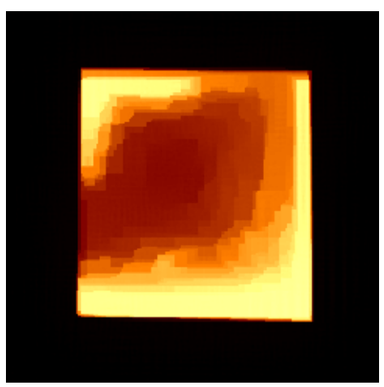
Zero-filled: in-order
MSE 0.00299 | PSNR 25.24 dB
SSIM 0.643 | HaarPSI 0.540



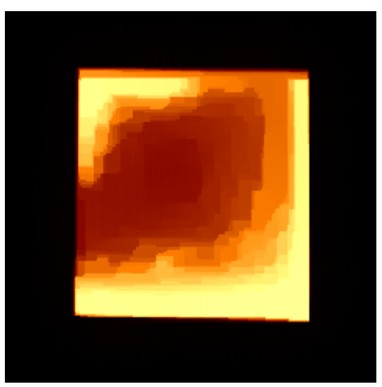
Zero-filled: in-order
MSE 0.00147 | PSNR 28.33 dB
SSIM 0.615 | HaarPSI 0.770



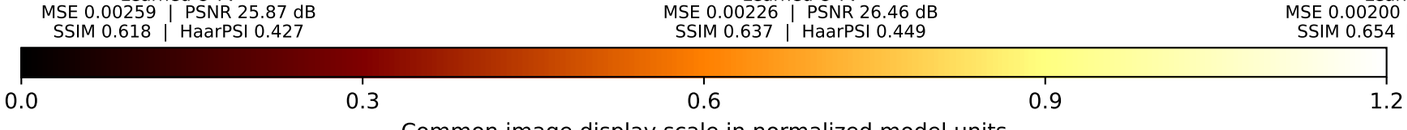
Learned U-TV
MSE 0.00259 | PSNR 25.87 dB
SSIM 0.618 | HaarPSI 0.427



Learned U-TV
MSE 0.00226 | PSNR 26.46 dB
SSIM 0.637 | HaarPSI 0.449



Learned U-TV
MSE 0.00200 | PSNR 26.98 dB
SSIM 0.654 | HaarPSI 0.458



Common image display scale in normalized model units

Perovskite / image0249 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

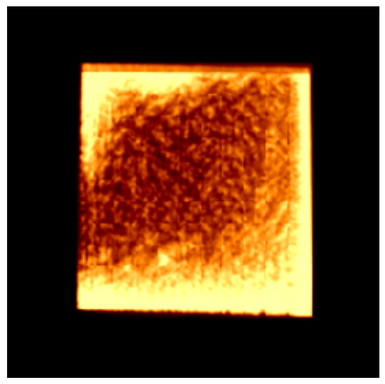
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

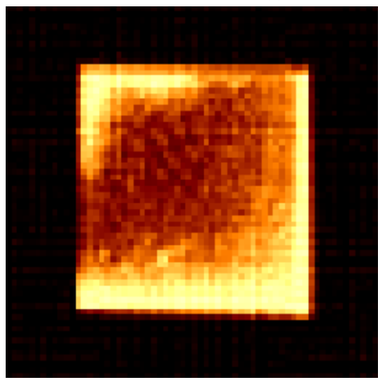
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

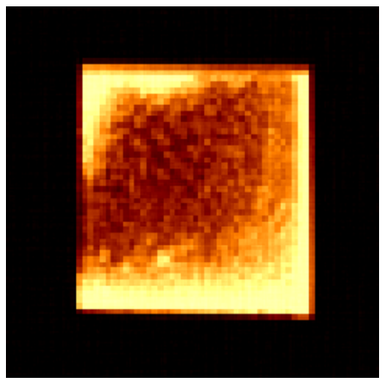


Ground truth



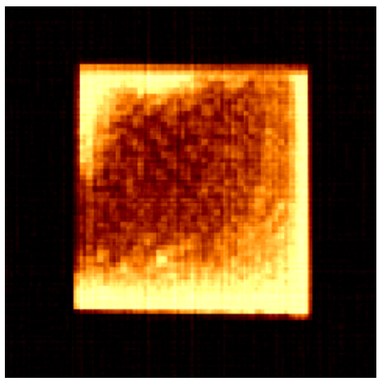
5% sampling

Zero-filled: matched multilevel
MSE 0.00501 | PSNR 23.00 dB
SSIM 0.429 | HaarPSI 0.336



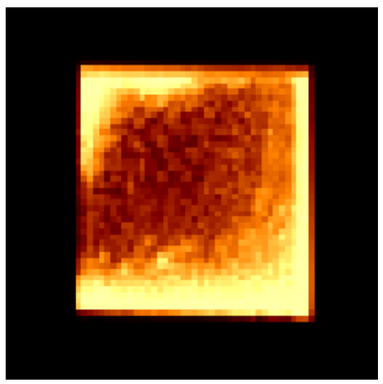
10% sampling

Zero-filled: matched multilevel
MSE 0.00341 | PSNR 24.67 dB
SSIM 0.750 | HaarPSI 0.486

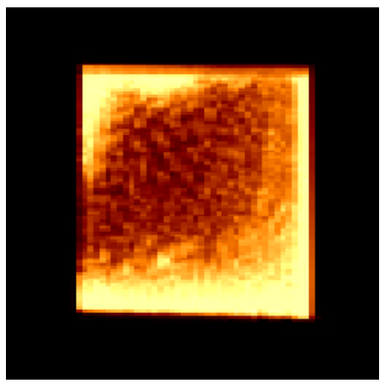


20% sampling

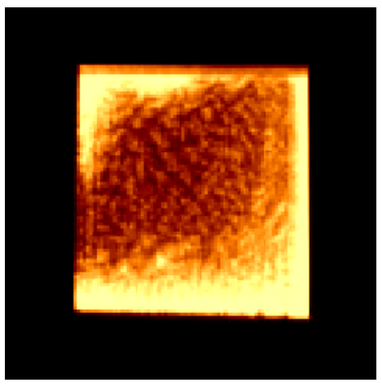
Zero-filled: matched multilevel
MSE 0.00260 | PSNR 25.86 dB
SSIM 0.615 | HaarPSI 0.569



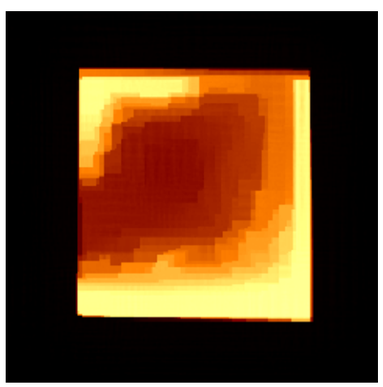
Zero-filled: in-order
MSE 0.00367 | PSNR 24.35 dB
SSIM 0.819 | HaarPSI 0.410



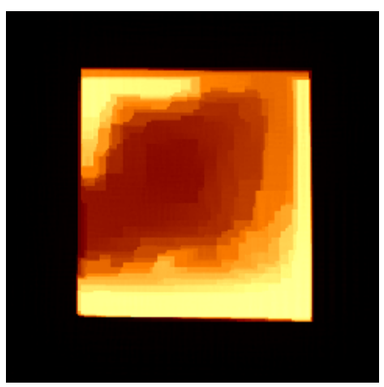
Zero-filled: in-order
MSE 0.00266 | PSNR 25.76 dB
SSIM 0.877 | HaarPSI 0.573



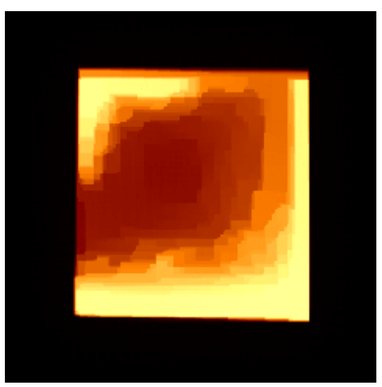
Zero-filled: in-order
MSE 0.00080 | PSNR 30.95 dB
SSIM 0.944 | HaarPSI 0.973



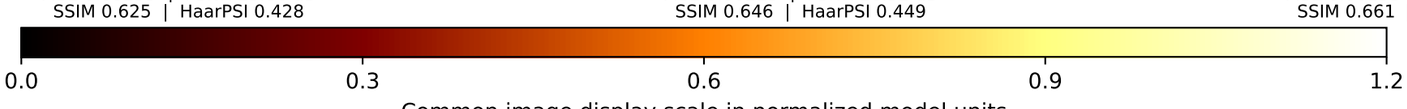
Learned U-TV
MSE 0.00259 | PSNR 25.86 dB
SSIM 0.625 | HaarPSI 0.428



Learned U-TV
MSE 0.00223 | PSNR 26.52 dB
SSIM 0.646 | HaarPSI 0.449



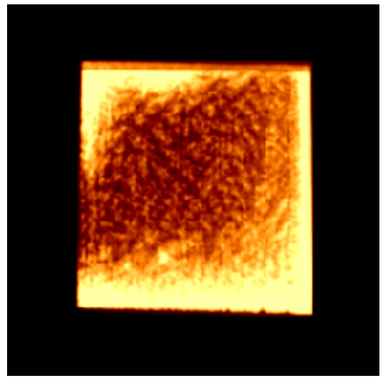
Learned U-TV
MSE 0.00193 | PSNR 27.15 dB
SSIM 0.661 | HaarPSI 0.465



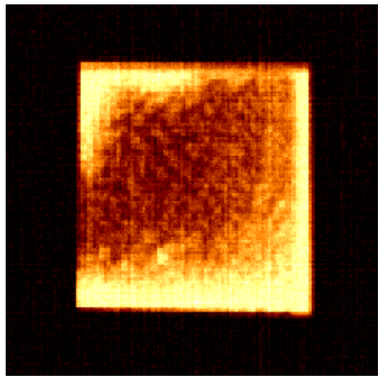
Perovskite / image0249 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

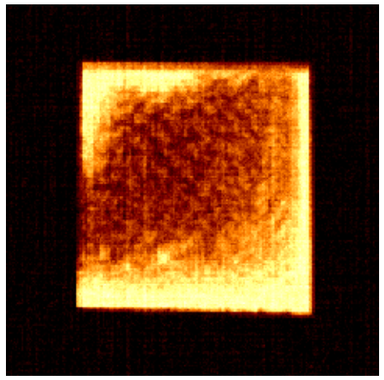


30% sampling



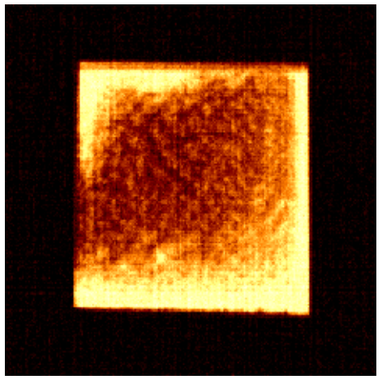
Zero-filled: matched multilevel
MSE 0.00306 | PSNR 25.14 dB
SSIM 0.451 | HaarPSI 0.605

40% sampling



Zero-filled: matched multilevel
MSE 0.00298 | PSNR 25.26 dB
SSIM 0.443 | HaarPSI 0.623

50% sampling



Zero-filled: matched multilevel
MSE 0.00288 | PSNR 25.40 dB
SSIM 0.443 | HaarPSI 0.658

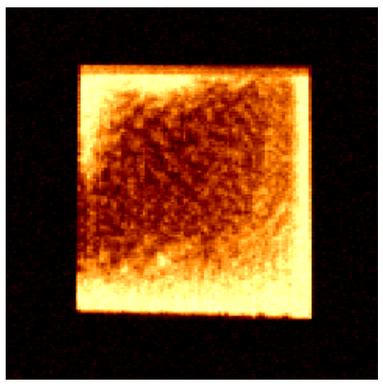
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

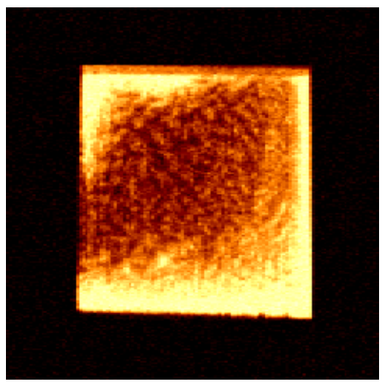
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

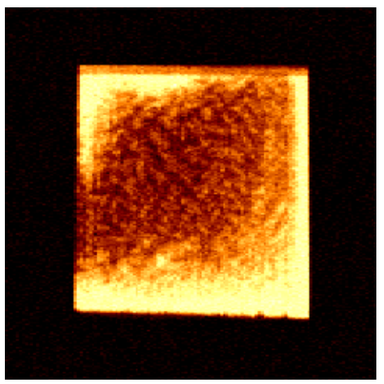
Single learned scale key; coarse J=6; PDHG=512.



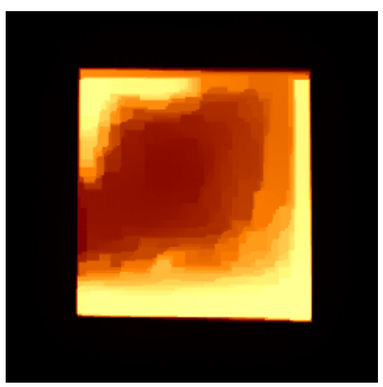
Zero-filled: in-order
MSE 0.00154 | PSNR 28.11 dB
SSIM 0.573 | HaarPSI 0.760



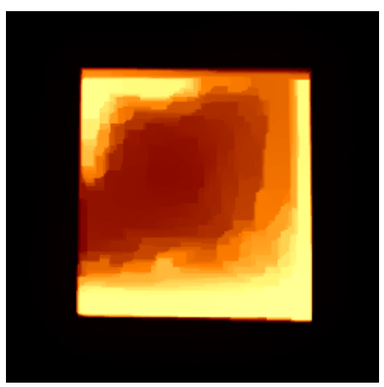
Zero-filled: in-order
MSE 0.00177 | PSNR 27.53 dB
SSIM 0.536 | HaarPSI 0.758



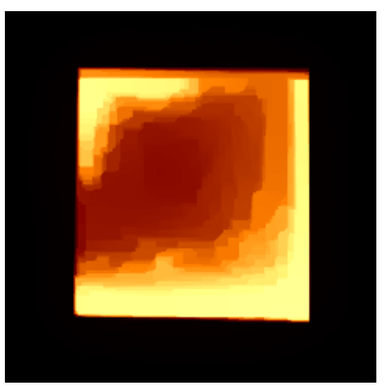
Zero-filled: in-order
MSE 0.00208 | PSNR 26.81 dB
SSIM 0.506 | HaarPSI 0.765



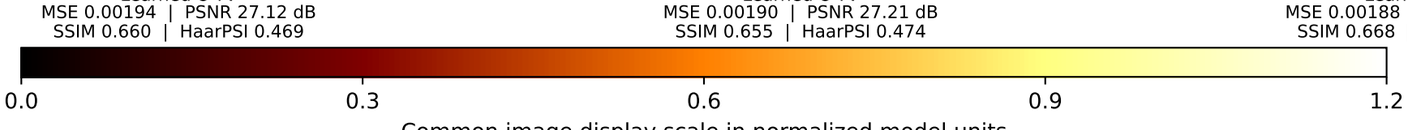
Learned U-TV
MSE 0.00194 | PSNR 27.12 dB
SSIM 0.660 | HaarPSI 0.469



Learned U-TV
MSE 0.00190 | PSNR 27.21 dB
SSIM 0.655 | HaarPSI 0.474



Learned U-TV
MSE 0.00188 | PSNR 27.25 dB
SSIM 0.668 | HaarPSI 0.480



Perovskite / image0249 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

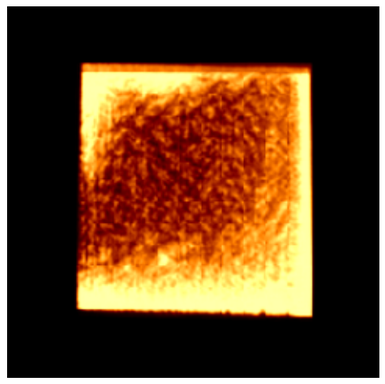
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

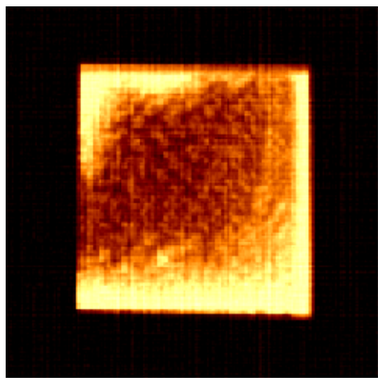
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

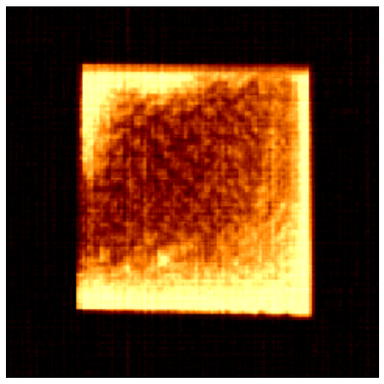


Ground truth



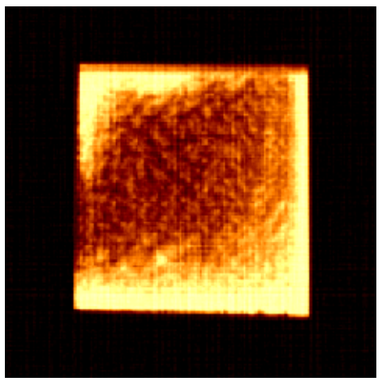
30% sampling

Zero-filled: matched multilevel
MSE 0.00206 | PSNR 26.85 dB
SSIM 0.603 | HaarPSI 0.634



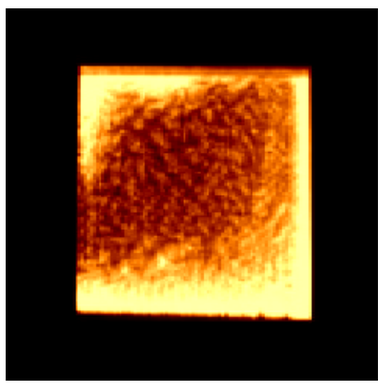
40% sampling

Zero-filled: matched multilevel
MSE 0.00164 | PSNR 27.85 dB
SSIM 0.616 | HaarPSI 0.676

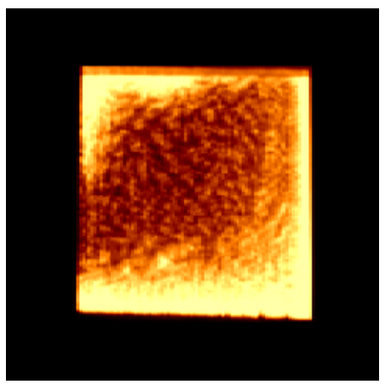


50% sampling

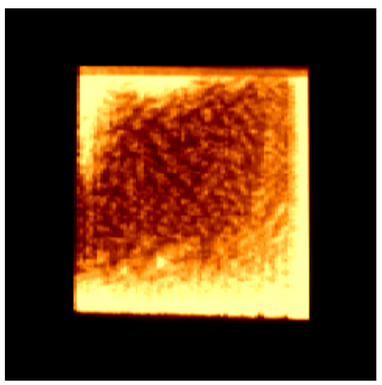
Zero-filled: matched multilevel
MSE 0.00121 | PSNR 29.17 dB
SSIM 0.666 | HaarPSI 0.726



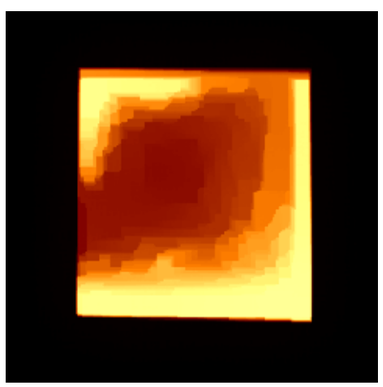
Zero-filled: in-order
MSE 0.00054 | PSNR 32.66 dB
SSIM 0.958 | HaarPSI 0.983



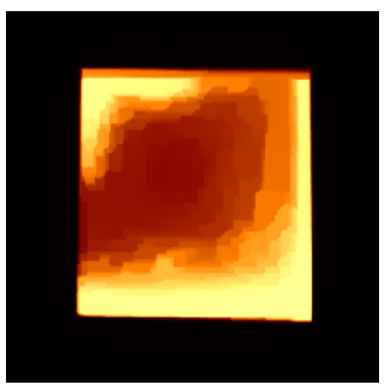
Zero-filled: in-order
MSE 0.00043 | PSNR 33.67 dB
SSIM 0.971 | HaarPSI 1.000



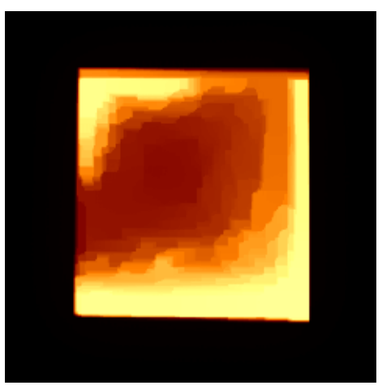
Zero-filled: in-order
MSE 0.00040 | PSNR 33.97 dB
SSIM 0.975 | HaarPSI 1.000



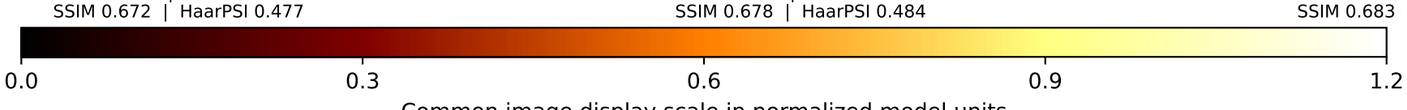
Learned U-TV
MSE 0.00184 | PSNR 27.36 dB
SSIM 0.672 | HaarPSI 0.477



Learned U-TV
MSE 0.00179 | PSNR 27.47 dB
SSIM 0.678 | HaarPSI 0.484



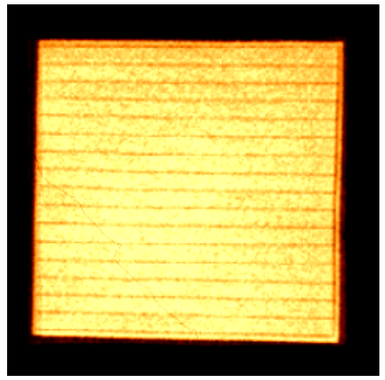
Learned U-TV
MSE 0.00175 | PSNR 27.57 dB
SSIM 0.683 | HaarPSI 0.497



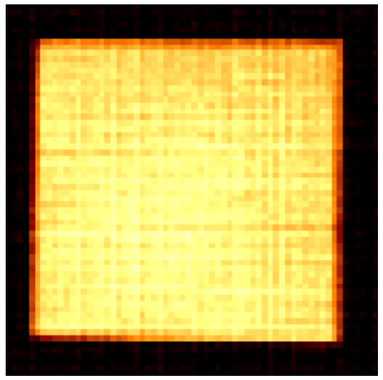
Reference / image0611 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

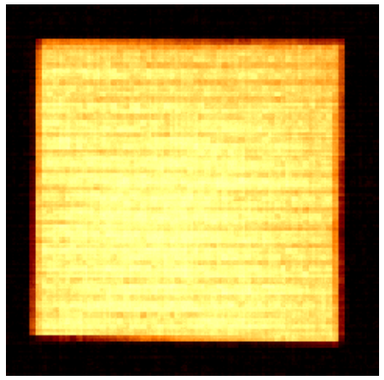


5% sampling



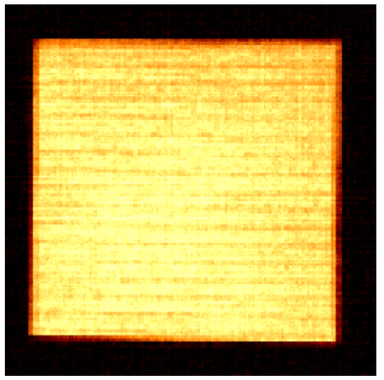
Zero-filled: matched multilevel
MSE 0.00526 | PSNR 22.79 dB
SSIM 0.401 | HaarPSI 0.331

10% sampling



Zero-filled: matched multilevel
MSE 0.00360 | PSNR 24.43 dB
SSIM 0.539 | HaarPSI 0.523

20% sampling



Zero-filled: matched multilevel
MSE 0.00360 | PSNR 24.44 dB
SSIM 0.505 | HaarPSI 0.557

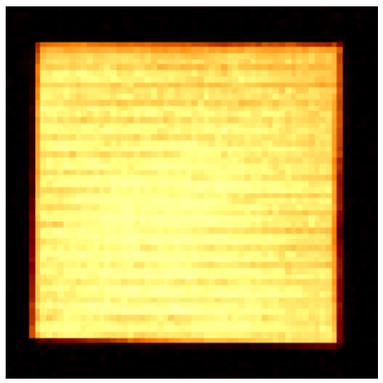
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

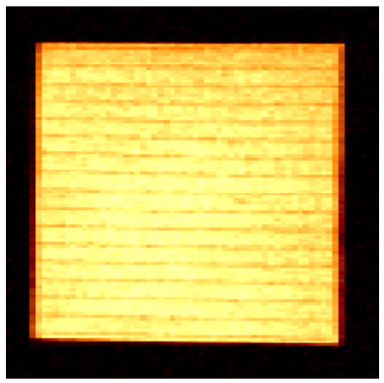
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

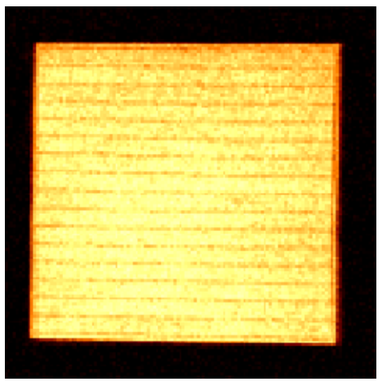
Single learned scale key; coarse J=6; PDHG=512.



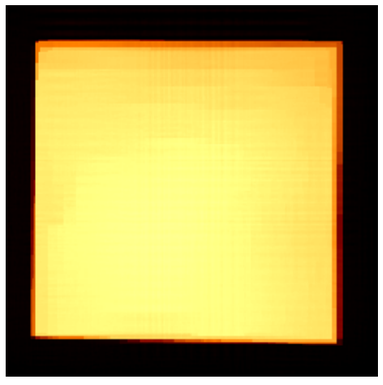
Zero-filled: in-order
MSE 0.00380 | PSNR 24.21 dB
SSIM 0.583 | HaarPSI 0.389



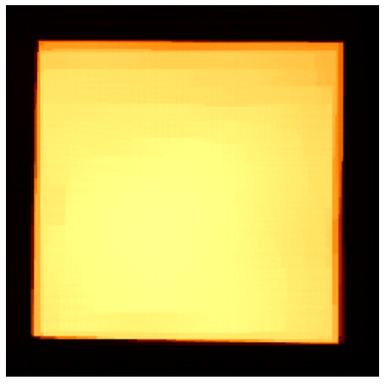
Zero-filled: in-order
MSE 0.00239 | PSNR 26.21 dB
SSIM 0.666 | HaarPSI 0.574



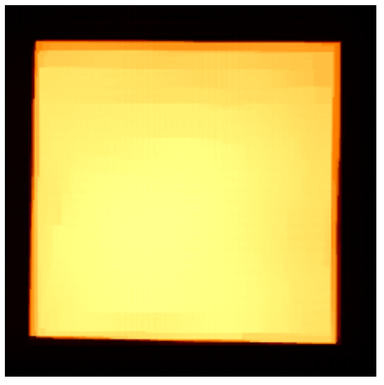
Zero-filled: in-order
MSE 0.00192 | PSNR 27.16 dB
SSIM 0.651 | HaarPSI 0.786



Learned U-TV
MSE 0.00358 | PSNR 24.46 dB
SSIM 0.417 | HaarPSI 0.370



Learned U-TV
MSE 0.00259 | PSNR 25.87 dB
SSIM 0.440 | HaarPSI 0.401



Learned U-TV
MSE 0.00234 | PSNR 26.31 dB
SSIM 0.450 | HaarPSI 0.407

Reference / image0611 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

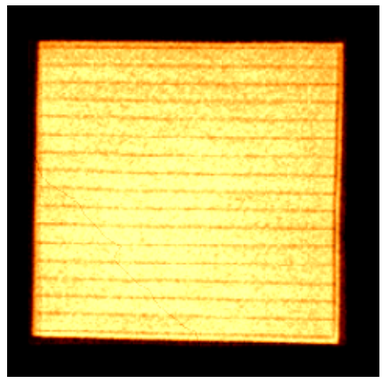
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

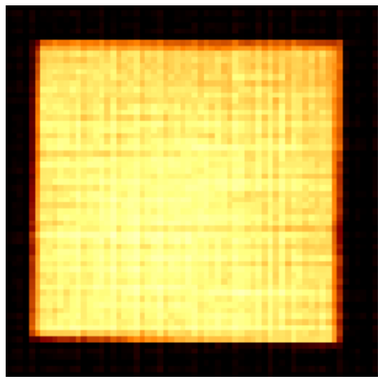
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

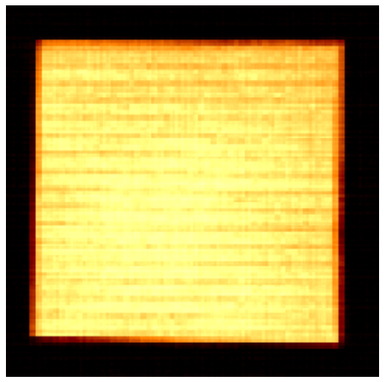


Ground truth



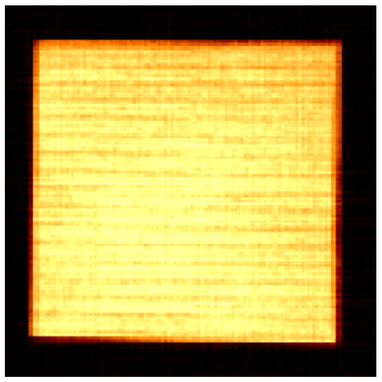
5% sampling

Zero-filled: matched multilevel
MSE 0.00507 | PSNR 22.95 dB
SSIM 0.421 | HaarPSI 0.334



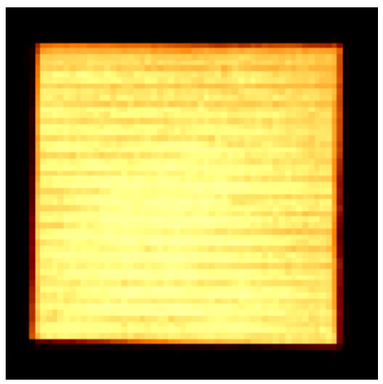
10% sampling

Zero-filled: matched multilevel
MSE 0.00323 | PSNR 24.90 dB
SSIM 0.631 | HaarPSI 0.536

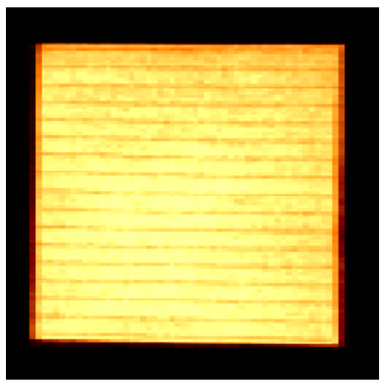


20% sampling

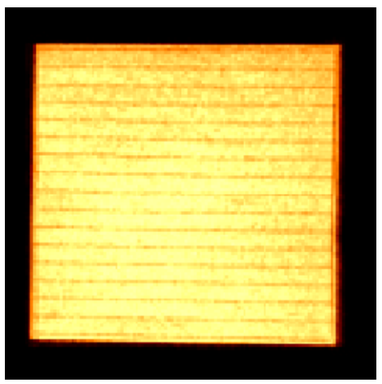
Zero-filled: matched multilevel
MSE 0.00282 | PSNR 25.49 dB
SSIM 0.633 | HaarPSI 0.579



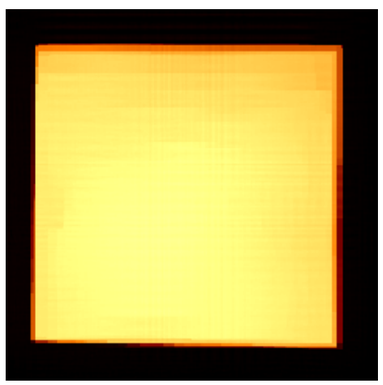
Zero-filled: in-order
MSE 0.00362 | PSNR 24.42 dB
SSIM 0.681 | HaarPSI 0.394



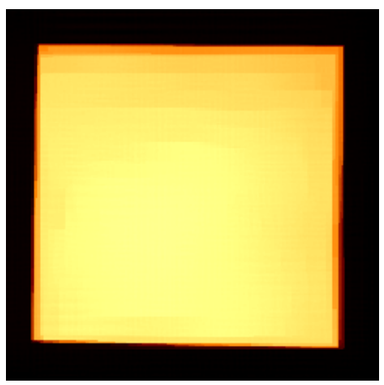
Zero-filled: in-order
MSE 0.00202 | PSNR 26.95 dB
SSIM 0.813 | HaarPSI 0.610



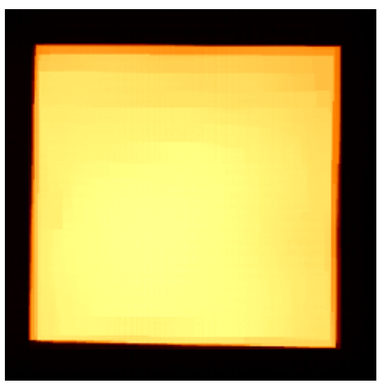
Zero-filled: in-order
MSE 0.00116 | PSNR 29.34 dB
SSIM 0.874 | HaarPSI 0.984



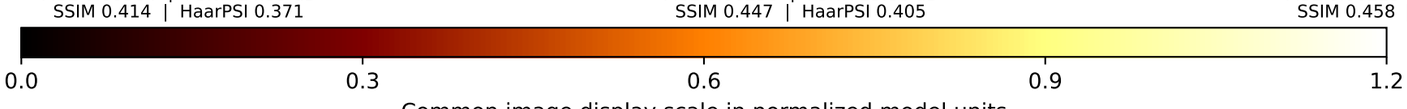
Learned U-TV
MSE 0.00364 | PSNR 24.39 dB
SSIM 0.414 | HaarPSI 0.371



Learned U-TV
MSE 0.00258 | PSNR 25.88 dB
SSIM 0.447 | HaarPSI 0.405



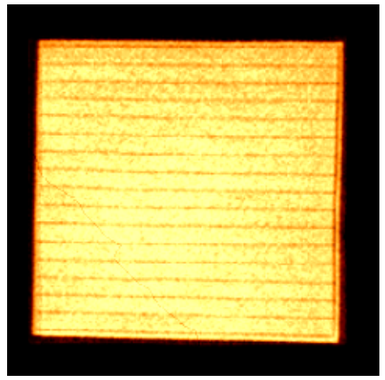
Learned U-TV
MSE 0.00229 | PSNR 26.39 dB
SSIM 0.458 | HaarPSI 0.411



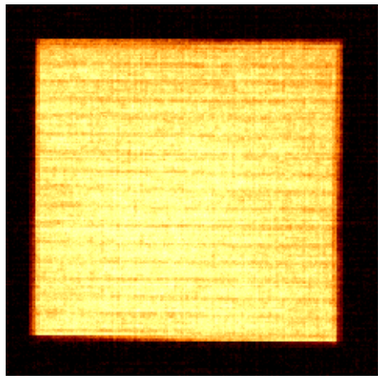
Reference / image0611 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

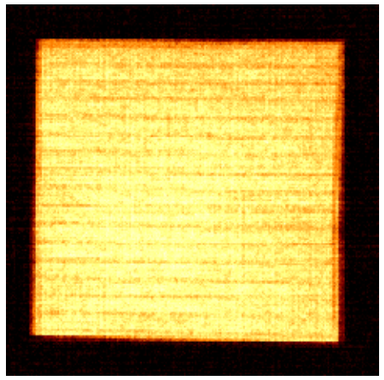


30% sampling



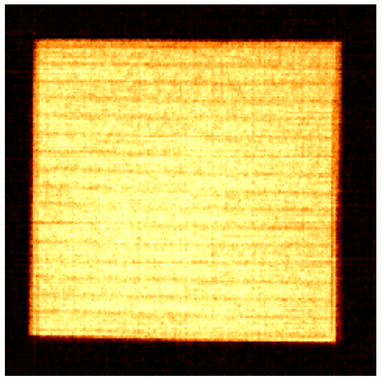
Zero-filled: matched multilevel
MSE 0.00341 | PSNR 24.67 dB
SSIM 0.514 | HaarPSI 0.632

40% sampling



Zero-filled: matched multilevel
MSE 0.00337 | PSNR 24.72 dB
SSIM 0.506 | HaarPSI 0.663

50% sampling



Zero-filled: matched multilevel
MSE 0.00327 | PSNR 24.85 dB
SSIM 0.513 | HaarPSI 0.679

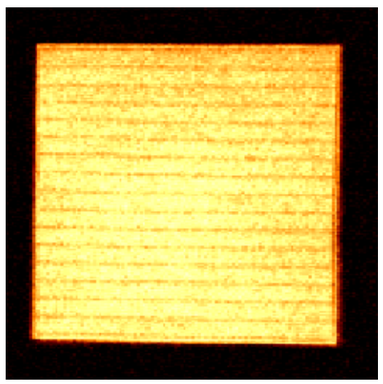
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

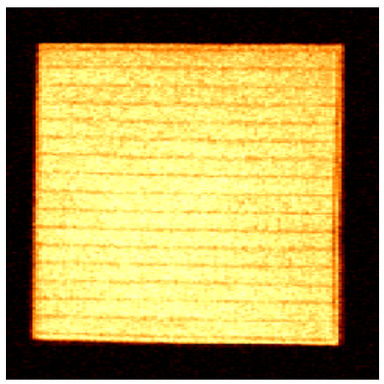
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

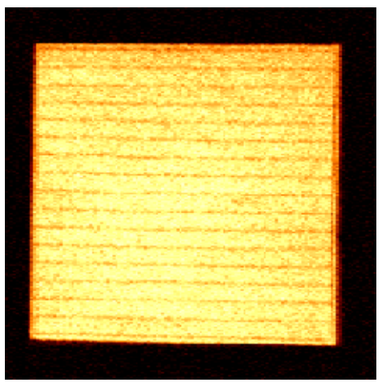
Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00195 | PSNR 27.10 dB
SSIM 0.628 | HaarPSI 0.779



Zero-filled: in-order
MSE 0.00195 | PSNR 27.11 dB
SSIM 0.630 | HaarPSI 0.781



Zero-filled: in-order
MSE 0.00230 | PSNR 26.38 dB
SSIM 0.601 | HaarPSI 0.774



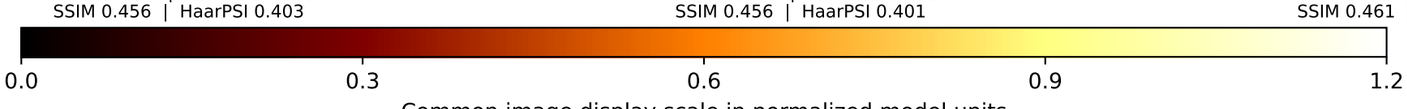
Learned U-TV
MSE 0.00226 | PSNR 26.45 dB
SSIM 0.456 | HaarPSI 0.403



Learned U-TV
MSE 0.00223 | PSNR 26.51 dB
SSIM 0.456 | HaarPSI 0.401



Learned U-TV
MSE 0.00220 | PSNR 26.57 dB
SSIM 0.461 | HaarPSI 0.405



Reference / image0611 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

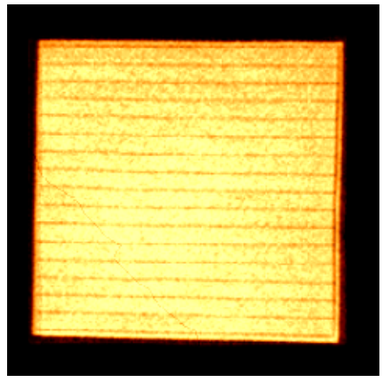
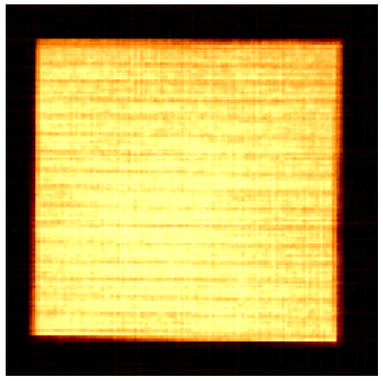
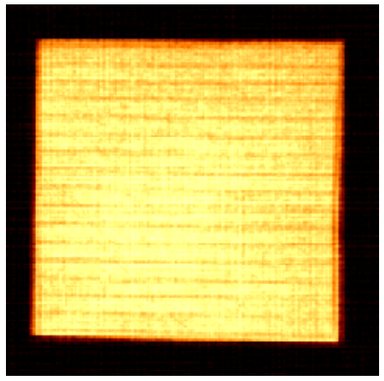
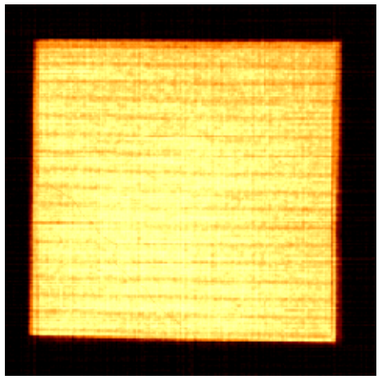
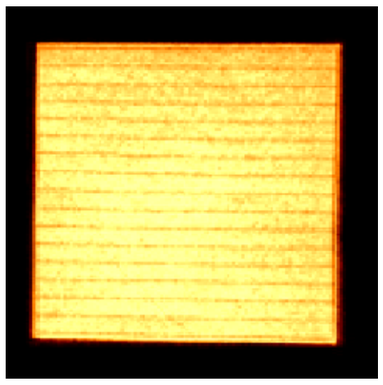
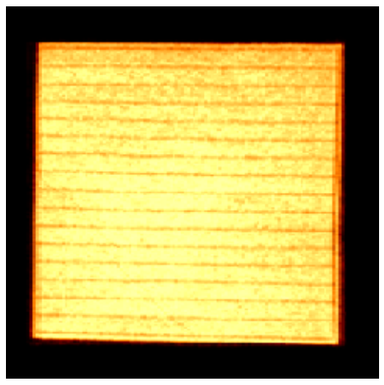
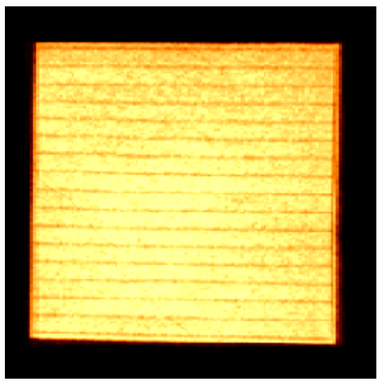
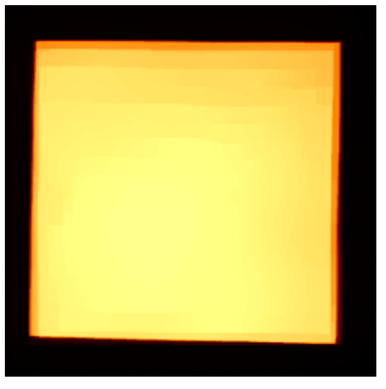
Columns: sampling ratios

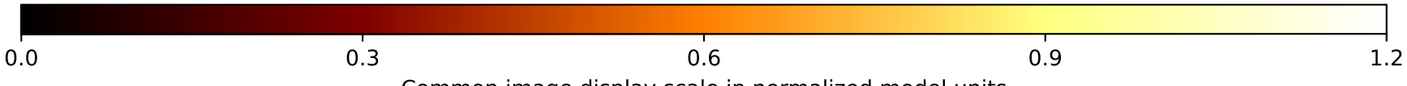
Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

Ground truth	30% sampling	40% sampling	50% sampling
			
	Zero-filled: matched multilevel MSE 0.00228 | PSNR 26.42 dB SSIM 0.661 | HaarPSI 0.677	Zero-filled: matched multilevel MSE 0.00186 | PSNR 27.29 dB SSIM 0.678 | HaarPSI 0.709	Zero-filled: matched multilevel MSE 0.00138 | PSNR 28.60 dB SSIM 0.719 | HaarPSI 0.762
			
	Zero-filled: in-order MSE 0.00081 | PSNR 30.92 dB SSIM 0.905 | HaarPSI 0.992	Zero-filled: in-order MSE 0.00044 | PSNR 33.55 dB SSIM 0.950 | HaarPSI 1.000	Zero-filled: in-order MSE 0.00039 | PSNR 34.11 dB SSIM 0.961 | HaarPSI 1.000
			
	Learned U-TV MSE 0.00222 | PSNR 26.54 dB SSIM 0.463 | HaarPSI 0.405	Learned U-TV MSE 0.00218 | PSNR 26.62 dB SSIM 0.464 | HaarPSI 0.400	Learned U-TV MSE 0.00217 | PSNR 26.65 dB SSIM 0.467 | HaarPSI 0.403



Reference / image0249 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

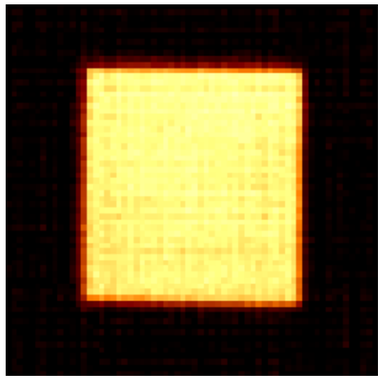
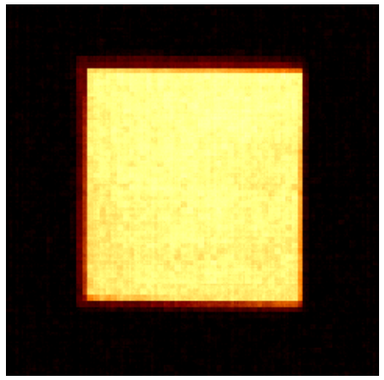
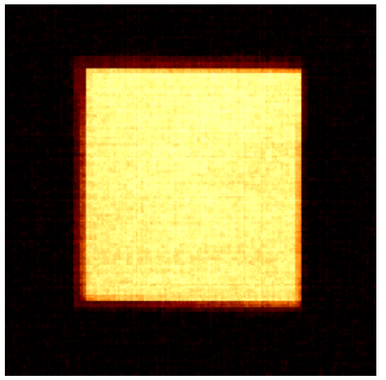
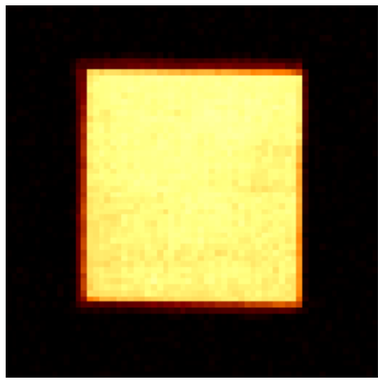
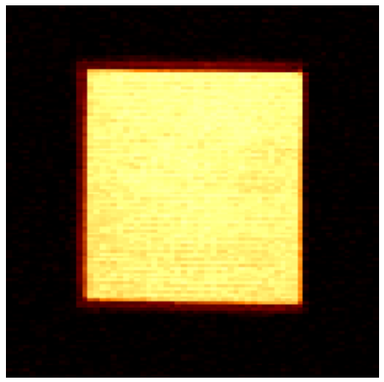
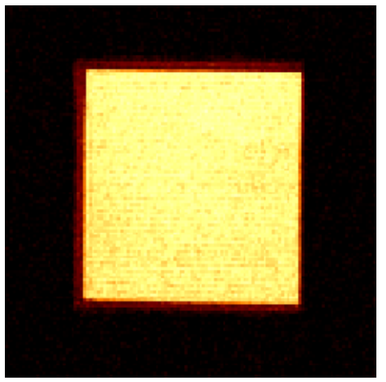
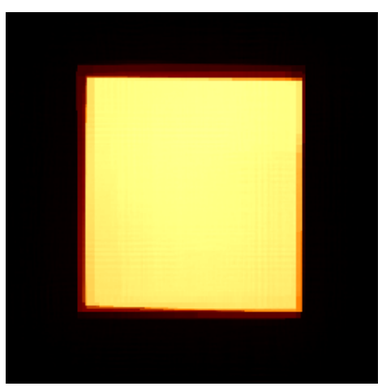
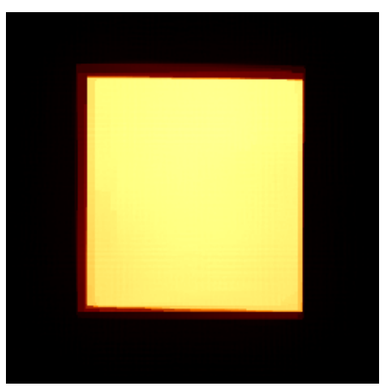
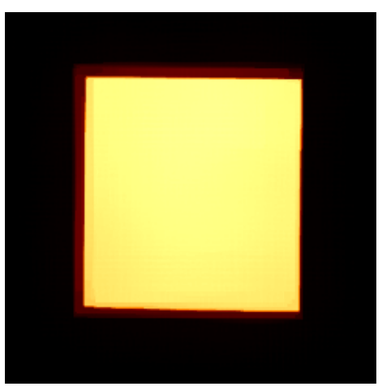
Columns: sampling ratios

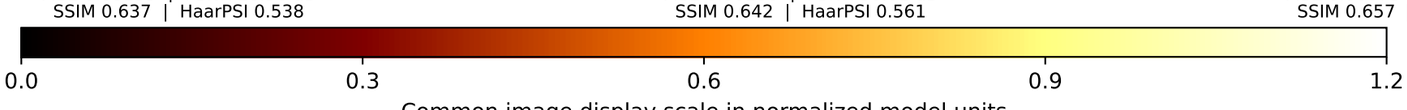
Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

Ground truth	5% sampling	10% sampling	20% sampling
			
	Zero-filled: matched multilevel MSE 0.00363 | PSNR 24.41 dB SSIM 0.376 | HaarPSI 0.317	Zero-filled: matched multilevel MSE 0.00256 | PSNR 25.92 dB SSIM 0.522 | HaarPSI 0.467	Zero-filled: matched multilevel MSE 0.00263 | PSNR 25.80 dB SSIM 0.446 | HaarPSI 0.519
			
	Zero-filled: in-order MSE 0.00255 | PSNR 25.94 dB SSIM 0.622 | HaarPSI 0.376	Zero-filled: in-order MSE 0.00184 | PSNR 27.36 dB SSIM 0.612 | HaarPSI 0.523	Zero-filled: in-order MSE 0.00151 | PSNR 28.20 dB SSIM 0.557 | HaarPSI 0.763
			
	Learned U-TV MSE 0.00139 | PSNR 28.56 dB SSIM 0.637 | HaarPSI 0.538	Learned U-TV MSE 0.00100 | PSNR 30.01 dB SSIM 0.642 | HaarPSI 0.561	Learned U-TV MSE 0.00075 | PSNR 31.26 dB SSIM 0.657 | HaarPSI 0.630



Reference / image0249 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

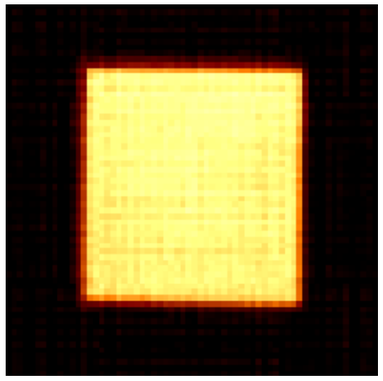
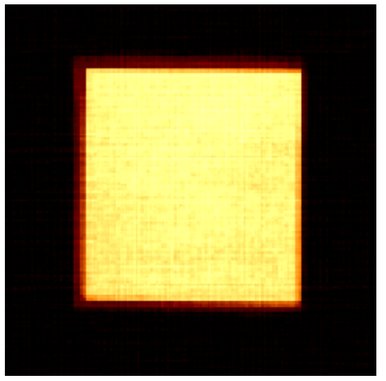
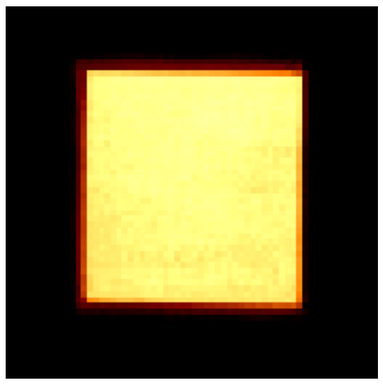
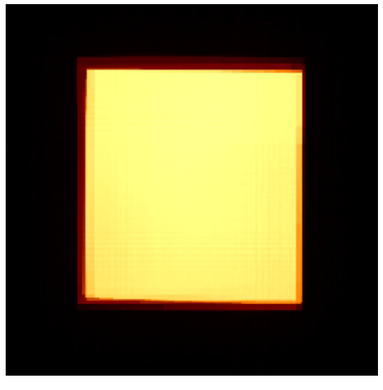
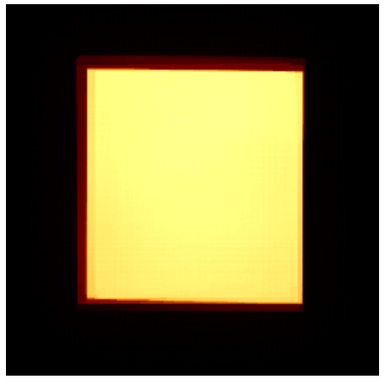
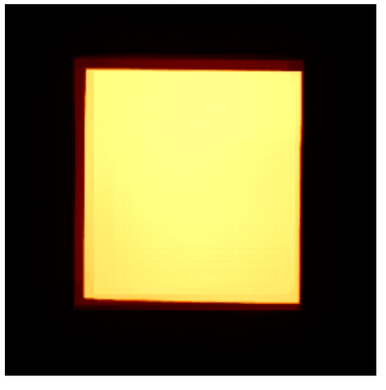
Columns: sampling ratios

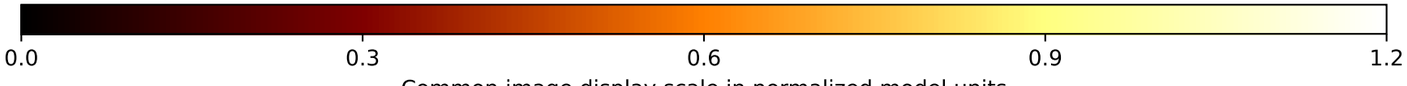
Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.

Ground truth	5% sampling	10% sampling	20% sampling
			
	Zero-filled: matched multilevel MSE 0.00345 | PSNR 24.62 dB SSIM 0.409 | HaarPSI 0.318	Zero-filled: matched multilevel MSE 0.00221 | PSNR 26.56 dB SSIM 0.710 | HaarPSI 0.476	Zero-filled: matched multilevel MSE 0.00195 | PSNR 27.10 dB SSIM 0.658 | HaarPSI 0.539
			
	Zero-filled: in-order MSE 0.00238 | PSNR 26.23 dB SSIM 0.792 | HaarPSI 0.373	Zero-filled: in-order MSE 0.00149 | PSNR 28.27 dB SSIM 0.867 | HaarPSI 0.555	Zero-filled: in-order MSE 0.00083 | PSNR 30.82 dB SSIM 0.912 | HaarPSI 0.984
			
	Learned U-TV MSE 0.00141 | PSNR 28.50 dB SSIM 0.633 | HaarPSI 0.538	Learned U-TV MSE 0.00096 | PSNR 30.16 dB SSIM 0.646 | HaarPSI 0.575	Learned U-TV MSE 0.00074 | PSNR 31.30 dB SSIM 0.659 | HaarPSI 0.623



Common image display scale in normalized model units

Reference / image0249 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

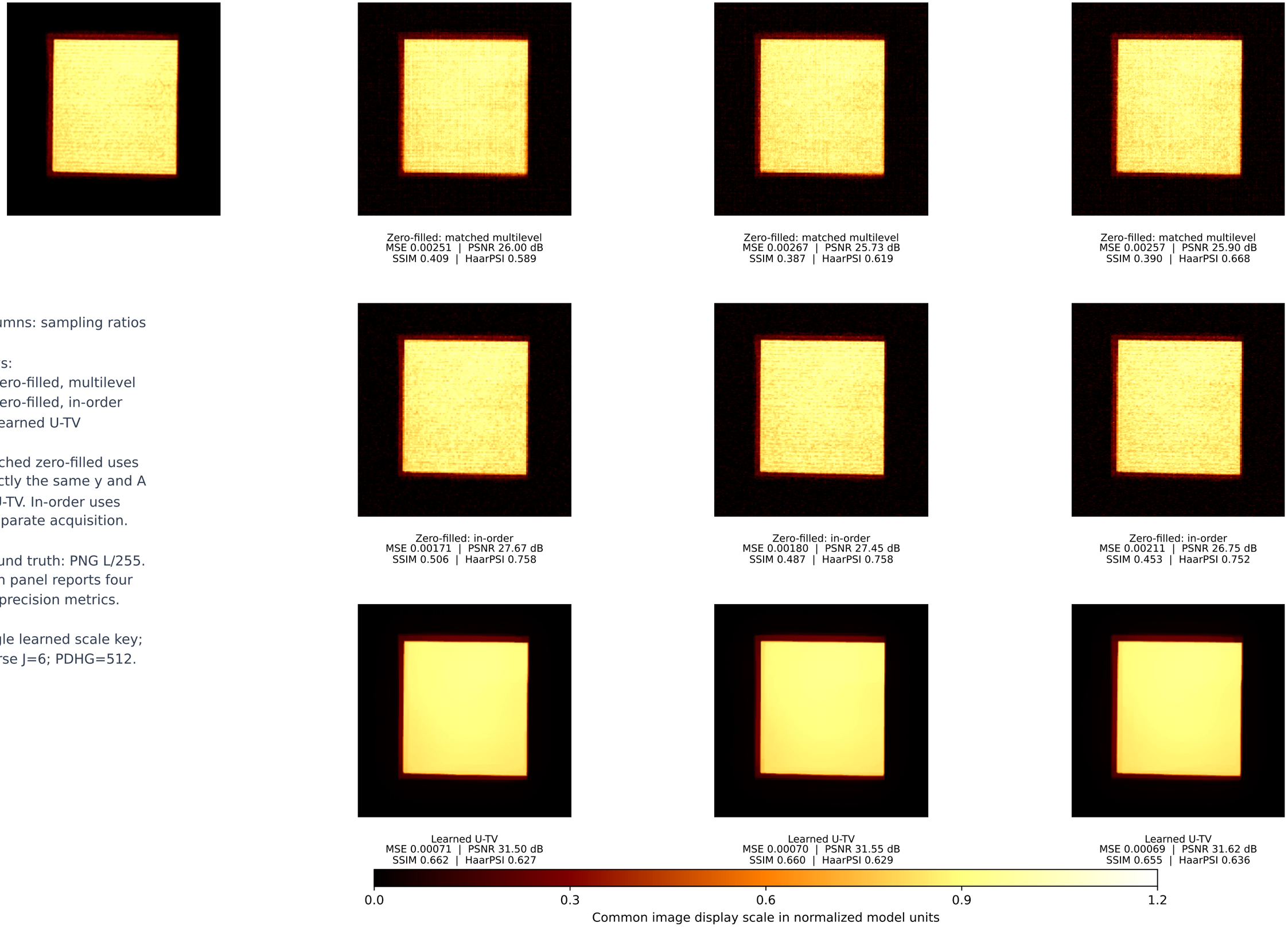
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.



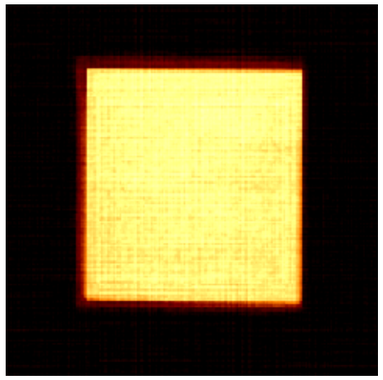
Reference / image0249 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

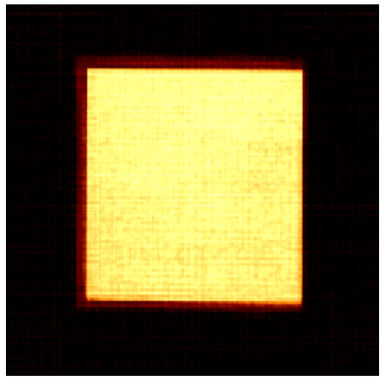


30% sampling



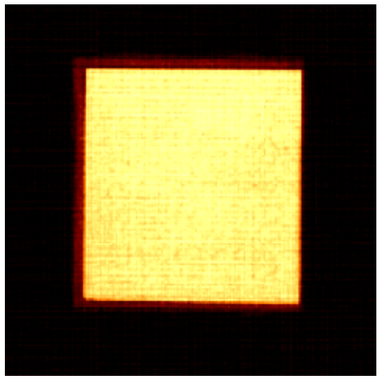
Zero-filled: matched multilevel
MSE 0.00149 | PSNR 28.27 dB
SSIM 0.636 | HaarPSI 0.622

40% sampling



Zero-filled: matched multilevel
MSE 0.00128 | PSNR 28.91 dB
SSIM 0.636 | HaarPSI 0.664

50% sampling



Zero-filled: matched multilevel
MSE 0.00085 | PSNR 30.69 dB
SSIM 0.697 | HaarPSI 0.775

Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

Single learned scale key; coarse J=6; PDHG=512.



Zero-filled: in-order
MSE 0.00068 | PSNR 31.71 dB
SSIM 0.928 | HaarPSI 1.000



Zero-filled: in-order
MSE 0.00043 | PSNR 33.63 dB
SSIM 0.964 | HaarPSI 1.000



Zero-filled: in-order
MSE 0.00041 | PSNR 33.87 dB
SSIM 0.971 | HaarPSI 1.000



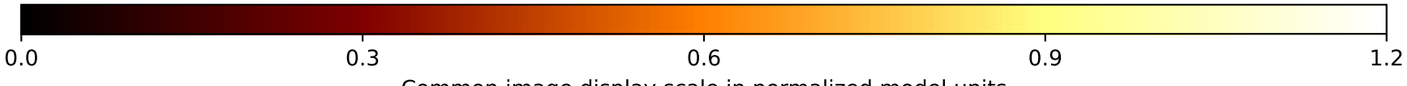
Learned U-TV
MSE 0.00068 | PSNR 31.67 dB
SSIM 0.664 | HaarPSI 0.636



Learned U-TV
MSE 0.00067 | PSNR 31.72 dB
SSIM 0.664 | HaarPSI 0.638



Learned U-TV
MSE 0.00066 | PSNR 31.80 dB
SSIM 0.668 | HaarPSI 0.647

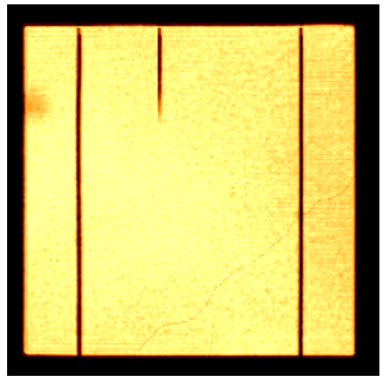


Common image display scale in normalized model units

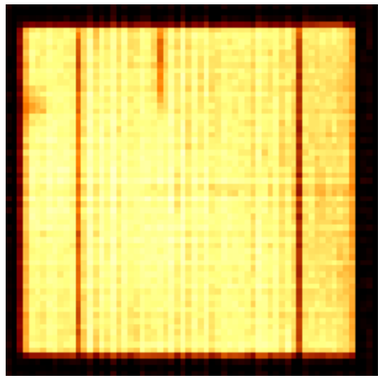
Silicone / image0611 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

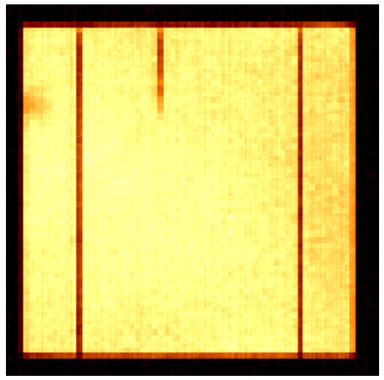


5% sampling



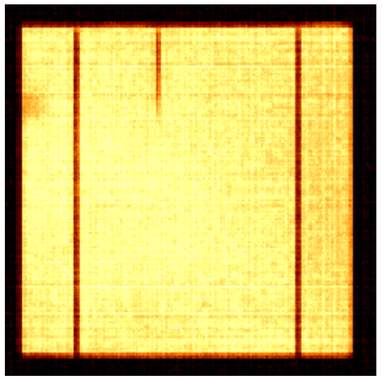
Zero-filled: matched multilevel
MSE 0.00899 | PSNR 20.46 dB
SSIM 0.396 | HaarPSI 0.314

10% sampling



Zero-filled: matched multilevel
MSE 0.00646 | PSNR 21.90 dB
SSIM 0.533 | HaarPSI 0.480

20% sampling



Zero-filled: matched multilevel
MSE 0.00574 | PSNR 22.41 dB
SSIM 0.434 | HaarPSI 0.513

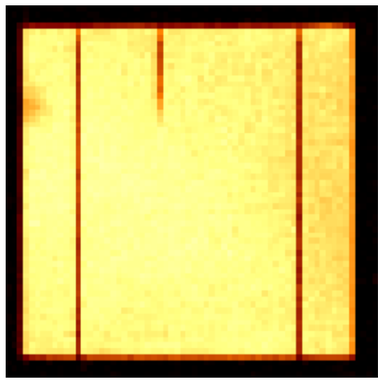
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

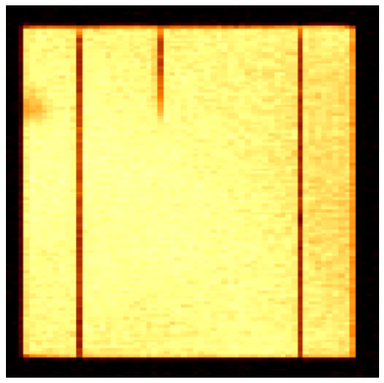
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

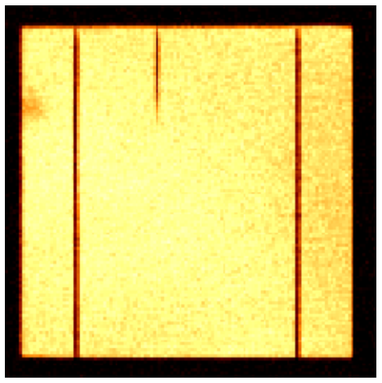
Single learned scale key; coarse J=6; PDHG=512.



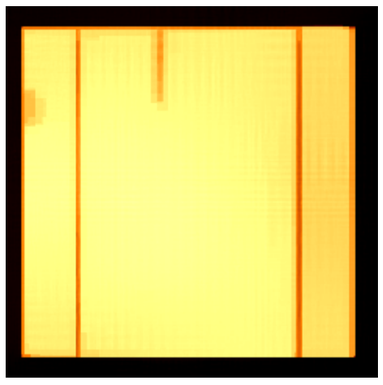
Zero-filled: in-order
MSE 0.00654 | PSNR 21.84 dB
SSIM 0.606 | HaarPSI 0.432



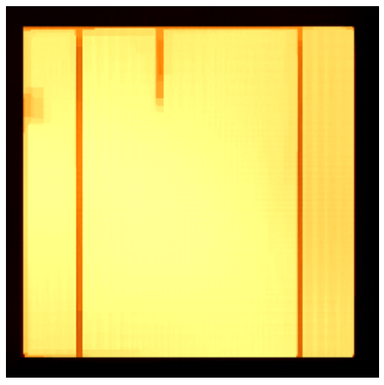
Zero-filled: in-order
MSE 0.00437 | PSNR 23.59 dB
SSIM 0.621 | HaarPSI 0.530



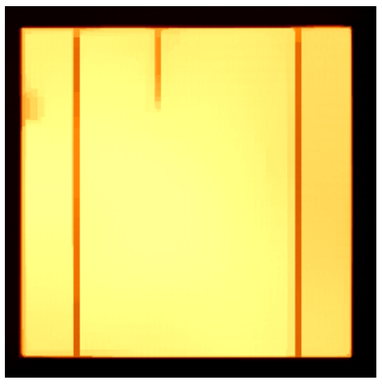
Zero-filled: in-order
MSE 0.00338 | PSNR 24.71 dB
SSIM 0.598 | HaarPSI 0.720



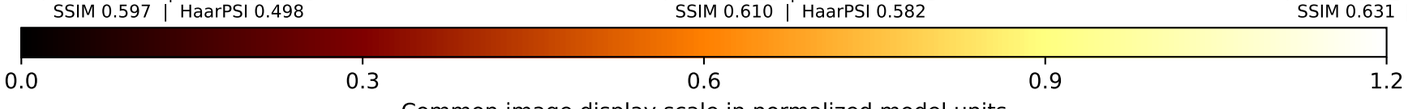
Learned U-TV
MSE 0.00467 | PSNR 23.31 dB
SSIM 0.597 | HaarPSI 0.498



Learned U-TV
MSE 0.00412 | PSNR 23.86 dB
SSIM 0.610 | HaarPSI 0.582



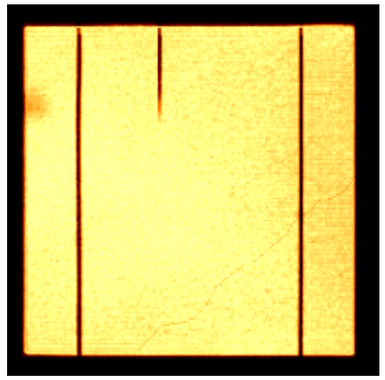
Learned U-TV
MSE 0.00342 | PSNR 24.67 dB
SSIM 0.631 | HaarPSI 0.659



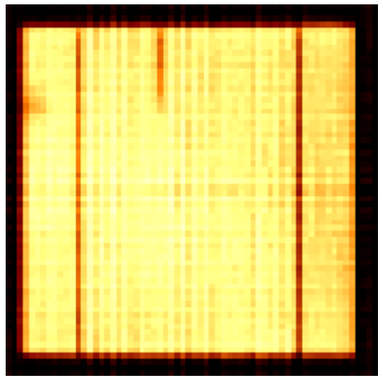
Silicone / image0611 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

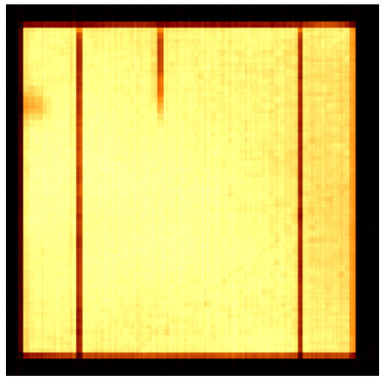


5% sampling



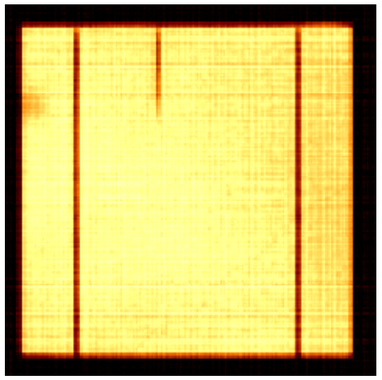
Zero-filled: matched multilevel
MSE 0.00879 | PSNR 20.56 dB
SSIM 0.407 | HaarPSI 0.314

10% sampling



Zero-filled: matched multilevel
MSE 0.00604 | PSNR 22.19 dB
SSIM 0.637 | HaarPSI 0.499

20% sampling



Zero-filled: matched multilevel
MSE 0.00491 | PSNR 23.09 dB
SSIM 0.530 | HaarPSI 0.540

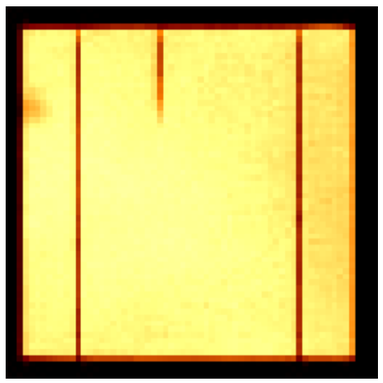
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

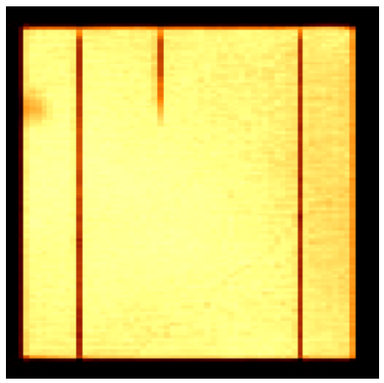
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

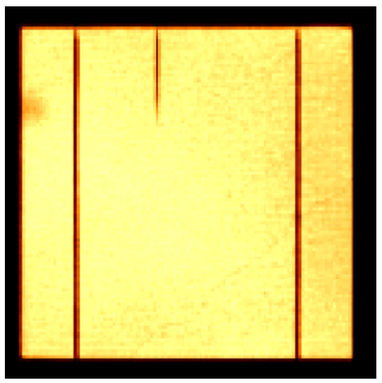
Single learned scale key; coarse J=6; PDHG=512.



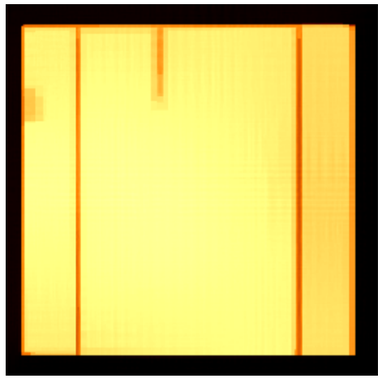
Zero-filled: in-order
MSE 0.00633 | PSNR 21.98 dB
SSIM 0.687 | HaarPSI 0.447



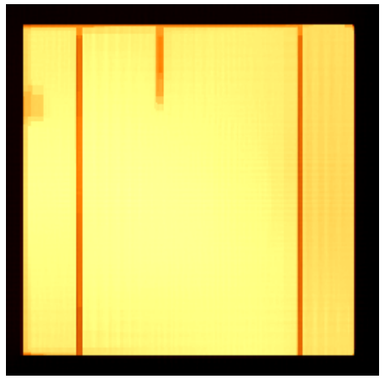
Zero-filled: in-order
MSE 0.00397 | PSNR 24.01 dB
SSIM 0.758 | HaarPSI 0.575



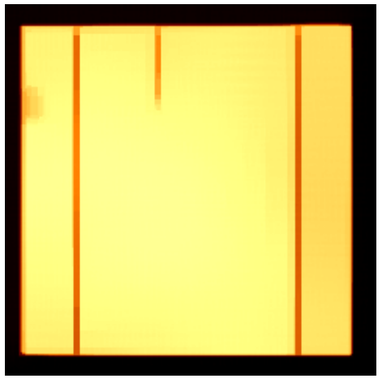
Zero-filled: in-order
MSE 0.00256 | PSNR 25.92 dB
SSIM 0.839 | HaarPSI 0.986



Learned U-TV
MSE 0.00461 | PSNR 23.37 dB
SSIM 0.598 | HaarPSI 0.502



Learned U-TV
MSE 0.00412 | PSNR 23.85 dB
SSIM 0.611 | HaarPSI 0.571

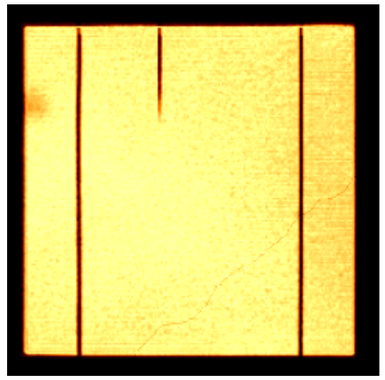


Learned U-TV
MSE 0.00328 | PSNR 24.84 dB
SSIM 0.635 | HaarPSI 0.671

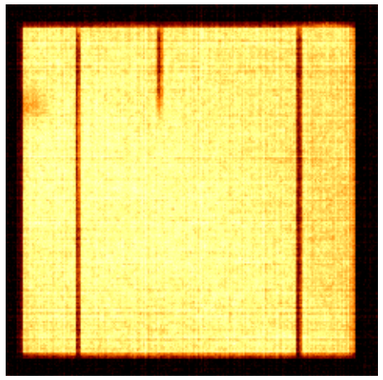
Silicone / image0611 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

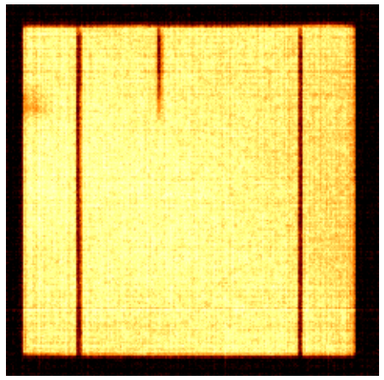


30% sampling



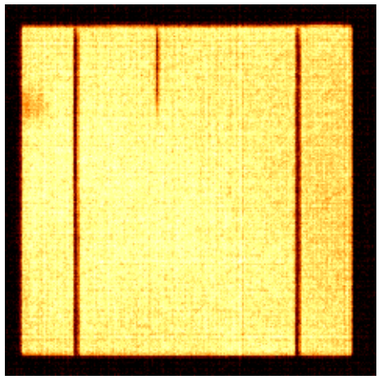
Zero-filled: matched multilevel
MSE 0.00540 | PSNR 22.67 dB
SSIM 0.414 | HaarPSI 0.560

40% sampling



Zero-filled: matched multilevel
MSE 0.00456 | PSNR 23.41 dB
SSIM 0.423 | HaarPSI 0.586

50% sampling



Zero-filled: matched multilevel
MSE 0.00426 | PSNR 23.70 dB
SSIM 0.430 | HaarPSI 0.636

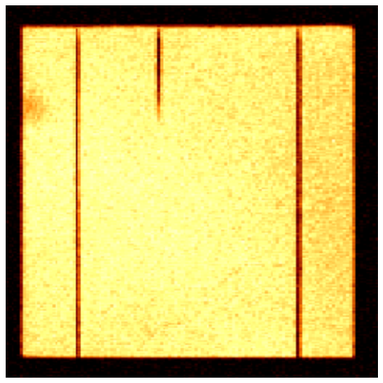
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

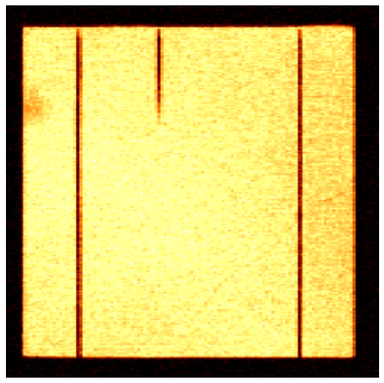
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

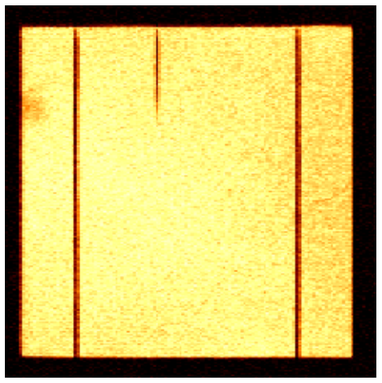
Single learned scale key; coarse J=6; PDHG=512.



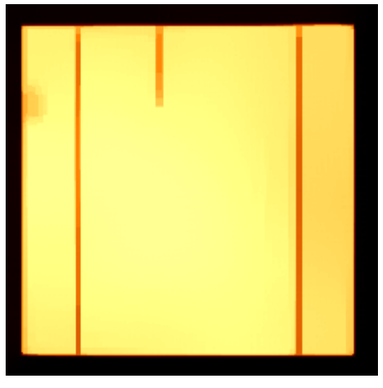
Zero-filled: in-order
MSE 0.00331 | PSNR 24.80 dB
SSIM 0.566 | HaarPSI 0.700



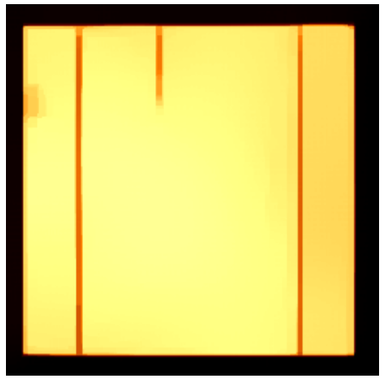
Zero-filled: in-order
MSE 0.00349 | PSNR 24.57 dB
SSIM 0.549 | HaarPSI 0.713



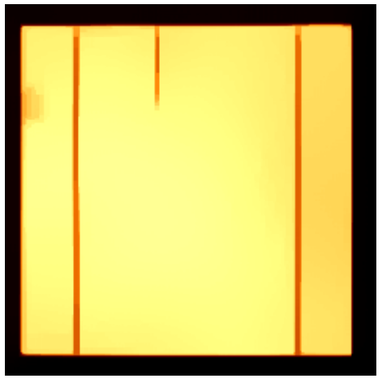
Zero-filled: in-order
MSE 0.00386 | PSNR 24.13 dB
SSIM 0.519 | HaarPSI 0.703



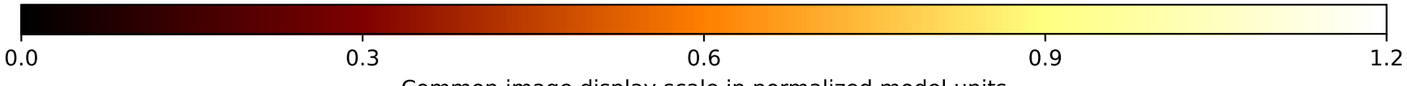
Learned U-TV
MSE 0.00336 | PSNR 24.73 dB
SSIM 0.634 | HaarPSI 0.663



Learned U-TV
MSE 0.00298 | PSNR 25.26 dB
SSIM 0.641 | HaarPSI 0.684



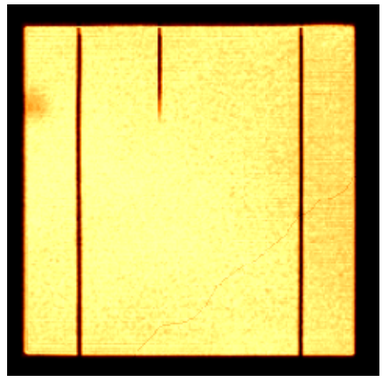
Learned U-TV
MSE 0.00286 | PSNR 25.43 dB
SSIM 0.642 | HaarPSI 0.677



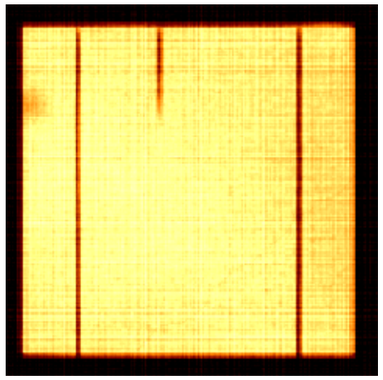
Silicone / image0611 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

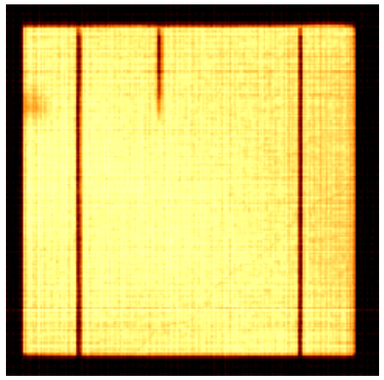


30% sampling



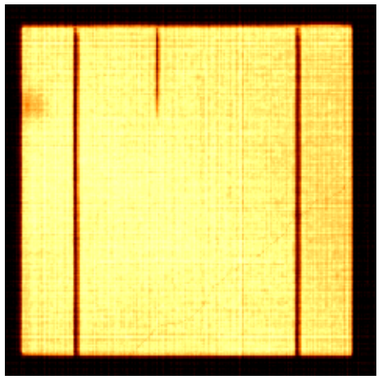
Zero-filled: matched multilevel
MSE 0.00416 | PSNR 23.81 dB
SSIM 0.532 | HaarPSI 0.605

40% sampling



Zero-filled: matched multilevel
MSE 0.00294 | PSNR 25.32 dB
SSIM 0.569 | HaarPSI 0.640

50% sampling



Zero-filled: matched multilevel
MSE 0.00221 | PSNR 26.56 dB
SSIM 0.625 | HaarPSI 0.722

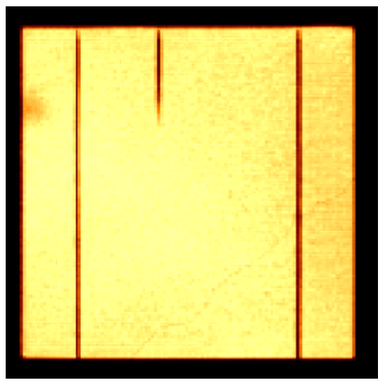
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

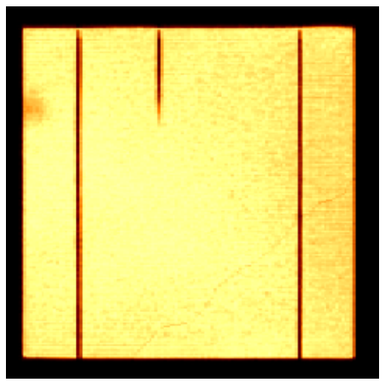
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

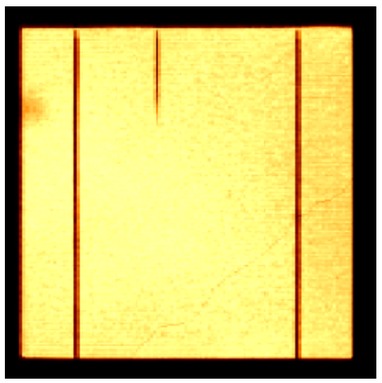
Single learned scale key; coarse J=6; PDHG=512.



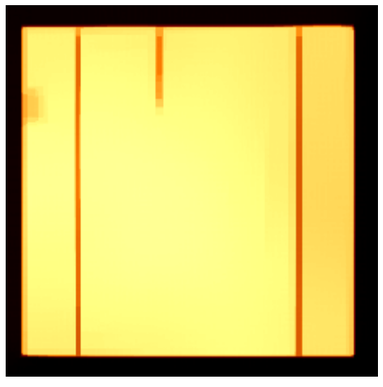
Zero-filled: in-order
MSE 0.00206 | PSNR 26.86 dB
SSIM 0.877 | HaarPSI 0.997



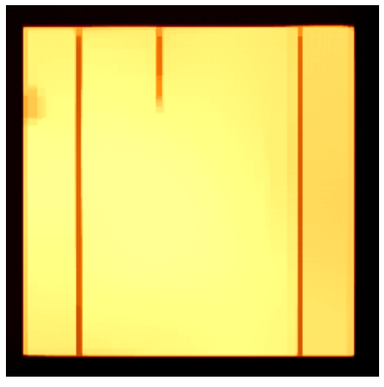
Zero-filled: in-order
MSE 0.00186 | PSNR 27.31 dB
SSIM 0.916 | HaarPSI 1.000



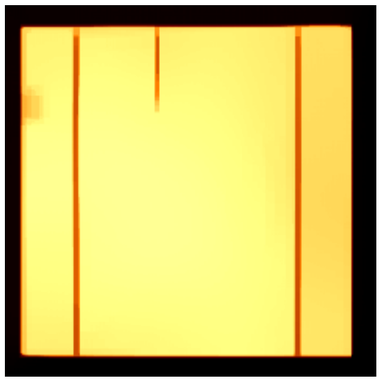
Zero-filled: in-order
MSE 0.00182 | PSNR 27.39 dB
SSIM 0.930 | HaarPSI 1.000



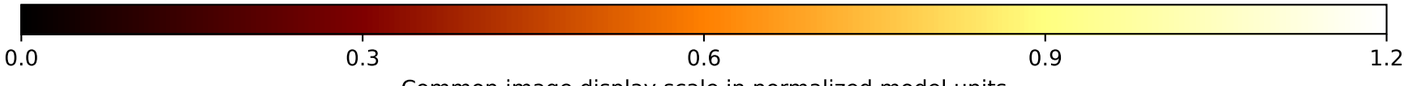
Learned U-TV
MSE 0.00314 | PSNR 25.03 dB
SSIM 0.637 | HaarPSI 0.670



Learned U-TV
MSE 0.00275 | PSNR 25.61 dB
SSIM 0.644 | HaarPSI 0.692



Learned U-TV
MSE 0.00249 | PSNR 26.04 dB
SSIM 0.650 | HaarPSI 0.693



Silicone / image0249 | Fixed NPL-default noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

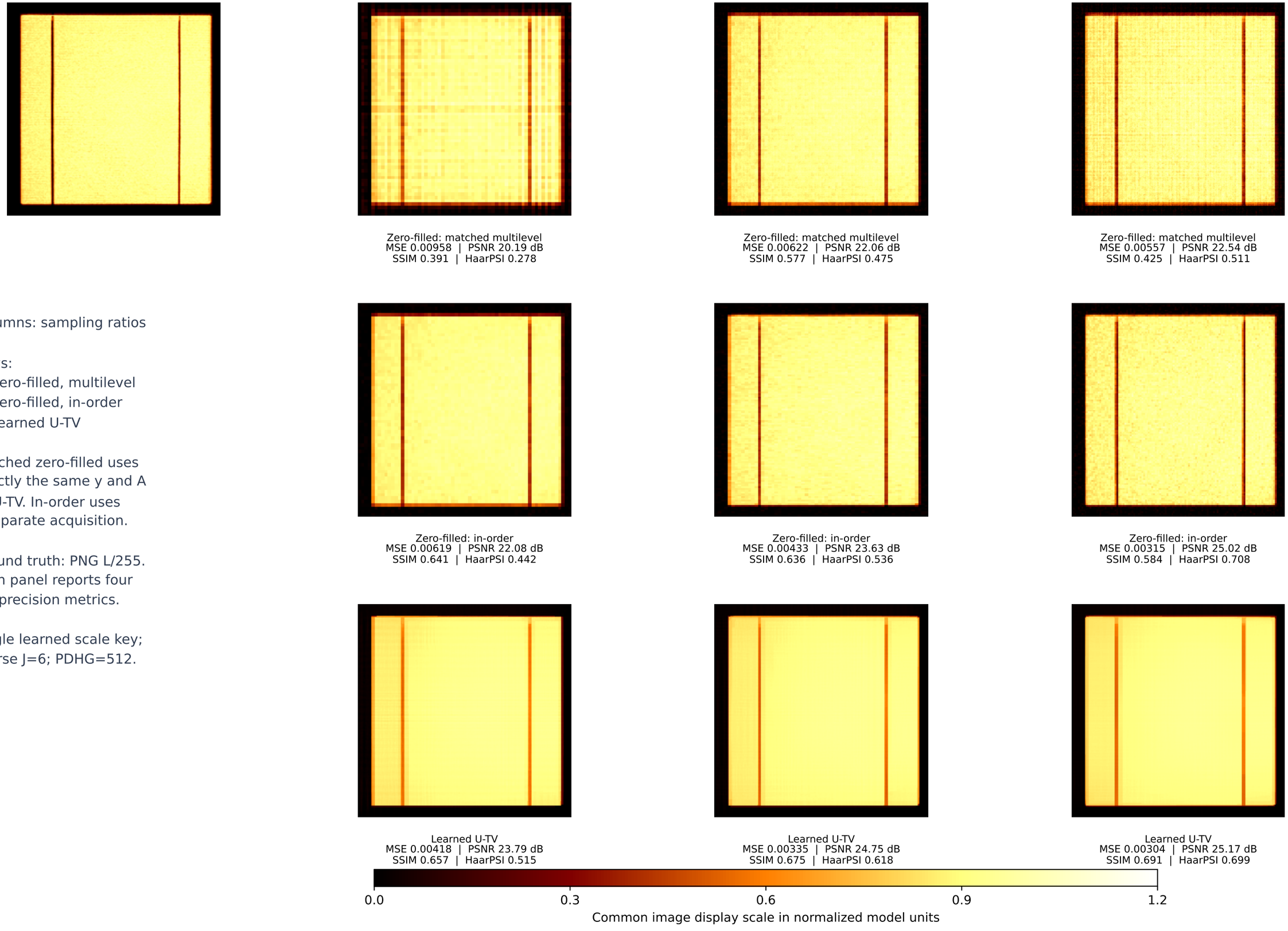
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

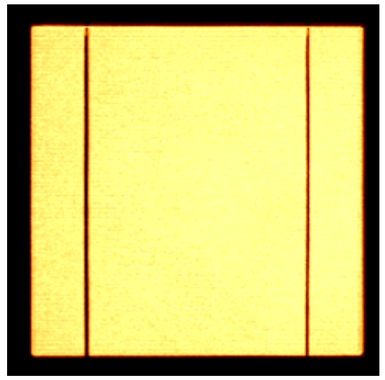
Single learned scale key; coarse J=6; PDHG=512.



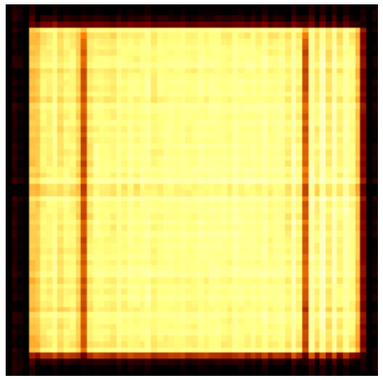
Silicone / image0249 | No simulated noise | 5-20% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

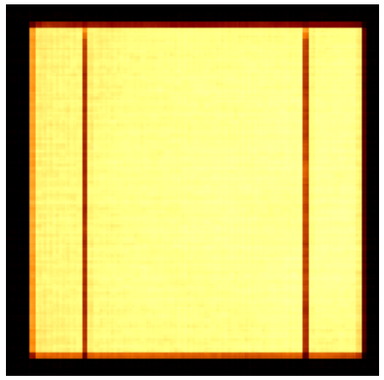


5% sampling



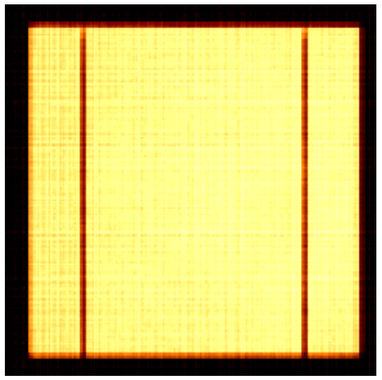
Zero-filled: matched multilevel
MSE 0.00938 | PSNR 20.28 dB
SSIM 0.406 | HaarPSI 0.279

10% sampling



Zero-filled: matched multilevel
MSE 0.00581 | PSNR 22.36 dB
SSIM 0.709 | HaarPSI 0.506

20% sampling



Zero-filled: matched multilevel
MSE 0.00475 | PSNR 23.24 dB
SSIM 0.533 | HaarPSI 0.549

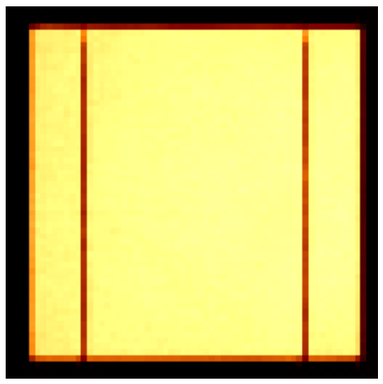
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

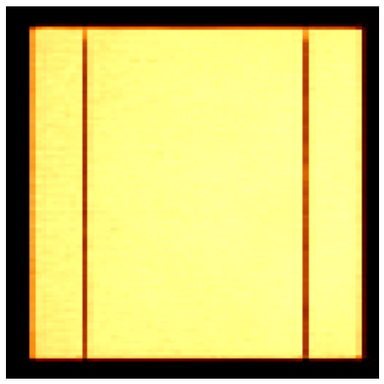
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

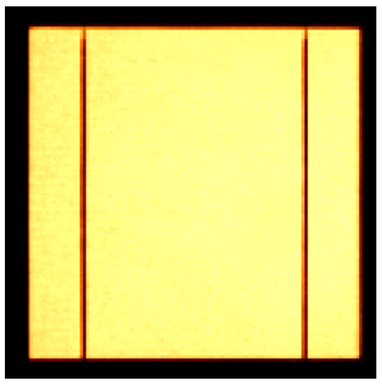
Single learned scale key; coarse J=6; PDHG=512.



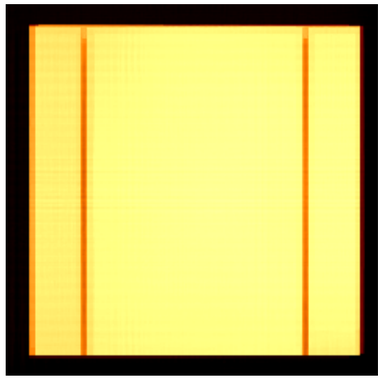
Zero-filled: in-order
MSE 0.00598 | PSNR 22.23 dB
SSIM 0.740 | HaarPSI 0.461



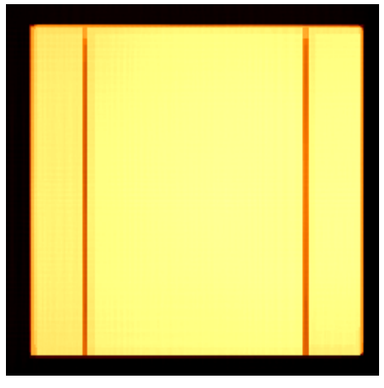
Zero-filled: in-order
MSE 0.00393 | PSNR 24.05 dB
SSIM 0.794 | HaarPSI 0.584



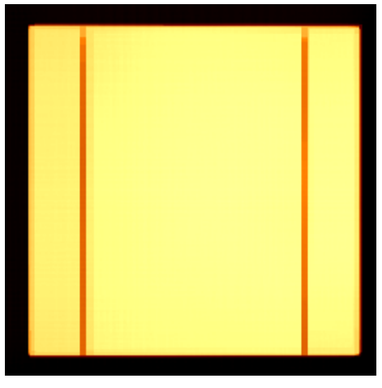
Zero-filled: in-order
MSE 0.00232 | PSNR 26.34 dB
SSIM 0.854 | HaarPSI 0.990



Learned U-TV
MSE 0.00425 | PSNR 23.72 dB
SSIM 0.658 | HaarPSI 0.507



Learned U-TV
MSE 0.00328 | PSNR 24.84 dB
SSIM 0.675 | HaarPSI 0.617

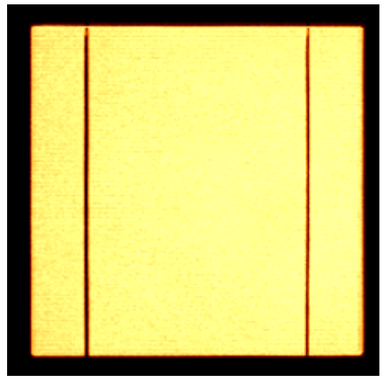


Learned U-TV
MSE 0.00296 | PSNR 25.28 dB
SSIM 0.693 | HaarPSI 0.700

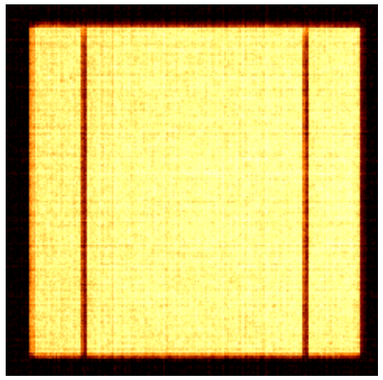
Silicone / image0249 | Fixed NPL-default noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

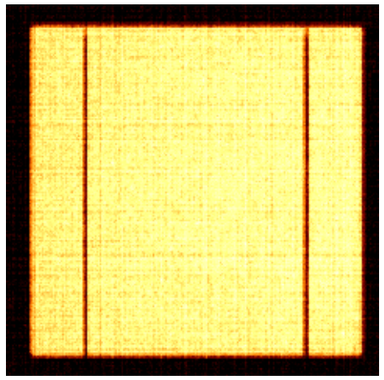


30% sampling



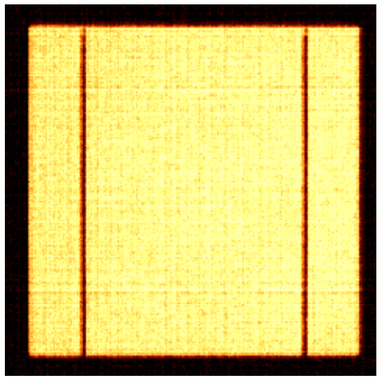
Zero-filled: matched multilevel
MSE 0.00513 | PSNR 22.90 dB
SSIM 0.402 | HaarPSI 0.521

40% sampling



Zero-filled: matched multilevel
MSE 0.00451 | PSNR 23.46 dB
SSIM 0.392 | HaarPSI 0.561

50% sampling



Zero-filled: matched multilevel
MSE 0.00451 | PSNR 23.46 dB
SSIM 0.389 | HaarPSI 0.591

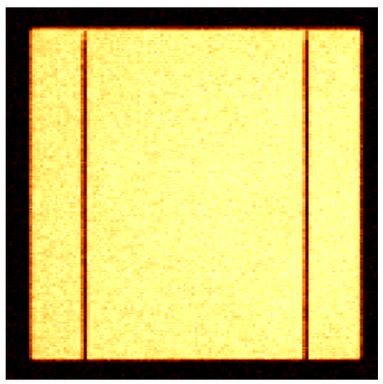
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

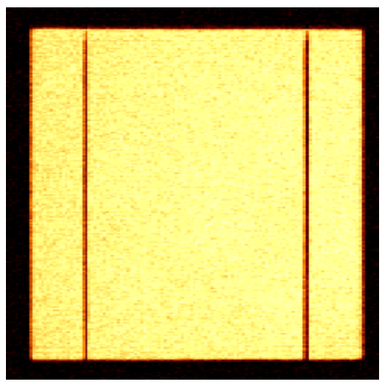
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

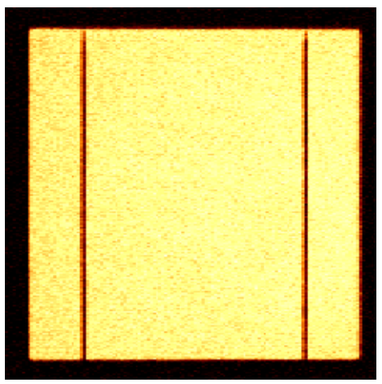
Single learned scale key; coarse J=6; PDHG=512.



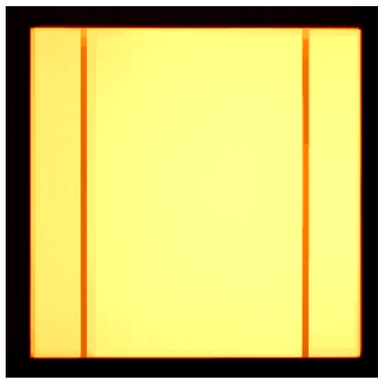
Zero-filled: in-order
MSE 0.00311 | PSNR 25.07 dB
SSIM 0.541 | HaarPSI 0.712



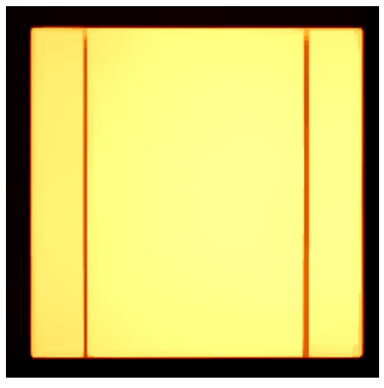
Zero-filled: in-order
MSE 0.00326 | PSNR 24.87 dB
SSIM 0.523 | HaarPSI 0.704



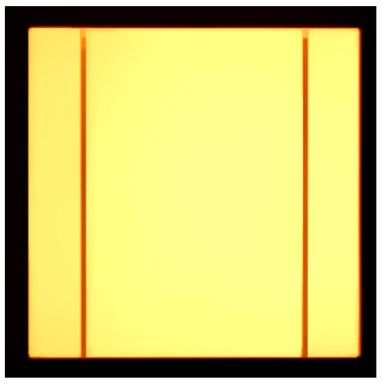
Zero-filled: in-order
MSE 0.00367 | PSNR 24.35 dB
SSIM 0.486 | HaarPSI 0.706



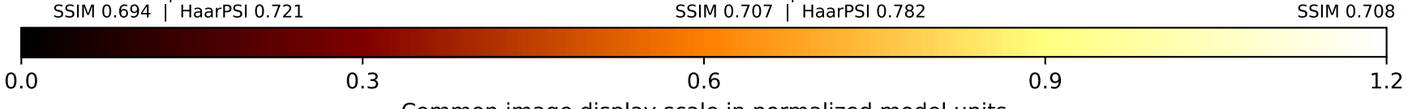
Learned U-TV
MSE 0.00291 | PSNR 25.36 dB
SSIM 0.694 | HaarPSI 0.721



Learned U-TV
MSE 0.00202 | PSNR 26.94 dB
SSIM 0.707 | HaarPSI 0.782



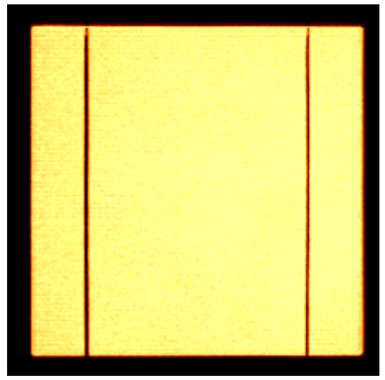
Learned U-TV
MSE 0.00200 | PSNR 26.98 dB
SSIM 0.708 | HaarPSI 0.791



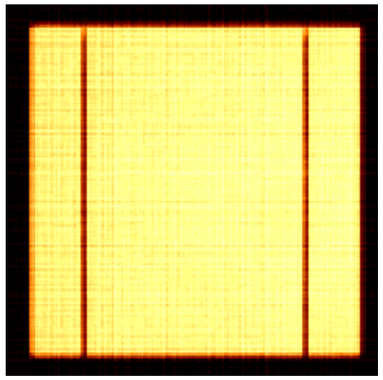
Silicone / image0249 | No simulated noise | 30-50% sampling

Shared display: afmhot [0,1.2]; metrics from original reconstruction tensors, NOT rendered PNGs.

Ground truth

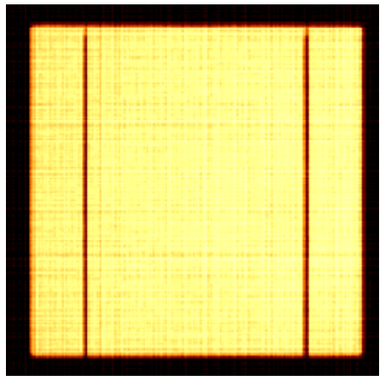


30% sampling



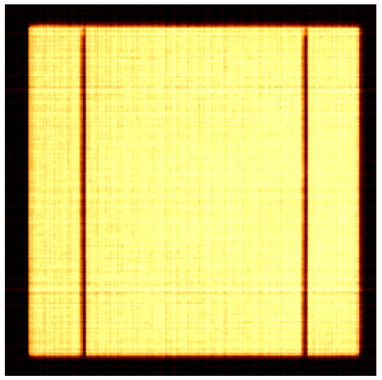
Zero-filled: matched multilevel
MSE 0.00391 | PSNR 24.08 dB
SSIM 0.529 | HaarPSI 0.558

40% sampling



Zero-filled: matched multilevel
MSE 0.00286 | PSNR 25.44 dB
SSIM 0.542 | HaarPSI 0.620

50% sampling



Zero-filled: matched multilevel
MSE 0.00246 | PSNR 26.09 dB
SSIM 0.579 | HaarPSI 0.666

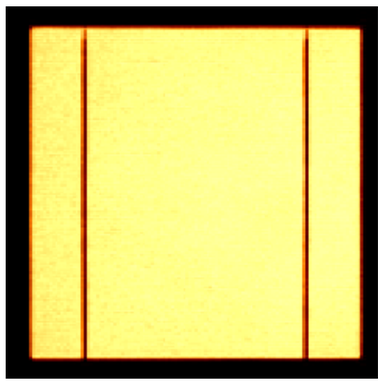
Columns: sampling ratios

Rows:
1. Zero-filled, multilevel
2. Zero-filled, in-order
3. Learned U-TV

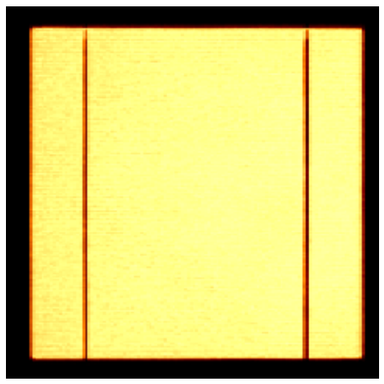
Matched zero-filled uses exactly the same y and A as U-TV. In-order uses a separate acquisition.

Ground truth: PNG L/255. Each panel reports four full-precision metrics.

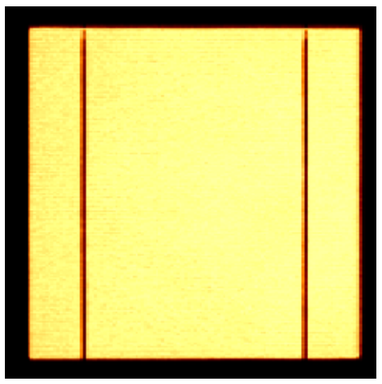
Single learned scale key; coarse J=6; PDHG=512.



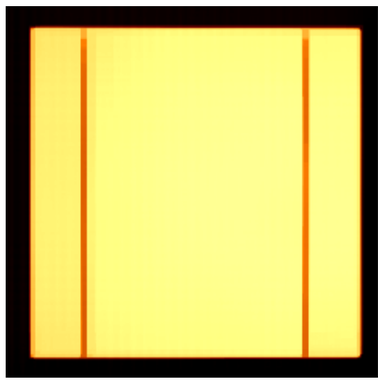
Zero-filled: in-order
MSE 0.00187 | PSNR 27.28 dB
SSIM 0.888 | HaarPSI 0.998



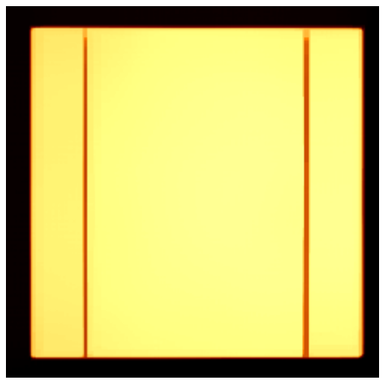
Zero-filled: in-order
MSE 0.00164 | PSNR 27.84 dB
SSIM 0.931 | HaarPSI 1.000



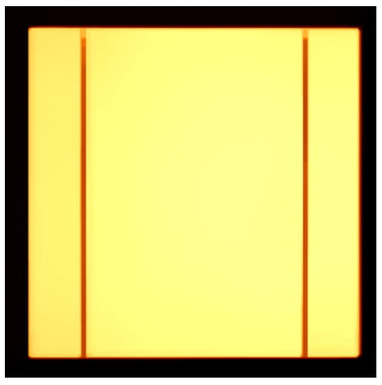
Zero-filled: in-order
MSE 0.00162 | PSNR 27.91 dB
SSIM 0.943 | HaarPSI 1.000



Learned U-TV
MSE 0.00273 | PSNR 25.63 dB
SSIM 0.697 | HaarPSI 0.722



Learned U-TV
MSE 0.00181 | PSNR 27.42 dB
SSIM 0.713 | HaarPSI 0.799



Learned U-TV
MSE 0.00178 | PSNR 27.49 dB
SSIM 0.713 | HaarPSI 0.809

